Introduction to Artificial Intelligence

Mihir Rao

Contents

1 Introduction to AI **4**
1.1 Rational decisions . . . 4
1.2 Agents and environments . . . 6
1.3 Methods and applications . . . 8
1.4 From an application to a computational problem . . . 9

2 Uninformed Search **10**
2.1 Planning and search problems . . . 10
2.2 State graphs and search trees . . . 11
2.3 Depth-first and breadth-first search . . . 12
2.4 Uniform-cost search . . . 14

3 A* Search and Heuristics **17**
3.1 Informed search . . . 17
3.2 Constructing heuristics . . . 19
3.3 Graph search and consistency . . . 20
3.4 Why closing states needs more care . . . 21

4 Constraint Satisfaction Problems I **23**
4.1 Representation and backtracking . . . 23
4.2 Filtering and arc consistency . . . 26
4.3 Ordering . . . 28

5 Constraint Satisfaction Problems II **29**
5.1 Consistency and structure . . . 29
5.2 Local search . . . 32
5.3 What local search keeps and what it loses . . . 32

6 Game Trees: Minimax **35**
6.1 Games and minimax values . . . 35
6.2 Alpha-beta pruning . . . 37
6.3 Depth limits and evaluation . . . 39

7 Game Trees: Expectimax and Utilities **41**
7.1 Chance nodes and opponent models . . . 41
7.2 Preferences over lotteries . . . 43
7.3 Money and risk . . . 45

8 Markov Decision Processes I **47**
8.1 States, rewards, and policies . . . 47
8.2 Bellman optimality and time-limited values . . . 49
8.3 Reading a Bellman backup . . . 50

9 Markov Decision Processes II **53**
9.1 Policy evaluation and extraction . . . 53
9.2 Policy iteration . . . 54
9.3 From planning to learning . . . 57

10 Reinforcement Learning I **58**
10.1 Learning a model or learning values . . . 58
10.2 Temporal differences . . . 60
10.3 Q-learning . . . 62

11 Reinforcement Learning II **63**
11.1 Exploration and regret . . . 63
11.2 Approximate Q-learning . . . 65
11.3 Policy search . . . 67

12 Probability **68**
12.1 Distributions, events, and conditioning . . . 68
12.2 Product, chain, and Bayes rules . . . 70
12.3 Inference by enumeration . . . 73

13 Bayes Nets: Representation **74**
13.1 Conditional independence . . . 74
13.2 Graph and local probabilities . . . 75
13.3 Representation and causality . . . 78

14 Bayes Nets: Independence **79**
14.1 Three local configurations . . . 79
14.2 D-separation . . . 81

15 Bayes Nets: Inference **84**
15.1 Factors and elimination . . . 84
15.2 Evidence and normalization . . . 86
15.3 Ordering and complexity . . . 87

16 Bayes Nets: Sampling **90**
16.1 Prior and rejection sampling . . . 90
16.2 Likelihood weighting . . . 91
16.3 Gibbs sampling . . . 94

17 Decision Networks and Value of Information **96**
17.1 Decisions and expected utility . . . 96
17.2 Value of information . . . 97
17.3 Partial observability . . . 100

18 Hidden Markov Models **102**
18.1 Markov chains and stationary distributions . . . 102
18.2 Hidden states and observations . . . 103
18.3 Filtering and the forward algorithm . . . 104
18.4 A complete weather-filtering calculation . . . 106

19 Particle Filtering and Applications of HMMs **108**
19.1 Particles as a belief representation . . . 108
19.2 Dynamic Bayes nets . . . 110
19.3 Most likely state sequences . . . 111

20 Machine Learning: Naive Bayes **114**
20.1 Classification and the model . . . 114
20.2 Parameter estimation and smoothing . . . 116
20.3 Generalization and evaluation . . . 118

21 Machine Learning: Perceptrons and Logistic Regression **120**
21.1 Linear scores and perceptron updates . . . 120
21.2 Logistic probabilities . . . 123
21.3 Multiclass logistic regression . . . 124

22 Machine Learning: Optimization and Neural Networks **126**
22.1 Gradient optimization . . . 126
22.2 Learning representations . . . 128
22.3 Approximation and automatic differentiation . . . 130

23 Machine Learning: Neural Networks II and Decision Trees **133**
23.1 Induction, consistency, and simplicity . . . 133
23.2 Decision trees and expressiveness . . . 134
23.3 Information gain and recursive learning . . . 135
23.4 Overfitting and pruning . . . 137

24 Advanced Applications: Robotics, Vision, and Control **140**
24.1 Combining search and learning . . . 140
24.2 Helicopter estimation and control . . . 141
24.3 Locomotion and rewards . . . 145
24.4 Autonomous driving and manipulation . . . 146

25 Advanced Topics and Final Contest **148**
25.1 Cooperative game strategies . . . 148
25.2 Speech as sequential inference . . . 150
25.3 Connecting inference to action . . . 153

Lecture 1: Introduction to AI

1.1 Rational decisions

- **The goal of the course.** Artificial intelligence can be approached through four goals: making machines think like people, act like people, think rationally, or act rationally. This course focuses on the last: choosing actions that best achieve specified goals. A decision can be rational even if it does not resemble human thinking.
- **The decision rule.** A rational decision maximizes expected utility, a numerical measure of the desirability of an outcome. If h denotes the information available when choosing an action, O is the set of possible outcomes, and $U(o)$ is the utility of outcome o, then for a finite outcome set the rule is

  $$a^* \in \arg\max_a \sum_{o \in O} P(o \mid h, a)\, U(o).$$

  For each action, weight each possible outcome's utility by its probability, then choose an action with the largest total. Rationality is relative to the goals and information available at decision time: an unlikely bad outcome can occur even when the chosen action had the highest expected utility. The criterion evaluates the decision, not whether its internal reasoning imitates a person.
- **Why uncertainty matters.** The word *expected* matters. An agent usually chooses before it knows which outcome will occur. It should compare the whole distribution of outcomes for each action, not assume that the most likely outcome is certain. Utility also matters: a high probability of success is not enough if the cost of failure is unacceptable.

Example: As a small illustration, suppose a robot can use a safe route worth 6 utility units for sure, or a shortcut worth 10 with probability 0.7 and -5 otherwise. The shortcut's expected utility is $0.7(10) + 0.3(-5) = 5.5$, so the safe route is preferred. If the success probability rises to 0.9, the shortcut's value becomes 8.5 and the decision changes. The same actions can therefore lead to different rational choices when the information changes.

Rationality is evaluated using information available at the time. A decision can be sensible and still end badly. Conversely, a poor decision can get lucky. Judging only the eventual outcome confuses decision quality with chance. The agent's goal is another input: a route that is best for speed may be poor for energy use or safety.

1.1.1 How an objective becomes a decision

A utility function is a modeling choice. Suppose the two routes reach the same destination, but one uses less energy and the other is faster. Merely saying "reach the destination" leaves the choice unresolved. The utility must express how the agent trades time, energy, and the risk of failing. Probability describes what the agent believes will happen; utility describes how much it values the outcomes. Mixing up these roles leads to mistakes: a rare outcome can still dominate a decision if its cost is large enough.

The route example can be solved for the probability at which the preferred action changes. Let p be the shortcut's success probability. Its expected utility is $10p - 5(1 - p) = 15p - 5$. It is better than the safe route exactly when

$$15p - 5 > 6, \qquad p > \frac{11}{15}.$$

At $p = 11/15$ both actions are optimal under this model. This threshold makes the role of information explicit: improving the estimated success probability from 0.7 to 0.72 does not change the decision, while improving it to 0.8 does. A more accurate prediction is useful for action only insofar as it can affect a consequential choice.

A utility number need not be a monetary amount. It is a way to compare outcomes, including tradeoffs that are difficult to express in a single physical unit. The agent does not choose the utility function by solving the expectation. That function is part of the problem specification. Optimizing the wrong objective very effectively can produce unwanted behavior. For example, maximizing the number of dots collected on the next move can send Pac-Man into a dead end even when surviving to collect more food later is the intended goal.

1.1.2 Human behavior and rational behavior

The four views of AI ask different questions. A system that thinks like a person is evaluated as a model of human mental processes. A system that acts like a person is evaluated by its outward behavior. A system that thinks rationally emphasizes valid reasoning, while a system that acts rationally is evaluated by the quality of its choices. These goals overlap, but one does not automatically establish another. A valid deduction can be useless for the action at hand, and a good action can be chosen without reproducing a human chain of thought.

Human brains motivate the importance of memory and computation, but an AI algorithm need not imitate their physical organization. A route planner can compare thousands of possibilities in a way that a human traveler does not. The useful question for this course is whether the resulting decision fits the stated information and objective, and whether the computation is feasible.

1.1.3 Worked example: the most likely outcome is not the whole decision

Consider an illustrative delivery robot. Route A takes 10 minutes with probability 0.9 and 60 minutes with probability 0.1. Route B always takes 16 minutes. The question is which route the stated objective prefers.

1. **Specify the outcomes.** Route A has two possible arrival times; route B has one. Comparing only 10 with 16 discards the delay event.
2. **Minimize expected time.** The expectations are $0.9(10) + 0.1(60) = 15$ minutes and 16 minutes, so this objective selects A. A bad realized journey does not retroactively make that calculation irrational.
3. **Change the objective explicitly.** Suppose arriving after 20 minutes incurs an additional penalty of 100 utility-cost units, and one minute also costs one unit. The expected costs become $0.9(10) + 0.1(60 + 100) = 25$ for A and 16 for B. Now B is preferred.
4. **Interpret the reversal.** The probabilities did not change. The consequences that matter to the task changed. An agent cannot choose rationally until both its uncertainty and its objective are specified.

This also separates a prediction problem from a decision problem. Predicting the likely arrival time is useful input, but does not by itself determine whether the risk of a large delay is acceptable.

1.2 Agents and environments

An AGENT perceives its environment and acts on it. Sensors deliver PERCEPTS, and actuators carry out actions. The environment then changes and supplies new percepts, so decisions form a continuing interaction. The nature of the observations, available actions, and environment determines which techniques are appropriate.

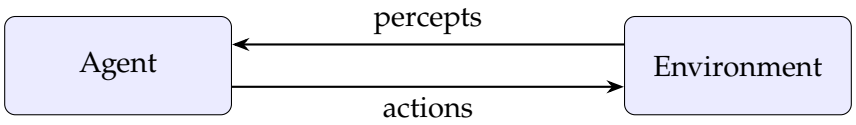


Figure 1: An agent interacts with its environment.

For Pac-Man, the environment is the game, percepts describe what the agent can observe, and actions move Pac-Man through the maze. The agent's objective specifies what counts as a good outcome; seeing the same game does not by itself determine what the agent should prefer.

A REFLEX AGENT chooses from its current percept, possibly using memory, without explicitly evaluating future consequences. A PLANNING AGENT uses a model to consider how candidate actions might change the world and selects a plan in light of its goal. A reflex rule can still be rational when

it selects an optimal action for the setting; planning is one way to make decisions, not the definition of rationality.

The distinction between reflex and planning agents concerns *how an action is selected*, not whether the agent stores anything. A reflex agent can maintain an internal estimate of the world and still apply a rule to that estimate. A planning agent goes further: it predicts possible future states and compares them before choosing.

For example, a rule that turns away from a nearby wall can work well in a simple corridor. At a junction, choosing the branch that eventually reaches food requires comparing consequences beyond the next observation. Planning also has a cost. Looking farther ahead takes computation, and a detailed model can be expensive to use.

An explicit policy stores an action for each relevant situation. Once such a policy has been computed, using it can look like a reflex even if extensive planning was needed to obtain it. Thus the cost of constructing a policy and the cost of executing it are separate questions.

1.2.1 Walking through one decision cycle

Consider Pac-Man approaching a junction. First, sensors provide the current observation: nearby walls, food, and any visible ghosts. The program turns that observation and its memory into an internal description. A reflex rule might choose the first legal direction containing food. A planning program instead creates hypothetical next states for each legal direction, considers what can happen afterward, and compares the resulting outcomes.

The hypothetical moves happen inside the model. The program does not move Pac-Man left in the actual game, undo that action, and then try right. After comparing possibilities, it executes one chosen action. The next observation may reveal that a ghost moved differently from the prediction, so the agent updates its description and makes another decision. This is why planning and replanning belong together: a computed plan is conditional on the model and on what was known when it was computed.

The model can fail even when the search is correct. If it assumes ghosts stay still, finding the best path in that model does not make the actual path safe. If its state forgets which food was collected, it may predict a reward that is no longer available. These are representation errors rather than errors in the rule used to compare candidate plans.

1.2.2 Worked example: one percept can represent different situations

Imagine a vacuum agent that currently observes “this square is clean.” There are two squares, left and right. The other square is outside its sensor.

1. **List compatible worlds.** In world w_1 both squares are clean. In world w_2 the current square is clean and the other is dirty. The current percept is identical in the two worlds.
2. **Identify the decision-relevant difference.** With a movement cost, stopping is attractive in w_1, while visiting the other square can be worthwhile in w_2. A reflex rule based only on the current percept cannot distinguish them.
3. **Use remembered information.** If the agent previously visited the other square and cleaned it, that history can support stopping. If dirt can reappear, the same memory is less decisive; elapsed time and the dirt model may matter.
4. **Choose a sufficient internal description.** The internal state can record location, remembered cleanliness, and uncertainty about changes. It should be updated after both observations and actions. It is a summary for future decisions, rather than a complete recording of every past event.

The example shows why an agent's input at one instant is not automatically a complete state. Observability is a property of what the sensors reveal about the task, not of how much computation the agent performs afterward.

1.3 Methods and applications

AI systems combine computation and data. Search and planning compare possible actions; constraint satisfaction finds assignments that obey specified rules; adversarial and uncertain settings require reasoning about other agents or uncertain outcomes. Probabilistic models represent uncertainty, decision theory connects uncertainty to utility, and machine learning uses data to improve predictions or decisions.

The same principles appear in different applications:

- **Language:** speech recognition maps audio to text, speech synthesis maps text to audio, and language systems support translation, question answering, and text classification.
- **Vision:** perception turns pixels into information, such as object identities, scene labels, or three-dimensional structure.
- **Robotics:** perception, planning, and control connect decisions to physical actions. The real world can differ from a simulator.
- **Games and logic:** game-playing systems reason over moves; logical systems use deduction, constraint satisfaction, and satisfiability to solve structured problems.

The field has used both knowledge-based and statistical approaches. Recent systems in the 2018 course draw on large datasets, computation, and neural networks; the appropriate representation and objective still depend on the task.

1.4 From an application to a computational problem

To build an agent, first make the decision problem explicit. What can it observe? Which actions can it take? What changes after an action? What outcomes does the objective reward? These choices determine what the agent must represent and what it can safely ignore.

In a maze, a route planner may need only locations and legal moves. If ghosts affect survival, their positions or a belief about their positions become relevant too. If the objective is to collect all food, remembering only the current location is insufficient because the remaining food depends on the past. Choosing a representation is therefore part of solving the problem, not merely a coding detail.

The major methods fit different gaps in this description. Search finds a good sequence when the model is known. Game search accounts for other decision makers. Probability represents unknown quantities. Learning estimates models or policies from examples and experience. A complete application often uses several of these methods together.

1.4.1 How computation and data complement each other

The distinction between computation and data is useful when choosing a method. If all roads and their travel costs are known, the remaining task may be to search efficiently for a route. If traffic costs are unknown, data are needed to estimate them before or during planning. More search cannot by itself reveal an unknown road closure, while more observations do not by themselves identify the cheapest route through a large network. A system needs a representation that allows the estimates and the planner to work together.

The field's historical changes reflect different ways of supplying this information. Early programs demonstrated search and symbolic reasoning. Knowledge-based systems emphasized explicit rules provided by people. Statistical approaches emphasized learning from observations and reasoning with uncertainty. The neural-network successes discussed in the 2018 course combine large datasets with substantial computation. These approaches do not remove the need to specify a task, choose a representation, and evaluate whether the decisions meet the objective.

For language, the observations may be audio or written words, while the output might be a transcript, translation, or label. For vision, pixels must be converted into information such as objects and geometry. For robotics, those estimates feed actions that change the physical world. For games, other players change which futures are plausible. The general techniques are valuable because these different applications share recurring structures: search over alternatives, uncertainty about hidden quantities, and learning from incomplete experience.

Source. Pieter Abbeel and Dan Klein, Introduction to Artificial Intelligence, Fall 2018, Lecture 1 and Note 1.

Lecture 2: Uninformed Search

2.1 Planning and search problems

- **Planning in a model.** A planning agent considers possible consequences before acting. It needs a model of how actions change the world and a goal test. A plan can be computed in advance or recomputed as new observations arrive. Search evaluates possible plans in the model, so its conclusions depend on the model's accuracy.
- **The search specification.** A search problem specifies a state space, a start state, a goal test, and a successor function. For a state, the successor function gives the available actions, resulting states, and step costs. A solution is a sequence of actions taking the start state to a goal; its cost is the sum of step costs. An optimal solution has minimum cost among all solutions.
- **Choosing the state.** A search state is an abstraction of the world: it retains the information needed to determine successors, costs, and goal status. For pathfinding in a maze, a position may suffice. To eat every dot, the state must also record which dots remain. If there are n possible positions and k dots, this representation has at most $n2^k$ states. Adding irrelevant details enlarges the search; omitting a detail that affects future actions or the goal makes the model inadequate.

The abstraction must contain enough information to predict the next search state. Two world situations may be merged only when their differences do not matter for the planned decisions. For pathfinding, facing direction can be omitted if turning is free and every direction is immediately available. If turning has a cost, it must be represented.

For the eat-all-dots problem, a state can be written (p, F), where p is the agent's position and F is the set of remaining food locations. Moving to p' gives the successor $(p', F \setminus \{p'\})$. Returning to the same position after eating a dot is a different state because F has changed. A search state should therefore describe the future-relevant situation, rather than merely where the agent is.

2.1.1 Counting states before choosing an algorithm

State counts multiply when independent pieces of information must be stored. In the maze illustration, suppose Pac-Man has 120 possible positions, there are 30 food locations, each of two ghosts has 12 possible positions, and Pac-Man faces one of four directions. An upper bound on world descriptions is

$$120 \cdot 2^{30} \cdot 12^2 \cdot 4.$$

The factor 2^{30} appears because each dot is either present or absent; it is not 30 possibilities for the total number of dots. Two configurations with the same count but different remaining locations can require different plans. Some combinations in this product may be unreachable, so this is a representation-size bound rather than an assertion that every combination occurs.

Pathfinding with stationary walls can discard the food, ghosts, and facing information if they do not affect the chosen route problem. Its state count then falls to 120. Eat-all-dots planning must retain the food subset, giving up to $120 \cdot 2^{30}$ states. If the task also requires keeping ghosts scared, remaining scared time and uneaten power pellets affect future safety and must be represented. The relevant abstraction changes with the objective and dynamics.

2.1.2 Worked example: a state must remember whether a key was collected

Take an illustrative corridor with locations S, K, D, G. The key is at K, and a locked door at D blocks the only route to goal G.

1. **Try a position-only state.** The agent can reach the square before the door once without the key and later after collecting it. Merging those situations gives the same state two different sets of legal successors. The successor function is then not well defined.
2. **Repair the representation.** Use (p, k), where p is location and $k \in \{0, 1\}$ says whether the key has been collected. Entering K sets $k' = 1$; other moves preserve k. Crossing the door is legal only if $k = 1$.
3. **Check repeated-state handling.** Reaching $(D, 0)$ does not justify discarding a later node for $(D, 1)$. Their future possibilities differ. Two nodes ending at $(D, 1)$ may be compared by path cost if the remaining model depends only on this state.
4. **Count the cost of the repair.** For n positions the product representation has at most $2n$ states. Extra memory is necessary here because it repairs a loss of information that affects feasibility.

If a key is consumed when used, the update must set its status back to zero. The representation is justified by the transition rules of the actual task; merely adding a variable with the right name is insufficient.

2.2 State graphs and search trees

In a STATE-SPACE GRAPH, each state is represented once, and an edge represents an action and its successor. A SEARCH TREE unfolds the possible plans from the start: its root is the start and each child extends its parent's plan by one action. A tree node represents a path as well as its end state. Thus different nodes can end at the same state, and a cycle in a finite state graph can generate an infinite search tree. Both structures are usually constructed only as needed.

Tree search maintains a FRINGE of partial plans. It removes a node according to an exploration strategy, checks whether its state is a goal, and otherwise expands it by adding successor plans. The strategy determines which plans are considered first. A goal is returned when selected from the fringe, which is especially important when path costs differ.

We compare strategies by COMPLETENESS, whether a solution is found when one exists, and OPTIMALITY, whether the returned solution has least cost. Let b be the maximum branching factor, m the maximum tree depth, and s the depth of the shallowest goal. These tree-search bounds count nodes, including repeated states; assume finite b.

A node normally stores its end state, a pointer to its parent, the action used to reach it, and its cumulative cost. Parent pointers reconstruct the answer after a goal is found: follow them to the root and reverse the actions. There is no need to copy an entire path into every node.

Algorithm 2.1: Tree search with a chosen fringe rule

```
Input: Start state s_0, successor function, and a fringe data structure
Insert a root node for s_0 into the fringe
while the fringe is not empty do
    Remove a node n according to the fringe rule
    if n ends at a goal then
        return the actions on n's parent chain
    for each legal successor (a, s', c) of n's state do
        Insert a child with state s', action a, and cost g(n) + c
return failure
```

The fringe represents alternatives still available. Expansion removes one alternative and replaces it with longer plans. A state can occur many times in this tree because different plans may reach it. This is why complexity in terms of tree depth can be large even when the underlying graph is small.

2.3 Depth-first and breadth-first search

DEPTH-FIRST SEARCH (DFS) expands a deepest fringe node first, using a last-in, first-out stack. It can pursue one branch indefinitely, so tree search is not complete for an infinite-depth tree. With finite depth it takes $O(b^m)$ time in the worst case. Its fringe holds at most the unexplored siblings along a root-to-leaf path, requiring $O(bm)$ space. DFS need not find a shallowest or least-cost solution.

BREADTH-FIRST SEARCH (BFS) expands a shallowest fringe node first, using a first-in, first-out queue. It explores by depth and is complete when branching is finite and a solution exists at finite depth. It finds a shallowest goal, which is optimal when all step costs are the same positive constant. Its time and space are exponential in solution depth; a conservative bound for the usual goal test on removal is $O(b^{s+1})$, because expansions at depth s before the goal can generate depth $s + 1$ nodes.

Iterative deepening runs depth-limited DFS repeatedly with increasing limits. It finds a shallowest goal with the space advantage of DFS. Although it revisits upper levels, for $b > 1$ most nodes lie near the deepest level, so the repeated work changes the asymptotic time by only a constant factor.

Example: Consider a root S with children A, B, and let A have children C, D. If successors are listed left to right, BFS selects $S, A, B, C, D, \ldots$. DFS follows one branch, selecting $S, A, C, \ldots$ before returning to the alternatives. A stack may need successors inserted in reverse order to realize that left-to-right traversal. The order among equal-priority nodes is a tie-breaking choice.

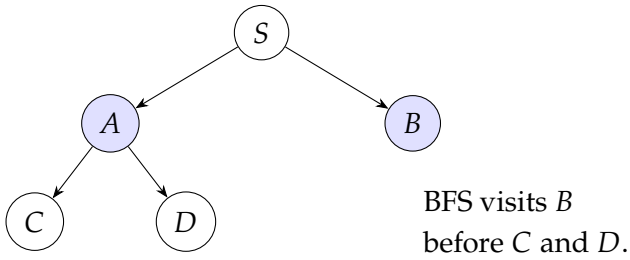


Figure 2: Depth and fringe order determine which partial plan is considered next.

BFS can be costly in memory because an entire wide layer remains available at once. DFS saves memory because it keeps only alternatives along its current branch, but that branch may never end. Iterative deepening trades repeated work near the root for a depth-first memory requirement. With a limit of L, a depth-limited search stops extending a branch at depth L; failure at that limit does not mean there is no deeper solution.

2.3.1 Tracing the fringe rather than only the visit order

Here is a complete small illustration of the two data structures. Let C have no children, let D be a goal, and let B have a child E. We show the next node to remove at the left. DFS inserts children so that the leftmost child is explored first.

Removed node	BFS fringe afterward	DFS fringe afterward
Before starting	$[S]$	$[S]$
S	$[A, B]$	$[A, B]$
A	$[B, C, D]$	$[C, D, B]$
Next removal	B, then fringe $[C, D, E]$	C, then fringe $[D, B]$
Next removal	C, then fringe $[D, E]$	D: stop
Next removal	D: stop	

The same successor function produces different search because insertion and removal preserve different alternatives. BFS reaches the depth-two goal only after examining the other depth-one node. DFS reaches it after finishing the chosen branch under A. If C instead led to an infinite chain, this same DFS ordering could postpone D forever.

For $b > 1$, a full tree through depth m contains

$$\sum_{i=0}^{m} b^i = \frac{b^{m+1} - 1}{b - 1}.$$

The last layer dominates this sum. BFS must retain a large part of a layer, whereas DFS keeps only pending siblings along its current path. For example, at depth 20 a binary layer alone has 2^{20} nodes, even though one root-to-leaf path has only 21 nodes. Low memory use says nothing by itself about finding a good answer: DFS saves memory by postponing alternatives.

Iterative deepening repeatedly pays for the shallow layers. Through final limit s, a node at depth i is generated in $s - i + 1$ iterations, so total work has the form

$$\sum_{i=0}^{s} (s - i + 1) b^i = b^s \sum_{j=0}^{s} \frac{j + 1}{b^j} = O(b^s) \quad (b > 1).$$

The convergent series explains why repeated work does not add a factor of s for a branching tree. For a single chain ($b = 1$), the cost instead is $1 + 2 + \cdots + (s + 1) = O(s^2)$. The branching assumption is doing real work.

2.4 Uniform-cost search

When step costs differ, the fewest actions need not yield the cheapest plan. Uniform-cost search (UCS) uses a priority queue ordered by the cumulative path cost g. It removes a cheapest partial plan, tests for a goal, and expands it if necessary. It explores in increasing cost contours, without using information about the direction of the goal.

Suppose branching is finite, every step costs at least $\epsilon > 0$, and an optimal goal has finite cost C^*. UCS is complete: only finitely many tree nodes can have cost at most C^*, since their depth is at most C^*/ϵ. It is also optimal. If a more expensive goal were removed first, a frontier prefix of an optimal path would have cost at most C^* and would have been selected before that goal. This argument relies on nonnegative step costs and on checking the goal when it is removed, not merely when it is generated. Time and space are exponential in the effective depth C^*/ϵ, with implementation-dependent boundary factors from generating the next level.

For a small cost illustration, suppose $S \to G$ costs 10, while $S \to A \to G$ costs $1 + 1$. BFS can prefer the single-edge route, but UCS first removes A at cost 1, then the goal through A at cost 2. The goal at cost 10 may already be in the queue; its presence does not justify stopping. The cheapest *complete* plan is known only when no cheaper unfinished plan can lead to a better answer.

The positive lower bound on step costs is needed for tree-search completeness: infinitely many ever-cheaper extensions could otherwise keep a goal waiting. Optimality and completeness are

distinct claims. Nonnegative costs support the optimality argument, while a positive lower bound plus finite branching ensures only finitely many relevant prefixes.

2.4.1 A full uniform-cost execution

Use the illustrative edges $S \to A$ of cost 2, $S \to B$ of cost 1, $B \to A$ of cost 0.5, $A \to G$ of cost 2, and $B \to G$ of cost 6. A queue entry records a whole path; write only its endpoint and total cost below, distinguishing repeated endpoints by their costs.

Removed path	Newly generated paths	Fringe in cost order
$S : 0$	$A : 2,\ B : 1$	$B : 1,\ A : 2$
$B : 1$	$A : 1.5,\ G : 7$	$A : 1.5,\ A : 2,\ G : 7$
$A : 1.5$	$G : 3.5$	$A : 2,\ G : 3.5,\ G : 7$
$A : 2$	$G : 4$	$G : 3.5,\ G : 4,\ G : 7$
$G : 3.5$	Stop	

The answer is $S \to B \to A \to G$, costing 3.5. Returning the first goal *generated* would give 7. Retaining only the first path generated to A would discard the better arrival of cost 1.5. A graph-search version can avoid expanding $A : 2$ once it knows that A was already expanded more cheaply; tree search keeps both paths. These are distinct improvements to the basic queue rule, not reasons to alter its goal test.

Why does removing a goal establish optimality? Every uncompleted route to a cheaper goal must have a prefix still on the fringe. Nonnegative costs make that prefix no more expensive than its eventual goal. Such a prefix would have been removed before the purported more expensive answer, a contradiction. Negative edges invalidate this argument: a currently expensive prefix might later receive a negative cost and become the best complete path. A positive minimum cost also rules out infinitely many cheap prefixes that could delay a reachable goal in a search tree.

Choosing a fringe rule is therefore a choice about what ordering can be justified. DFS favors depth and memory; BFS favors number of actions; UCS favors accumulated cost. None can compensate for a successor function that leaves out a legal action or assigns the wrong cost.

2.4.2 Worked example: when finding a goal is too early

In a new illustrative graph, $S \to G$ costs 9, $S \to A$ costs 2, and $A \to G$ costs 3. There are no other edges.

1. **Generate the first alternatives.** Expanding S creates goal node $(G, 9)$ and nonterminal node $(A, 2)$. The goal has been discovered, but its cost is only an upper bound on the best answer.
2. **Select by accumulated cost.** UCS removes $(A, 2)$ first and creates a new route to G of cost $2 + 3 = 5$.

3. **Return on removal.** The next cheapest available goal is $(G, 5)$. Its route is the optimal answer. Returning when $(G, 9)$ was generated would have failed despite using a priority queue elsewhere.
4. **Handle stale entries.** A queue may still contain $(G, 9)$. A best-cost table can mark that entry as obsolete. The table records the best known route, while permanent closure requires the removal-based argument.

BFS chooses the one-edge route here because it minimizes the number of edges, not their total cost. Its result is consistent with its depth objective but incorrect for this weighted problem. Always state both the priority rule and the point at which the goal test can terminate search.

Source. Pieter Abbeel and Dan Klein, Introduction to Artificial Intelligence, Fall 2018, Lecture 2.

Lecture 3: A* Search and Heuristics

3.1 Informed search

- **Greedy guidance.** A heuristic $h(s)$ estimates the remaining cost from state s to a nearest goal. Greedy search expands the node with smallest h, ignoring the cost already paid. It can be fast when the estimate points toward a goal, but can follow a misleading route and need not find a least-cost solution.
- **Balancing past and future cost.** A* search instead orders the fringe by $f(n) = g(n) + h(n)$, where $g(n)$ is the cost of the path represented by node n. With $h = 0$ it is uniform-cost search. A goal is tested when removed from the fringe: the first goal generated need not be the cheapest.
- **The lower-bound guarantee.** A heuristic is admissible if $0 \le h(s) \le h^*(s)$, where h^* is the true least remaining cost. In particular, h is zero at goals. For finite branching and step costs bounded below by a positive constant, A* tree search with an admissible heuristic is complete and optimal. To see optimality, suppose a suboptimal goal of cost $C > C^*$ is on the fringe. Some prefix n of an optimal path is also on the fringe until the optimal goal is removed. Admissibility gives $f(n) \le C^* < C = f(\text{goal})$, so the suboptimal goal cannot be removed first.

The score $g(n)$ looks backward: it is the cost already paid to reach this node. The score $h(n)$ looks forward: it estimates the cost still needed. Thus $f(n)$ estimates the total cost of a solution passing through this particular plan. Two nodes ending in the same state can have the same heuristic but different g values.

As an illustration, a frontier node with $g = 4$, $h = 3$ has $f = 7$, whereas one with $g = 1$, $h = 8$ has $f = 9$. A* selects the former even though reaching the latter has so far been cheaper. Greedy search would ignore that difference in past cost; UCS would ignore the estimated distance still to go.

The proof of optimality uses a *lower bound*, not a promise that the heuristic is accurate. For a prefix n of an optimal path,

$$g(n) + h(n) \le g(n) + h^*(n) = C^*.$$

For a suboptimal goal B, $f(B) = g(B) > C^*$. While that goal waits in the fringe, some unfinished prefix of the optimal path has a smaller score. This is the reason an admissible estimate cannot hide every optimal route behind a more expensive goal.

3.1.1 A complete A* trace

Consider the same illustrative cost graph with edges $S \to A : 2$, $S \to B : 1$, $B \to A : 0.5$, $A \to G : 2$, and $B \to G : 6$. Choose $h(S) = 3$, $h(A) = 2$, $h(B) = 2.5$, and $h(G) = 0$. These estimates obey the triangle inequality on every edge. Queue entries below show (g, h, f).

Node removed	Generated or improved	Remaining priority order
S $(0, 3, 3)$	A $(2, 2, 4)$, B $(1, 2.5, 3.5)$	B, A
B $(1, 2.5, 3.5)$	A $(1.5, 2, 3.5)$, G $(7, 0, 7)$	$A_{1.5}$, A_2, G_7
$A_{1.5}$ $(1.5, 2, 3.5)$	G $(3.5, 0, 3.5)$	$G_{3.5}$, A_2, G_7
$G_{3.5}$	Return cost 3.5	

Unlike UCS, A* does not need to remove the cost-2 arrival at A before finding the answer. Its $f = 4$ already says that this particular route cannot beat the goal costing 3.5. An admissible heuristic prunes work through ordering: it does not change which actions are legal.

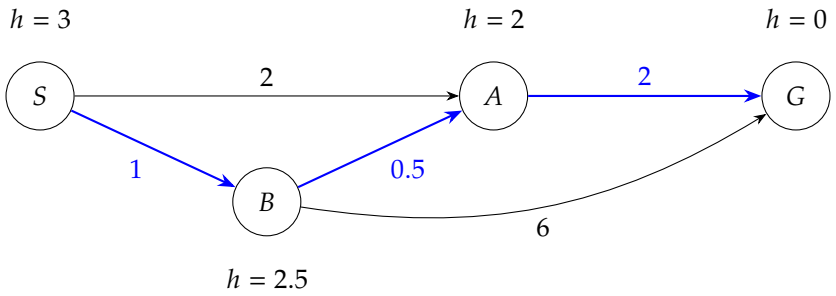


Figure 3: Blue edges form the cost-3.5 route; the heuristic helps compare unfinished routes.

Admissibility is essential to the comparison. If $h(B)$ were set to 100, B would wait behind A and the goal costing 4, despite leading to the best solution. The large estimate would hide the useful unfinished route. Greedy search has a different problem even with sensible estimates: it can prefer a nearby goal reached by an expensive route because it ignores g. An optimistic estimate becomes safe only when combined with the cost already paid and the correct stopping rule.

3.1.2 Worked example: greedy search can chase a cheap-looking remainder

Consider an illustrative graph with edges $S \to A$ of cost 2, $A \to G$ of cost 2, $S \to B$ of cost 1, and $B \to G$ of cost 100. Choose heuristic values $h(A) = 2$, $h(B) = 1$, $h(G) = 0$, and $h(S) = 2$.

1. **Check admissibility.** The true remaining costs at A, B are 2, 100, so both estimates are lower bounds. Being a lower bound does not mean being equally informative at every state.
2. **Run greedy search.** It chooses B because $1 < 2$, then chooses the generated goal because $h(G) = 0$. The returned route costs 101.

3. **Run A*.** Initially $f(A) = 4$ and $f(B) = 2$, so B is still expanded first. Its goal node has $f = 101$. The fringe still has A with $f = 4$, so A* expands A and returns the goal of cost 4.

4. **Explain the difference.** A* does not promise that every expansion is on a good route. It promises that the accumulated cost remains part of the comparison, preventing an expensive completed plan from winning solely because its remaining cost is zero.

The same example distinguishes expansion efficiency from answer quality. A weak but admissible estimate can waste work while preserving optimality.

3.2 Constructing heuristics

A relaxed problem removes restrictions on actions; its optimal cost is a lower bound on the original cost and therefore an admissible heuristic. In the sliding-tile puzzle, the number of misplaced tiles is a lower bound on moves. The sum of the tiles' Manhattan distances to their goal positions is another: each move changes one tile's distance by at most one. It corresponds to relaxing the restrictions imposed by the other tiles and the blank.

If $h_1(s) \geq h_2(s)$ everywhere, h_1 DOMINATES h_2. A more informative admissible estimate generally avoids expanding nodes with $f > C^*$, though ties and computation cost matter. The maximum of admissible heuristics remains admissible. The exact remaining cost is ideal as a bound, but computing it can be as hard as the original problem.

For the sliding-tile puzzle, the blank is not counted as a tile. A misplaced tile must move at least once, which explains the first lower bound. For Manhattan distance, a tile two columns and one row away contributes three. Since a legal move shifts just one tile by one grid step, the total distance can decrease by at most one per move. That is why the sum is a lower bound.

Taking $\max(h_1, h_2)$ keeps the strongest available lower bound at each state. Adding two admissible heuristics is not generally safe: they may both count the same necessary work. The heuristic itself also costs time, so fewer expansions need not imply less total computation.

3.2.1 Deriving a heuristic from a precise relaxation

In an eight-puzzle state, the locations of the numbered tiles and the blank specify the legal moves. A move slides a tile adjacent to the blank into it, usually at unit cost. For the illustrative board

$$\begin{array}{ccc} 1 & 2 & 3 \\ 4 & \square & 6 \\ 7 & 5 & 8 \end{array} \quad \longrightarrow \quad \begin{array}{ccc} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & \square \end{array},$$

tiles 5 and 8 are misplaced, so the misplaced-tile heuristic is 2. Their Manhattan distances are also 1 each, giving 2. Sliding 5 upward and then 8 left solves this state in two moves. Here both lower bounds are exact; in general they need not be.

The relaxation must make every original solution remain possible at no higher cost. If any tile may move directly to its target square, ignoring adjacency and occupancy, one move per misplaced tile suffices. If tiles may instead move one grid step at a time while ignoring occupancy and the blank, each tile needs its Manhattan distance, and the independent distances add. The original puzzle cannot be easier than either relaxed version. This is a construction of a lower bound, not merely a guess based on visual distance.

For a tile whose target is two rows and one column away, the misplaced-tile contribution is 1 but the Manhattan contribution is 3. Every misplaced tile has distance at least one, so the Manhattan sum dominates the misplaced count. Taking a maximum preserves admissibility because each candidate is at most h^*, hence their maximum is too. Taking a sum has no such argument: if both heuristics equal the true remaining cost 2, their sum is 4 and overestimates. Independent-looking formulas can still count the same moves.

The pancake example illustrates that the action cost matters. If flipping the top k pancakes costs k, the size of the largest pancake out of place is a lower bound: that pancake must participate in a flip whose size is at least its current or required involvement in correcting the unsettled prefix. More directly, for the usual numbered stack with a correct suffix, a pancake of size k outside position k must at some point be moved using a flip of length at least k to place it there. If every flip instead cost one, this same numerical heuristic would not have the same justification. A lower bound is always relative to the specified actions and costs.

3.3 Graph search and consistency

Tree search may revisit a state through many paths. Graph search keeps a closed set of expanded states and skips later occurrences. With costs, closing a state reached by a poor path can discard a better path discovered later; admissibility alone does not prevent this in A*.

A heuristic is CONSISTENT if for every edge $s \to s'$ of cost c,

$$h(s) \le c(s, s') + h(s'), \qquad h(g) = 0 \text{ at goals.}$$

This triangle inequality implies f cannot decrease along a path. It also implies admissibility by summing along a path to a goal. With a consistent heuristic, the first expansion of a state is along a cheapest path to it: a frontier prefix of any cheaper path would have no greater f than that cheaper arrival, and hence would be expanded first. Consequently A* graph search that closes states on expansion is optimal. An alternative for an admissible but inconsistent heuristic is to reopen a state when a cheaper path is found.

3.4 Why closing states needs more care

A closed set should record states already *expanded*, not simply every state ever generated. For cost-based search it is also useful to record the best known g for each state and ignore stale queue entries.

The consistency inequality has a direct operational meaning:

$$f(n') = g(n) + c(n, n') + h(n') \geq g(n) + h(n) = f(n).$$

A child's priority cannot be better than its parent's. An optimal arrival at a state therefore cannot be overtaken by a more expensive arrival that causes the optimal one to be discarded.

An admissible but inconsistent estimate can misorder intermediate arrivals. For example, let $S \to A$ and $S \to B$ each cost 1, $A \to C$ cost 1, $B \to C$ cost 2, and $C \to G$ cost 3. The values $h(A) = 4$, $h(B) = 1$, $h(C) = 1$ are admissible. A* can expand B, then close C with $g(C) = 3$, before expanding A. If the later arrival at C with cost 2 is discarded, the answer costs 6 instead of 5. Reopening C when its cost improves fixes this failure. Consistency would forbid $h(A) = 4 > 1 + h(C) = 2$.

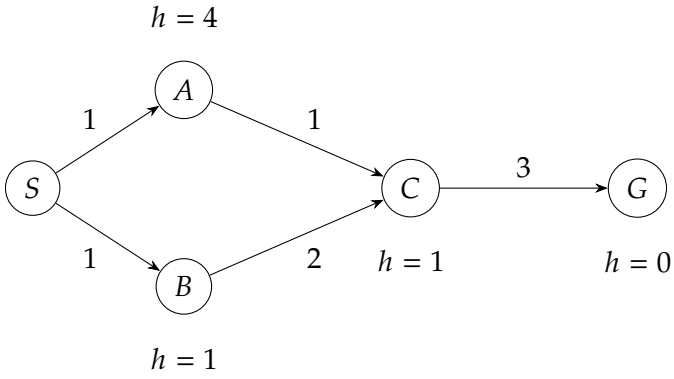


Figure 4: An inconsistent heuristic can let the costlier route to C arrive first.

3.4.1 Why consistency makes permanent closure safe

Follow a path $s_0, s_1, \ldots, s_k$ to a goal with edge costs c_i. Repeated use of consistency gives

$$h(s_0) \leq c_0 + h(s_1) \leq c_0 + c_1 + h(s_2) \leq \cdots \leq \sum_{i=0}^{k-1} c_i + h(s_k).$$

Since $h(s_k) = 0$, this proves admissibility. The converse fails: being below the total cost to a goal does not prevent a large drop across a single edge. That is exactly what happens at $A \to C$ in the inconsistent example.

To prove safe closure, suppose a state X is about to be expanded through a suboptimal path with cost $g(X) > g^*(X)$. On a cheapest path to X, take the first state not yet expanded along that path; an optimal prefix reaching it must be on the frontier. Call it P. Consistency summed from P to X gives $g^*(P) + h(P) \leq g^*(X) + h(X)$. Therefore

$$f(P) \leq g^*(X) + h(X) < g(X) + h(X) = f(X).$$

So P would be selected before this suboptimal arrival, contradicting the assumption that X is the first such mistake. This argument uses the same heuristic value for both arrivals at X, and closes states when expanded.

A practical implementation stores the cheapest known g per state. When a cheaper arrival is found, update its parent and queue priority, or insert a new queue entry and later discard stale entries. With an inconsistent but admissible heuristic, an improved closed state must become eligible for expansion again. With a consistent heuristic it never needs such reopening. A hash set or map supports these state checks without scanning a growing list.

3.4.2 Worked example: an admissible estimate that requires reopening

Take edges $S \to A : 3$, $S \to B : 1$, $B \to A : 1$, and $A \to G : 3$. Let $h(S) = 0$, $h(A) = 0$, $h(B) = 4$, and $h(G) = 0$.

1. **Separate the two conditions.** These estimates are admissible: the remaining cost from B is 4. They are not consistent on $B \to A$, since $h(B) = 4 > 1 + h(A) = 1$.
2. **Follow the priorities.** After expanding S, A* removes A with $g = 3$, $f = 3$ before B with $g = 1$, $f = 5$. Expanding A creates G at cost 6.
3. **Discover a better route to a closed state.** Removing B finds A with $g = 2$. A rule that permanently rejects all later routes to A would discard this improvement and eventually return cost 6.
4. **Reopen and repair the descendants.** Put the improved A back on the fringe. Its expansion produces goal cost 5, the cost of $S \to B \to A \to G$. Updating only a table entry without allowing the improved route to influence successors would not perform this repair.

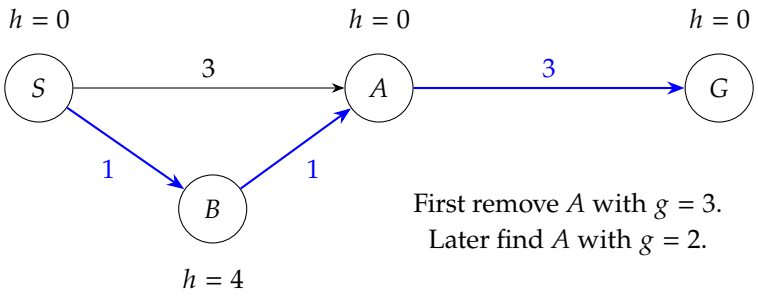


Figure 5: An inconsistent lower bound can hide a better route to an already expanded state.

Source. Pieter Abbeel and Dan Klein, Introduction to Artificial Intelligence, Fall 2018, Lecture 3.

Lecture 4: Constraint Satisfaction Problems I

4.1 Representation and backtracking

- **The assignment problem.** A constraint satisfaction problem (CSP) consists of variables $X_1, \ldots, X_n$, domains D_i of possible values, and constraints on allowed combinations of values. A solution assigns every variable a value and satisfies all constraints. Unlike path planning, the assignment itself is the goal. A constraint may be unary, binary, or involve more variables; it can be stated as a rule or a list of allowed tuples.
- **The constraint graph.** For map coloring, variables are regions, domains are colors, and adjacent regions must differ. In a binary CSP the constraint graph has a vertex for each variable and an edge for each pair constrained together. Disconnected components are independent subproblems. With n variables and at most d values each, there can be d^n complete assignments.
- **The basic search procedure.** Backtracking search performs depth-first search over partial assignments. At each step it picks one unassigned variable and tries its values, rejecting an assignment as soon as it violates a constraint. Assignments commute, so it is unnecessary to branch over all possible orders in which the same variables could be assigned. If no value works, undo the most recent choice and try another. Filtering and ordering can reduce the search without changing its set of solutions.

Example: For n-queens, one useful variable is X_i, the row of the queen in column i, with domain $\{1, \ldots, n\}$. Different columns are built into this representation. The constraints for $i \neq j$ are $X_i \neq X_j$ and $|X_i - X_j| \neq |i - j|$, excluding shared rows and diagonals. This gives n variables instead of representing every square independently.

A backtracking call starts with a consistent partial assignment. It chooses one variable, tentatively assigns a value, checks constraints, and recurses. If recursion fails, it must undo both that assignment and any domain deletions caused by it. Failing to restore domains incorrectly rules out solutions in sibling branches.

Algorithm 4.1: Backtracking for a CSP

```
Input: Consistent partial assignment A and current domains D
if every variable is assigned then
    return A
Choose an unassigned variable X
for each candidate value v for X do
    if A ∪ {X = v} satisfies its applicable constraints then
        Save the current domains
        Assign X = v and propagate constraints
        if no domain is empty then
            Recursively search from the new assignment
            if a solution is found then
                return that solution
        Undo the assignment and restore saved domains
return failure
```

4.1.1 Representations change the search space

A CSP exposes structure that an arbitrary successor function hides. A unary constraint such as $X \neq r$ can be handled by deleting r from D_X. A binary relation specifies allowed pairs. A higher-order relation checks several variables together, such as the digits in one addition column. An *all-different* constraint requires every value in a collection to be distinct. It can be expressed as pairwise inequalities, although a solver that understands the collection as a whole can sometimes propagate more.

For Sudoku, an empty square is a variable with domain $\{1, \ldots, 9\}$. Each row, column, and block has an all-different condition, and a given digit restricts a domain to one value. For queens, using one row variable per column builds the "one queen in each column" rule into the representation. A model with a Boolean variable per square must express that rule separately. Both can describe the same solutions, but their domains and constraints lead to different intermediate search and propagation.

Cryptarithmetic shows why extra variables can simplify constraints. For $SEND + MORE = MONEY$, introduce a digit variable for each letter and carry variables $C_1, \ldots, C_4$. The units and tens columns require

$$D + E = Y + 10C_1, \qquad N + R + C_1 = E + 10C_2.$$

The hundreds and thousands columns require

$$E + O + C_2 = N + 10C_3, \qquad S + M + C_3 = O + 10C_4, \qquad C_4 = M.$$

Also require $S, M \neq 0$, distinct letter digits, and carries in $\{0, 1\}$. Each column checks a local relation rather than redoing the entire addition for every partial assignment. The leading carry

gives $C_4 = M = 1$. This illustrates using arithmetic structure to rule out values before a complete candidate is built.

4.1.2 Why one variable per recursive level is enough

If naïve search chooses both an unassigned variable and a value at every step, a complete assignment can be reached in $n!$ different variable orders. Assigning $X = r$ and then $Y = g$ has the same final effect as reversing those assignments. Backtracking chooses a single next variable within each call, so the leaves correspond to at most d^n value combinations instead of $n!d^n$ ordered construction histories. The chosen variable may depend on the current domains; it need not be fixed globally.

As an illustrative trace, take X, Y, Z with pairwise inequality constraints and domains $\{r, g\}$. Try $X = r$, then $Y = g$. Neither value remains for Z, so undo $Y = g$ and all pruning caused by it. The alternative $Y = r$ conflicts immediately with X. The branch $X = r$ fails. Trying $X = g$ leads to the symmetric failure. Search has proved that the triangle has no two-coloring. A solver that merely finds no legal value at one leaf has proved only that one partial assignment cannot be completed, not that the whole CSP fails.

4.1.3 Worked example: solving a complete small scheduling CSP

Suppose three one-hour meetings A, B, C have domains $\{1, 2, 3\}$. Meeting A must precede B, B must precede C, and A cannot use slot 2. The numbers and scheduling instance are illustrative.

1. **Apply the unary restriction.** Replace D_A by $\{1, 3\}$. This removes an impossible choice without making any speculative assignment.
2. **Try an unhelpful value.** If backtracking chooses $A = 3$, the constraint $A < B$ leaves no value for B. Reject this branch immediately; there is no need to assign C.
3. **Undo the choice and try $A = 1$.** Then $D_B = \{2, 3\}$. Trying $B = 3$ would leave no value of $C > 3$, so reject it. Trying $B = 2$ leaves $C = 3$.
4. **Check all original constraints.** The complete assignment $(A, B, C) = (1, 2, 3)$ satisfies both ordering constraints and the unary restriction. It is a solution because it is complete and consistent, not merely because no conflict has yet been encountered.

A domain deletion caused by a trial assignment must be restored when that trial is undone. Otherwise the failed branch $A = 3$ could incorrectly remove values needed by the successful branch. Constraint propagation and restoration are therefore part of the search state, not incidental bookkeeping.

4.2 Filtering and arc consistency

Forward checking removes from each unassigned neighbor's domain values inconsistent with a new assignment. An empty domain proves failure, but forward checking does not detect every conflict between unassigned variables.

A directed arc $X \to Y$ is arc consistent if every value in D_X has a supporting value in D_Y satisfying the constraint between them. To revise an arc, delete unsupported values from the *tail* domain. To propagate constraints, repeatedly revise arcs; whenever D_X shrinks, recheck arcs from neighbors into X. An empty domain proves that the current branch has no solution. Domains must be restored on backtracking. Arc consistency can be enforced before search and after each assignment.

Arc consistency is not sufficient for global satisfiability. For example, three pairwise unequal variables forming a triangle, each with domain $\{\text{red}, \text{blue}\}$, are arc consistent but cannot all be assigned. Propagation therefore complements rather than replaces search. A basic implementation takes $O(n^2 d^3)$ time in the worst case; more efficient support tracking can improve this bound.

Consider $X \neq Y$ with $D_X = \{r, g\}$ and $D_Y = \{r\}$. The value r in D_X has no supporting value in D_Y, so revising $X \to Y$ removes it. The surviving g does have support, namely $Y = r$. Revising an arc means checking *each candidate in the tail*, not deleting matching values arbitrarily from either endpoint.

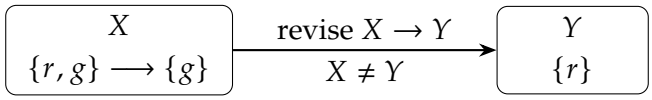


Figure 6: The head domain supplies support; unsupported values are deleted from the tail.

A queue makes propagation systematic. Start with arcs needing revision. When revising $X \to Y$ shrinks D_X, insert incoming arcs $Z \to X$ from other neighbors Z: a value in D_Z may have relied on a value just removed from D_X. Continue until the queue is empty or a domain becomes empty. No removed value could occur in a solution extending the current branch, because it had no compatible support.

Each arc can be reconsidered after domain changes. With at most d values, a simple revision checks up to d^2 pairs and an arc may be requeued $O(d)$ times. Over $O(n^2)$ possible arcs, this yields the basic $O(n^2 d^3)$ bound. Sparse graphs have fewer arcs, but the exponential worst case of the surrounding search remains.

4.2.1 A propagation chain that forward checking misses

Let a chain $X \neq Y \neq Z$ have domains $D_X = \{r\}$, $D_Y = \{r, g\}$, $D_Z = \{g\}$. There is no edge between X and Z. Once $X = r$ is fixed, forward checking deletes r from Y, leaving $Y = g$. Forward checking from that original assignment does not by itself revise the relation between the still-unassigned Y

and Z. Arc propagation does: revise $Z \to Y$, and $Z = g$ has no unequal support, so D_Z becomes empty. The assignment is impossible without any further branching.

The relevant events are therefore

$$D_X = \{r\} \quad \Rightarrow \quad D_Y : \{r, g\} \to \{g\} \quad \Rightarrow \quad D_Z : \{g\} \to \varnothing.$$

Information travels through a domain change, not just through a newly assigned variable. This explains the incoming arcs placed back on the queue.

Algorithm 4.2: AC-3 constraint propagation

```
   Input: Current domains and a queue Q of directed constraint arcs
1  while Q is not empty do
2      Remove an arc (X, Y) from Q
3      Delete each x ∈ D_X with no compatible y ∈ D_Y
4      if D_X changed then
5          if D_X = ∅ then
6              return failure
7          Add (Z, X) for each other neighbor Z of X to Q
8  return the reduced domains
```

The invariant is that no value belonging to a solution extending the current branch has been deleted. If a deleted value were in such a solution, the solution's value at its neighbor would be a supporting value, contradicting the deletion rule. Termination follows because domains only shrink and have finitely many values. A nonempty result means that the local support tests have stopped changing; it does not certify a global solution.

4.2.2 Worked example: supported pairs need not form a global solution

Let X, Y, Z each have domain $\{r, g\}$, with all three constraints $X \neq Y$, $Y \neq Z$, and $X \neq Z$.

1. **Inspect one directed arc.** For $X = r$, the value $Y = g$ supports $X \neq Y$; for $X = g$, the value $Y = r$ supports it. Both values survive.
2. **Inspect every arc.** The same argument applies to all six directed arcs. Arc consistency finishes without deleting a value.
3. **Try to build a complete assignment.** Choose $X = r$. Its two neighbors must both be g, violating $Y \neq Z$. Choosing $X = g$ produces the symmetric contradiction. No solution exists.
4. **Locate the limitation.** The supporting value for one constraint is checked separately from support for another. Arc consistency does not require all these separate supports to agree in one complete assignment.

This is why successful propagation is generally not a success test for an arbitrary cyclic CSP. An empty domain proves failure, but nonempty arc-consistent domains may still require search.

Tree-structured problems have an additional structural property that makes local support sufficient for forward assignment.

4.3 Ordering

The MINIMUM REMAINING VALUES rule chooses an unassigned variable with the smallest current domain, exposing failure early. The LEAST CONSTRAINING VALUE rule tries the value that eliminates the fewest choices from other variables. The first rule chooses where to branch; the second chooses which branch to try first. Evaluating these choices and propagating constraints has a cost, traded against the search it saves.

MRV and least-constraining value serve different purposes. Choosing a nearly impossible variable first avoids spending time on choices that will later be undone. Once a variable is chosen, trying a flexible value first makes a successful completion more likely. The former asks where failure is most likely; the latter asks which value leaves the most room for success. Filtering changes the current domains, so these choices should use the updated domains rather than only their original sizes.

4.3.1 Using ordering rules on actual domains

Suppose $D_X = \{r, g, b\}$, $D_Y = \{r, g\}$, and $D_Z = \{g\}$. MRV chooses Z first because it has the fewest remaining values. If assigning $Z = g$ forces $Y = r$, the next selection should use the newly reduced domain, not its original size. A zero-size domain is an immediate failure rather than another variable to branch over.

For a separate least-constraining-value illustration, let X differ from neighbors with domains $D_Y = \{r, g\}$ and $D_Z = \{r\}$, while $D_X = \{r, g, b\}$. Trying $X = r$ deletes one value from each neighbor and makes Z empty. Trying $X = g$ deletes g from Y only. Trying $X = b$ deletes nothing. LCV therefore tries b first. Notice that this is the opposite attitude from MRV for a good reason: choose a difficult *variable* to expose failure, then a flexible *value* to help the chosen branch succeed.

Ordering changes the amount of search, not the solutions admitted. Propagation also preserves solutions, while detecting impossible partial assignments. The combination is powerful because each new assignment changes the domains on which both decisions rely. Their bookkeeping must be restored when the assignment is undone. Copying domains is simple; recording a trail of deletions can save copying, provided each recursive call undoes exactly its own changes.

Source. Dan Klein and Pieter Abbeel, Introduction to Artificial Intelligence, Fall 2018, Lecture 4.

Lecture 5: Constraint Satisfaction Problems II

5.1 Consistency and structure

- **Consistency strength.** A CSP is k-consistent if any consistent assignment to $k-1$ variables can be extended to any kth variable. It is strongly k-consistent if it is also j-consistent for every $1 \leq j < k$. Strong n-consistency permits a solution to be built one variable at a time without backtracking, but enforcing higher-order consistency can be expensive.
- **Independent components.** The constraint graph can make a problem easier. Separate connected components can be solved independently. If n variables split into components of at most c variables, each with domain size at most d, exhaustive search costs roughly $O((n/c)d^c)$ for equal-sized components rather than $O(d^n)$.
- **Tree structure.** A binary CSP whose constraint graph is a tree can be solved in $O(nd^2)$. Choose a root and order parents before children. Process children in reverse order, making each parent-to-child arc consistent by deleting unsupported parent values. Then assign the root and proceed forward, choosing for each child a value compatible with its parent. The backward pass guarantees support for each surviving parent value; once a child has been processed, later revisions do not remove its support. Thus the forward pass needs no backtracking. An empty domain signals failure.

For a nearly tree-structured graph, choose a cutset whose removal leaves a tree or forest. Try each assignment to the cutset, reject inconsistent ones, restrict the remaining domains, and solve the resulting forest. A cutset of size c gives a bound of order $d^c(n-c)d^2$, apart from checking the cutset constraints. Tree decompositions use overlapping groups of variables arranged in a tree, with agreement on shared variables.

Arc consistency checks support one neighbor at a time. It does not require all those supports to coexist in a single assignment. In the two-color triangle, $X = r$ has support $Y = g$ and support $Z = g$, but those two supports violate $Y \neq Z$. Higher-order consistency addresses this gap by requiring compatible extensions of assignments to several variables together.

The tree algorithm avoids that conflict because different child subtrees do not reconnect. Suppose a rooted chain has domains $D_X = \{r, g\}$, $D_Y = \{r, g\}$, $D_Z = \{r\}$ with inequality constraints. Processing $Y \to Z$ leaves $D_Y = \{g\}$; processing $X \to Y$ then leaves $D_X = \{r\}$. Assigning forward gives (r, g, r). There is no later edge between X and Z that could invalidate this choice.

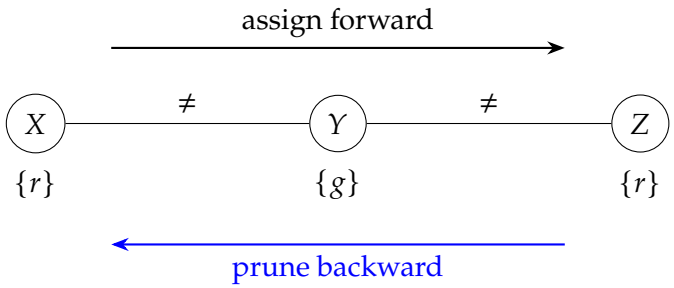


Figure 7: Pruning from leaves to root makes a forward assignment safe.

For cutset conditioning, assigning a variable means fixing its value and reducing its neighbors' domains before solving the remaining graph. Try all consistent cutset assignments; the method is still complete. A small cutset is valuable because the exponential dependence is on its size rather than on every variable.

Tree decomposition uses a different representation: groups of original variables become larger variables. Overlapping groups must agree on their shared assignments, and groups containing a given original variable must stay connected in the decomposition tree. Otherwise distant groups could give that variable conflicting values without detecting it. The cost moves into the domains of these larger groups.

5.1.1 Proving the tree algorithm by following the dependencies

Let Y be a child of X. The backward pass processes every descendant of Y before revising $X \to Y$. Thus when it checks support for values of X, D_Y already excludes values that cannot be supported below Y. No later backward step changes D_Y: later steps revise domains closer to the root. This is why a value of X retained by the pass continues to have a valid choice for Y.

Now assign the root any surviving value. Suppose all variables before Y have been assigned consistently. Its only already assigned neighbor is its parent X, because the graph is a tree and parents precede children. Arc consistency provides a compatible value for Y. Choosing it introduces no conflict with some other previously assigned branch because there is no edge to that branch. Induction gives a full solution. There are $n - 1$ arcs to revise, each requiring at most d^2 pair checks, which gives $O(nd^2)$. The guarantee comes from graph structure, not from exceptionally good luck in choosing values.

In a cycle, a variable can have two assigned neighbors. A value supporting one neighbor might conflict with the other. The backward pass cannot protect against that situation using one parent arc per variable. This explains both the failure of ordinary arc consistency on the two-color triangle and the value of removing a cutset.

5.1.2 Conditioning on a cutset, one case at a time

Take a three-color triangle X, Y, Z. Choose X as the cutset. For the case $X = r$, remove r from Y and Z; the residual graph is the single edge $Y \neq Z$, with domains $\{g, b\}$. The tree algorithm can choose $Y = g, Z = b$. For $X = g$ or $X = b$, it solves different residual cases. If every case fails, no global assignment exists, because every solution would have some value of X that was examined. If a case succeeds, combine it with the fixed cutset assignment to obtain the original CSP's solution.

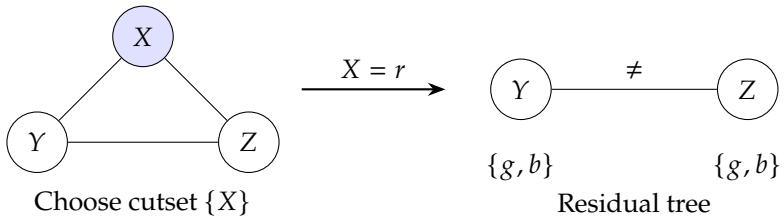


Figure 8: Fixing the cutset removes a cycle while its constraints remain as domain restrictions.

For a cutset of c variables, there are at most d^c cases. Each residual forest can be solved in $O((n-c)d^2)$, yielding the stated bound. Fixing the cutset does not mean deleting its constraints: their effect survives as domain restrictions on the residual variables. Forgetting those restrictions would solve a different, easier problem.

A tree decomposition instead makes a large domain for each bag. For bags $\{X, Y\}$ and $\{Y, Z\}$, candidate bag assignments (r, g) and (g, b) are compatible because they agree on $Y = g$. Candidates (r, g) and (b, r) are incompatible even if each satisfies its internal constraint. Shared-variable agreement is the new edge constraint. A bag of k original variables can have d^k assignments, so moving complexity into bags helps only when the bags stay small. The connectedness condition for every original variable ensures that agreement is carried along the tree rather than lost between separated occurrences.

5.1.3 Worked example: solve two disconnected components separately

Consider two independent binary CSP components. The first has $X \neq Y$ with $D_X = \{r\}$ and $D_Y = \{r, g\}$. The second has $U = V$ with $D_U = \{0, 1\}$ and $D_V = \{1\}$. No constraint joins the components.

1. **Solve the color component.** The fixed value $X = r$ forces $Y = g$. Its unique local solution is (r, g).
2. **Solve the bit component.** The fixed value $V = 1$ forces $U = 1$. Its unique local solution is $(1, 1)$.
3. **Combine the assignments.** Their union is a complete solution because every original constraint lies wholly inside one component. There is no cross-component compatibility test to perform.

4. **Change one domain.** If $D_U = \{0\}$ instead, the second component fails. The whole CSP then fails, regardless of the first component's solution. Independence makes solving separable; it does not let one ignore a failed part.

If components have several solutions, global solutions are combinations of local ones. Thus their counts multiply, while the work needed to find one solution can be split across the components. Adding even one connecting constraint invalidates this decomposition unless its effect is handled.

5.2 Local search

Local search keeps a single complete assignment and changes it rather than maintaining a fringe of partial assignments. For a CSP, MIN-CONFLICTS selects a variable involved in a violated constraint and assigns it a value that minimizes the number of conflicts. It can be effective on large problems, but can become stuck or cycle and is not generally complete.

More generally, HILL CLIMBING repeatedly moves to a better neighbor and stops when none is better. Local optima and plateaus can prevent it from reaching a global optimum. SIMULATED ANNEALING sometimes accepts a worse move: for objective decrease $\Delta < 0$, a common acceptance probability is $e^{\Delta/T}$. As temperature T falls, downhill moves become less likely. Slow cooling has theoretical convergence guarantees, but escape can still be slow. Genetic algorithms select a population by fitness and create candidates using crossover and mutation. Their usefulness depends on the representation.

5.3 What local search keeps and what it loses

Backtracking remembers alternatives; local search normally remembers only the current complete assignment. For queens, a move changes the row of a queen within its column. Min-conflicts selects a queen currently attacking another queen and tests which row would leave the fewest attacks. Unlike backtracking, it may temporarily keep violated constraints while repairing the assignment.

A local optimum is a state with no improving neighbor even though a better state exists elsewhere. On a plateau, many neighbors have equal scores, so the score gives little direction. Random choices, restarts, or occasional worse moves can help escape these situations, but ordinary hill climbing does not prove that no solution exists when it stops.

In simulated annealing, let Δ be the new score minus the current score for a maximization problem. Accept improving moves; for $\Delta < 0$, accept with probability $e^{\Delta/T}$. If a move loses 2 score units, its acceptance probability is e^{-1} at $T = 2$, but only e^{-4} at $T = 0.5$. The same loss becomes less acceptable as the search cools. A theoretical slow-cooling guarantee should not be confused with a fast practical algorithm.

5.3.1 A repair step in min-conflicts

For four queens represented by rows $(1, 1, 3, 4)$ in columns $1, 2, 3, 4$, consider moving the queen in column 1. It conflicts with the queen in column 2 through a row and with those in columns 3 and 4 through diagonals. Evaluate each candidate row by counting attacks involving the moved queen:

Candidate row	1	2	3	4
Attacks involving column 1	3	1	1	1

Rows 2, 3, and 4 tie for the minimum. Picking row 2 reduces the total number of attacking pairs from four to two; the existing conflict between columns 3 and 4 remains. A repair move improves the current assignment without making it a solution. Random tie-breaking and later repairs may succeed, but a step limit can be reached without a proof of impossibility. The objective should count attacking *pairs*, so a pair is not counted twice merely because both queens participate.

5.3.2 Neighborhoods, randomness, and populations

A local optimum depends on the allowed moves. If a state has score 8 and all one-variable changes score at most 7, hill climbing stops even if changing two variables together would reach score 10. Enlarging the neighborhood can make escape possible, but checking all such moves costs more. A plateau has equal-score alternatives rather than a strict barrier; a sideways move can help, or can create a cycle. Random restarts begin new searches from different complete assignments and sacrifice the work of the previous attempt in return for another basin of attraction.

Annealing samples a candidate neighbor rather than necessarily checking every neighbor. With fixed positive temperature and suitable symmetric proposals, the acceptance rule favors states in proportion to $e^{\text{score}/T}$. Lower temperature concentrates this distribution around good states. But crossing a barrier can require several unfavorable moves in succession, so lowering the temperature too quickly can freeze the search in a poor region. The cooling schedule trades exploration against concentration.

For queens, a genetic representation is a vector of row choices, one per column. Crossing $(2, 4 \mid 1, 3)$ with $(3, 1 \mid 4, 2)$ can form $(2, 4, 4, 2)$. It still has one queen per column, but can violate row or diagonal constraints. Mutation might randomly replace one row. Fitness can be the number of nonattacking pairs, or the negative number of attacks. Crossover is useful only if pieces of good candidates combine meaningfully: arbitrary vector boundaries can destroy relationships between interacting variables. The metaphor alone gives no guarantee of completeness or optimality.

5.3.3 Worked example: a local minimum under single-variable repairs

Take three Boolean variables and constraints $X = Y$, $Y = Z$, and $Z = X$. Let the cost be the number of violated constraints.

1. **Evaluate a complete assignment.** At $(X, Y, Z) = (0, 0, 1)$, the first equality holds and the other two fail, so the cost is 2.
2. **Evaluate every one-bit move.** Flipping Z gives $(0, 0, 0)$ at cost 0. Flipping X gives $(1, 0, 1)$ at cost 2, and flipping Y gives $(0, 1, 1)$ at cost 2. A repair method should choose Z here.
3. **Change the objective to expose a trap.** As a separate illustration, let an objective on two bits have costs $c(0, 0) = 1$, $c(0, 1) = c(1, 0) = 2$, and $c(1, 1) = 0$. At $(0, 0)$ every one-bit neighbor is worse even though $(1, 1)$ is globally best.
4. **Match the escape rule to the problem.** A restart can begin elsewhere; a stochastic move can cross a worse intermediate point; a larger neighborhood can change both bits together. Each changes which routes through the assignment space are accessible.

The first problem shows an ordinary successful repair; the second shows why absence of an improving neighbor is not a proof of global optimality. A local search stopping rule must be interpreted relative to its neighborhood.

Source. Dan Klein and Pieter Abbeel, Introduction to Artificial Intelligence, Fall 2018, Lecture 5.

Lecture 6: Game Trees: Minimax

6.1 Games and minimax values

- **The game specification.** A deterministic game specifies states, legal actions, a transition function, whose turn it is, a terminal test, and terminal utilities for each player. A solution is a policy that chooses an action in each state. In a two-player zero-sum game, one player's gain is the other's loss. Write all values from MAX's perspective; MIN minimizes that same value.
- **The guarantee against opposition.** The minimax value is the best utility MAX can guarantee against an optimal adversary. For successors $\mathrm{Succ}(s)$,

$$V(s) = \begin{cases} U(s), & s \text{ terminal}, \\ \max_{s' \in \mathrm{Succ}(s)} V(s'), & \text{MAX's turn}, \\ \min_{s' \in \mathrm{Succ}(s)} V(s'), & \text{MIN's turn}. \end{cases}$$

  Choose a successor attaining the maximum at MAX's root. The recursion follows from each player choosing its best continuation. A minimax strategy is safe against optimal opposition, but can be excessively cautious against an opponent whose errors could be exploited. For branching factor b and depth m, full minimax takes $O(b^m)$ time and $O(bm)$ space with depth-first traversal.
- **Reading a backed-up value.** A terminal value is an outcome specified by the game. A nonterminal minimax value is computed from the values below it. At MAX nodes ask which move MAX would choose; at MIN nodes ask which move the opponent would choose. Do not average an adversary's moves unless the model assigns probabilities to those moves.

For a root with two MIN children, suppose the left child leads to utilities $8, 2$ and the right child to $5, 6$. Their values are $\min(8, 2) = 2$ and $\min(5, 6) = 5$. MAX chooses the right branch and can guarantee 5. Choosing the branch containing 8 would assume that MIN cooperates.

If the opponent has several consecutive turns, use one minimizing layer for each turn. The recursion depends on whose choice a node represents, rather than assuming that every second level must always be MAX.

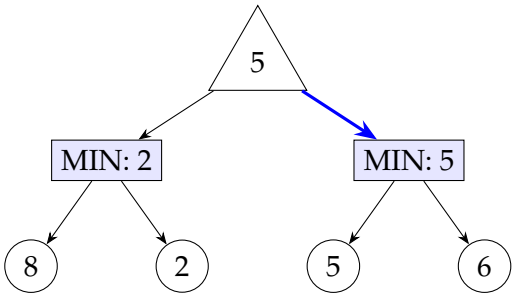


Figure 9: Back values up from leaves before choosing the root action.

6.1.1 Backward induction and the strategy it produces

The value calculation starts at leaves because only their outcomes are already known. Suppose the root has three MIN children with leaf lists $(3, 12, 8)$, $(2, 4, 6)$, and $(14, 5, 2)$. The three backed-up values are $3, 2, 2$. The root chooses the first branch. It does not choose the third branch for its leaf 14, because that outcome is under the opponent's control; the opponent can choose 2 there.

This is a proof by induction on remaining depth. At a terminal node the stated utility is achievable because play has ended. If all child values correctly describe what each player can force, a MAX node can select a child with largest value and guarantee that amount. A MIN node can select a child with smallest value, so MAX cannot guarantee more there. Repeating the argument up to the root establishes the minimax guarantee for a finite game tree. A policy records which action achieves the appropriate extremum at each reachable decision state, including states reached after unexpected replies.

A single path is not a complete strategy. A chess instruction such as "move this piece, then that piece" ignores the reply that occurs between those moves. The tree describes conditional plans: if the opponent chooses one reply, act one way; if it chooses another, act differently. Minimax evaluates those contingencies even when only the current root move is executed.

The guarantee is relative to a zero-sum model. If the opponent plays worse than its minimizing strategy, MAX may obtain more than the guaranteed value. It is not obliged to receive the worst case, but the calculation cannot rely on the opponent making a mistake. Perfect-information deterministic games also differ from games with hidden information or random outcomes: the same recursion cannot silently stand in for those missing parts of the model.

6.1.2 Worked example: a strategy specifies contingent responses

MAX first chooses L or R. After L, MIN can choose outcomes 4 or 7; after R, MIN can choose outcomes 2 or 10. These illustrative payoffs are all measured for MAX.

1. **Solve MIN's choices.** The left continuation has value $\min(4, 7) = 4$ and the right has value $\min(2, 10) = 2$.
2. **Solve MAX's choice.** MAX chooses L and guarantees 4. The tempting payoff 10 does not justify choosing R, because MIN controls whether that outcome occurs.
3. **Describe the policy.** At every reachable MAX decision, record the action maximizing the backed-up value. At later MAX nodes this action may depend on the opponent's earlier move. The result is a contingent strategy, rather than a single fixed path through the tree.
4. **Interpret an opponent mistake.** If MIN chooses 7 after L, MAX receives more than the guarantee. Minimax's value is the worst outcome under optimal opposition, not a prediction that every actual play ends at 4.

The interpretation also explains why MIN values must use MAX's utility scale. Minimizing one player's utility represents the other's interests only under the stated zero-sum assumption. General games require the relevant player's own utility when choosing a move.

6.2 Alpha-beta pruning

Alpha-beta pruning avoids branches that cannot affect the root's minimax decision. Carry α, the best alternative already available to MAX along the current path, and β, the best alternative available to MIN. Initialize them to $-\infty$ and $+\infty$.

At a MAX node, accumulate the largest child value, update α, and stop if the running value is at least β: an ancestor MIN already has an option no larger than β. At a MIN node, accumulate the smallest child value, update β, and stop if it is at most α. Unexamined children cannot change the ancestor's choice. A returned value after a cutoff can be only a bound on that internal node's true value; it should not be treated as an exact evaluation of every root child.

Pruning preserves the root value. Its effectiveness depends on ordering: examine promising moves first so useful bounds become available early. The worst case remains $O(b^m)$; ideal ordering reduces the leaf count to order $b^{m/2}$, permitting roughly twice the depth for the same budget.

To see a cutoff, suppose MAX has already found a root move worth 3. It begins exploring another MIN child and sees a leaf worth 2. That MIN child's final value is at most 2, whatever its other leaves contain. MAX can already obtain 3, so this entire branch is irrelevant to its decision. The bound is enough; the exact minimum is unnecessary.

The symmetric case occurs below a MIN ancestor. If MIN can already choose a branch worth 5 and another MAX branch reaches a child worth 7, MAX's final value is at least 7. MIN will avoid it, so its other children need not be searched. Always compare values in the same utility convention, from MAX's point of view.

The root action should be stored when its running best value improves. A cutoff in some other child need not produce that child's exact value, but it does establish that the child cannot improve the retained choice. Good move ordering finds strong alternatives early and creates more cutoffs.

6.2.1 An alpha-beta execution with actual bounds

Search the three-branch example from left to right. Initially $(\alpha, \beta) = (-\infty, +\infty)$ at the root.

1. Evaluate all leaves $3, 12, 8$ of the first MIN node. Its running minimum is 3 after the first leaf and remains 3. There is no useful root alternative yet, so its other leaves cannot be pruned. Return 3; the root sets $\alpha = 3$.
2. Enter the second MIN node carrying $\alpha = 3$. Its first leaf is 2, so its running minimum and local upper bound become 2. Since $2 \leq 3$, stop: the unseen leaves $4, 6$ cannot raise a minimum above 2.
3. Enter the third MIN node with $\alpha = 3$. The first two leaves give running minima 14 and 5, neither low enough to prune. The final 2 makes its value 2. The root retains the first branch and value 3.

Only seven leaves were needed instead of nine. If the third branch's leaf 2 were examined first, two more leaves could be omitted. Search order changes the saved work while preserving the minimax root value.

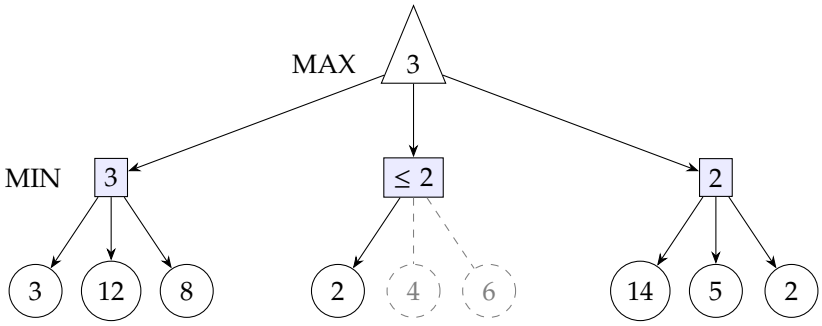


Figure 10: The middle branch cannot improve MAX's known value of 3; dashed leaves are not evaluated.

Algorithm 6.1: Alpha-beta value for a finite or depth-limited tree

```
Input: State s, remaining depth d, inherited bounds α, β
if s is terminal then
    return U(s)
if d = 0 then
    return Eval(s)
if MAX moves at s then
    v ← −∞
    for each successor s′ do
        v ← max(v, AB(s′, d − 1, α, β))
        if v ≥ β then
            return v
        α ← max(α, v)
else
    v ← +∞
    for each successor s′ do
        v ← min(v, AB(s′, d − 1, α, β))
        if v ≤ α then
            return v
        β ← min(β, v)
return v
```

The bounds belong to the current path of decision makers, not to every node in the tree globally. A MIN cutoff returns an upper bound on its true value; a MAX cutoff returns a lower bound. Treating every returned number as exact can produce incorrect analysis of unchosen actions. At the root, retain the action whose searched value improves the current best rather than attempting to reconstruct all action values from pruned subtrees.

6.3 Depth limits and evaluation

Realistic trees are too large to reach terminal states. At a depth limit, replace terminal utility by an EVALUATION FUNCTION, often $\text{Eval}(s) = \sum_i w_i f_i(s)$ for features such as material advantage or distance to food. This removes the optimal-play guarantee: search now optimizes the truncated model and its estimates.

Greater depth can reduce sensitivity to evaluation error, but expensive features leave less time to search. Iterative deepening provides a completed shallower result while deeper search proceeds. Evaluation also guides move ordering for alpha-beta. Replanning with a short horizon can postpone a good action forever when taking it now and taking it just beyond the horizon look equally valuable; the evaluation must distinguish useful progress from delay.

A PLY is one player's move. Cutting off after two plies means considering one's action and one reply, not necessarily two complete rounds. At the cutoff, an evaluation function tries to summarize what happens beyond the horizon. It should reward progress that the search will otherwise postpone and recognize danger that lies just outside the search depth.

Suppose eating a dot now and eating it one move later produce the same score at the cutoff. A replanning agent may repeatedly choose to delay, since every new search again makes delay look harmless. Adding a feature that values proximity to remaining food can distinguish those choices. This is an evaluation problem, not a failure of the minimax recursion itself.

6.3.1 Allocating a fixed computation budget

If $b \approx 35$, increasing depth by one multiplies the potential leaf count by about 35. This is why an evaluation function that takes ten times longer per node can be harmful even if it is more accurate locally. Conversely, a better evaluation can order children so that alpha-beta prunes much more, recouping some of its cost. Evaluation quality, move ordering, and depth must be considered together.

Depth-limited minimax is exact for the truncated tree with the supplied cutoff scores. It is approximate for the original game. A mistake beyond the horizon can change which current action is good; searching deeper can reveal it, but does not make arbitrary cutoff scores trustworthy. A terminal win should be recognized as an actual outcome rather than replaced by a heuristic estimate merely because the depth limit was also reached.

Iterative deepening searches depth one, then two, and so on. If time runs out during a deeper pass, use the root move from the last completed pass. An unfinished pass may have examined only one root alternative and offers no fair comparison with the others. This gives a usable answer at interruption while allowing a deeper completed search when time permits.

6.3.2 Worked example: a cutoff can reverse a move ranking

Suppose an illustrative search can examine one MAX move but not the following MIN reply. Its evaluation function gives resulting states A, B scores $8, 5$. At one additional level, A has MIN replies with terminal utilities $-10, 8$, while B has replies $4, 5$.

1. **Apply the original cutoff.** Search chooses A because $8 > 5$. This is the best decision for the estimated leaf values available to that search.
2. **Extend the tree.** The backed-up values become $V(A) = \min(-10, 8) = -10$ and $V(B) = \min(4, 5) = 4$. Search now chooses B.
3. **Locate the error.** Minimax correctly propagated the supplied values at both depths. The shallow evaluation failed to represent the opponent's imminent threat; an exact recurrence cannot repair information that was absent from its leaves.
4. **Assess pruning separately.** Alpha-beta with the same leaf values and depth gives the same move as full minimax. It saves work without correcting an inaccurate evaluation function.

Source. Dan Klein and Pieter Abbeel, Introduction to Artificial Intelligence, Fall 2018, Lecture 6.

Lecture 7: Game Trees: Expectimax and Utilities

7.1 Chance nodes and opponent models

- **Chance-node values.** Uncertain outcomes may come from random events, action failures, or an opponent whose behavior is modeled probabilistically. Expectimax uses MAX nodes for choices and chance nodes for outcomes:

$$V(s) = \max_a V(s, a), \qquad V(s, a) = \sum_{s'} P(s' \mid s, a) V(s').$$

  At terminal nodes use utility; at a depth cutoff use an estimate of future expected utility. For example, outcomes $8, 24, -12$ with probabilities $1/2, 1/3, 1/6$ have expectation $4 + 8 - 2 = 10$.

- **The opponent model.** The probabilities encode a model, even when the opponent does not literally randomize. A model of an opponent that usually chooses a minimax move can put most mass on that move and the remainder on alternatives. Computing such a model may itself require search. A wrong model can be costly: optimism about an adversarial opponent can be dangerous, while worst-case reasoning about a random opponent can miss worthwhile opportunities.

- **Pruning and other players.** An unexplored chance outcome can have an arbitrarily large value, so ordinary alpha-beta cutoffs do not apply without bounds on utilities. Games with both opposition and chance use MAX, MIN, and chance layers as appropriate. With multiple independent utilities, a node can carry a utility vector and the player to move chooses the child maximizing its own component, with a specified rule for ties.

In a chance node, every outcome contributes to the value. Probabilities must be nonnegative and sum to one. They describe outcomes conditional on the current state and selected action, so using the same probabilities in every state can be a serious modeling error.

For a small illustration, action A gives 4 for sure. Action B gives 10 with probability $1/2$ and 0 otherwise. Expectimax prefers B, whose value is 5; a worst-case MIN model prefers A, since B's worst outcome is 0. Neither recursion determines which model is appropriate. That depends on whether the uncertainty actually behaves like the stated distribution or like an opponent selecting the worst outcome.

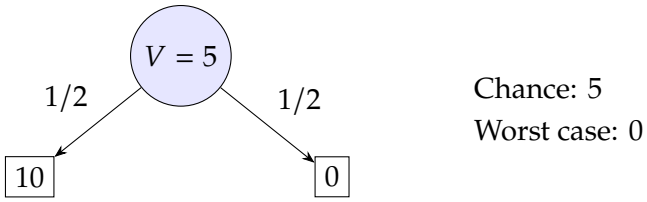


Figure 11: Chance takes a probability-weighted average; MIN takes the worst child.

At a depth limit, the evaluation should estimate expected future utility under the same opponent or environment model. A function tuned for worst-case danger can produce a cautious expectimax agent even though its recursion averages outcomes.

Ordinary pruning fails because an unfinished weighted average can rise or fall as remaining outcomes are revealed. If utilities are known to lie in $[L, H]$ and unexamined probability mass is p, their remaining contribution lies in $[pL, pH]$. Bounds can then be used, but the first bad child alone does not justify an alpha-beta-style cutoff.

7.1.1 Building and evaluating a chance model

Suppose an opponent has two legal moves, L and R, and a depth-two minimax simulation recommends L. If the model says it follows that recommendation with probability 0.8 and chooses uniformly at random otherwise, then

$$P(L \mid s) = 0.8 + 0.2/2 = 0.9, \qquad P(R \mid s) = 0.2/2 = 0.1.$$

The random portion can choose the recommended move too. Assigning probability 0.8 to L and 0.2 to R would describe a different model. At another state the simulated recommendation may change, so these probabilities must be recomputed. Modeling a strategic opponent probabilistically can therefore require substantial computation inside each chance node.

As a complete small backup, suppose MAX can choose A or B. Under A, chance gives a future MAX state with actions worth $3, 9$ with probability 0.6, and a state with actions worth $2, 4$ with probability 0.4. The inner MAX values are 9 and 4, so $V(A) = 0.6(9) + 0.4(4) = 7$. If B gives 6 for sure, choose A. Averaging the four leaves uniformly would give 4.5 and the wrong answer: MAX's later choices are not random, and the two chance outcomes do not have equal probabilities.

A chance cutoff needs a bound on what remains. Suppose another root action already gives 8, and a candidate chance action has examined probability mass 0.7 with weighted contribution 3. If every utility is at most 10, the candidate's final value is at most $3 + 0.3(10) = 6$, so it cannot beat 8. Without the upper bound, an unseen outcome worth 100 would change the contribution to 30 and could reverse the decision. This identifies exactly what ordinary MIN-node pruning knows that a chance-node calculation lacks.

For non-zero-sum games, values are vectors. If player 2 chooses between terminal utilities $(8, 1)$ and $(3, 7)$, it selects $(3, 7)$ because its own component is 7. Returning the entire vector preserves player 1's resulting utility 3. Replacing the node by the minimum of player 1's component would happen to agree in this example but is not the general rule. With options $(8, 9)$ and $(3, 7)$, player 2 instead chooses $(8, 9)$. Ties in the choosing player's component require a specified tie rule because the other components may differ.

7.1.2 Worked example: a useful cutoff at a bounded chance node

A root action already has exact value 7. Another action has three chance outcomes with probabilities $0.2, 0.3, 0.5$. All outcome utilities are known to lie in $[0, 10]$.

1. **Evaluate two outcomes.** Suppose their values are 1 and 2. The observed contribution is $0.2(1) + 0.3(2) = 0.8$. Unexplored probability mass is 0.5.
2. **Bound the remaining contribution.** Whatever the final utility is, it contributes between 0 and $0.5(10) = 5$. The action's complete expected utility therefore lies in $[0.8, 5.8]$.
3. **Compare the upper bound.** Since $5.8 < 7$, this action cannot win at the root. Its last outcome can be skipped without changing the choice. The value 5.8 is an upper bound, not its exact expectation.
4. **Remove the assumption.** If the final outcome were allowed to have utility 100, the expectation could be 50.8, reversing the decision. The cutoff is justified by the utility bound, not by the small probabilities or by a direct reuse of MIN-node pruning.

At a chance node, retain both the contribution already accumulated and the probability mass still unexplored. A favorable or unfavorable single child alone cannot generally determine the expected value.

7.2 Preferences over lotteries

A LOTTERY is a probability distribution over outcomes. Rational preferences satisfy constraints including completeness, transitivity, continuity, substitutability, monotonicity in the probability of a preferred prize, and reduction of compound lotteries to their outcome probabilities. Under the expected-utility representation, an outcome utility U can be chosen so that a lottery L is valued by

$$U(L) = \sum_o P_L(o)U(o).$$

The maximum expected utility principle chooses the action with greatest expected utility given the available information. A rational policy need not explicitly calculate this expression; its decisions can implement the same preferences by another mechanism.

Minimax decisions are unchanged by any strictly increasing transformation of terminal utilities because only order matters. Expectimax depends on the relative spacings: only positive affine transformations $U' = aU + b$, $a > 0$, preserve all lottery preferences. Normalize best and worst outcomes to 1 and 0; an outcome indifferent to a lottery paying the best outcome with probability p then has utility p.

The preference assumptions rule out contradictory choices. Completeness lets the agent compare two lotteries. Transitivity prevents a cycle in which it prefers A to B, B to C, and C to A. Continuity allows an intermediate prize to be matched by an appropriate lottery between better and worse prizes. Substitution allows indifferent lotteries to replace one another inside larger lotteries.

Why does utility scale matter? In the example above, taking square roots changes the sure outcome's value from 4 to 2, while the risky action's expected value becomes $\frac{1}{2}\sqrt{10} \approx 1.58$. The preferred action reverses. This increasing transformation keeps outcome rankings but changes preferences between lotteries. For a positive affine change,

$$\mathbb{E}[aU + b] = a\mathbb{E}[U] + b,$$

so the action ordering is preserved. This calculation is the reason the allowed transformations are narrower for expected utility.

7.2.1 Why lottery utilities are weighted sums

Let B be a best outcome and W a worst outcome, normalized to utilities 1 and 0. For an intermediate outcome A, continuity allows an indifference

$$A \sim [p, B;\ 1 - p, W].$$

Define $U(A) = p$. Better intermediate outcomes correspond to larger p by monotonicity. If another outcome C has utility q, substitution allows A and C to be replaced by their equivalent standard lotteries inside any larger lottery.

For $L = [r, A;\ 1 - r, C]$, that substitution creates a two-stage lottery. The total probability of B is $rp + (1 - r)q$, because B can be reached by either first-stage branch. Reduction of compound lotteries makes this equivalent to a direct lottery with that probability of B. Therefore

$$U(L) = rp + (1 - r)q = rU(A) + (1 - r)U(C).$$

The same argument extends to more outcomes. The weighted sum follows from consistent preferences and the ability to substitute equivalent lotteries; it is not an arbitrary averaging convention.

For example, if $U(A) = 0.8$ and $U(C) = 0.2$, a lottery giving each with probability $1/2$ has utility 0.5. It is indifferent to a 50% chance of the best outcome and a 50% chance of the worst. A person may

rank A above C without having those particular utility spacings. Deterministic rankings alone do not identify how that person trades their probabilities.

Transitivity prevents another kind of inconsistency. If $A \succ B$, $B \succ C$, and $C \succ A$, an agent can be induced to pay a small amount for each successive "improvement" and end with its original prize and less money. This illustrates why the preference conditions concern whole patterns of choices, not whether any single action sounds reasonable in isolation.

7.3 Money and risk

Expected money and expected utility are different objectives. A concave utility of wealth expresses risk aversion: the utility of a sure mean can exceed the mean utility of a lottery. A CERTAINTY EQUIVALENT is the sure amount judged equivalent to a lottery; its difference from expected money measures the amount the agent would pay to remove the risk. Human preferences need not satisfy the axioms. In the Allais example, the common choices \$3000 for sure over an 80% chance of \$4000, and a 20% chance of \$4000 over a 25% chance of \$3000, imply contradictory inequalities for any single expected-utility function with $U(0) = 0$.

For the Allais choices, preferring the certain \$3000 gives $U(3000) > 0.8U(4000)$. Preferring the 20% chance of \$4000 gives $0.2U(4000) > 0.25U(3000)$, or $0.8U(4000) > U(3000)$. These cannot both hold. The inconsistency concerns the combination of preferences, not whether either individual gamble is inherently sensible.

Risk aversion does not mean always avoiding uncertain actions. It means evaluating them with a utility curve whose gains from additional money become smaller. Expected utility can still favor a risky action when its potential benefits are large enough.

7.3.1 Computing a certainty equivalent

Use an illustrative wealth utility $U(w) = \sqrt{w}$ for $w \geq 0$. A fair lottery between wealth 100 and wealth 0 has expected wealth 50 but expected utility

$$\tfrac{1}{2}\sqrt{100} + \tfrac{1}{2}\sqrt{0} = 5.$$

Its certainty equivalent c solves $\sqrt{c} = 5$, giving $c = 25$. The utility of receiving 50 for sure is $\sqrt{50} \approx 7.07$, greater than 5. The agent would therefore give up as much as $50 - 25 = 25$ in expected wealth to replace this risk by its equivalent sure amount. These numbers illustrate the curve; they are not measurements of anyone's actual preferences.

Concavity explains this result geometrically. On a concave utility curve, the point above the mean wealth is higher than the straight chord joining the endpoint utilities. Averaging utility follows the chord, whereas taking utility of average wealth follows the curve. Linear utility makes the two equal and represents risk neutrality. A convex region reverses the inequality and can express

risk-seeking behavior there. Risk attitude is thus part of utility, while probability continues to describe uncertainty about outcomes.

7.3.2 Worked example: which utility transformations preserve choices?

Compare a certain monetary outcome 4 with a fair lottery over 0 and 10.

1. **Use money as utility.** Their utilities are 4 and $\frac{1}{2}(0) + \frac{1}{2}(10) = 5$, so the lottery wins.
2. **Change only the utility scale.** Under $U'(x) = 3U(x) + 2$, the values become 14 and 17. A positive affine transformation preserves all expected-utility rankings because expectation is linear and every lottery has total probability one.
3. **Change the preference for risk.** Under $U(x) = \sqrt{x}$, the values are 2 and $\sqrt{10}/2 \approx 1.581$. Now the certain amount wins. This transformation is increasing, so it preserves rankings of certain amounts, but it changes preferences over uncertain amounts.
4. **Convert back to money carefully.** The lottery's certainty equivalent solves $\sqrt{c} = \sqrt{10}/2$, giving $c = 2.5$. Its expected monetary amount remains 5. These numbers answer different questions.

The order of operations matters: compute the utility of each outcome before averaging. Taking utility after averaging money would replace the lottery by a certain payment and erase the risk preference being modeled.

Source. Dan Klein and Pieter Abbeel, Introduction to Artificial Intelligence, Fall 2018, Lecture 7.

Lecture 8: Markov Decision Processes I

8.1 States, rewards, and policies

- **The model components.** A Markov decision process (MDP) specifies states S, actions $A(s)$, transitions $T(s, a, s') = P(s' \mid s, a)$, rewards $R(s, a, s')$, a start state, and possibly terminal states. The Markov assumption says that the current state and action determine the distribution of the next state, without additional dependence on history.
- **A policy reacts to outcomes.** In a noisy grid world, a requested move can go in the intended direction with probability 0.8 and each perpendicular direction with probability 0.1; a blocked move leaves the state unchanged. A living reward and terminal rewards express preferences between speed, safety, and remaining in the grid. The solution is a policy $\pi : S \to A$ rather than a fixed sequence: the action must respond to the state actually reached.
- **The return being optimized.** For discount $0 \le \gamma < 1$, the return from rewards $r_0, r_1, \ldots$ is $G = \sum_{t \ge 0} \gamma^t r_t$. With bounded rewards $|r_t| \le R_{\max}$, $|G| \le R_{\max}/(1 - \gamma)$. Discounting favors earlier rewards and makes infinite-horizon values finite. Finite horizons instead end after a fixed number of steps, and the best action can depend on time remaining. Undiscounted episodic problems need conditions ensuring termination and finite returns.

A policy must cover outcomes the agent did not intend to produce. If a move north can slip west, the agent needs an action for the westward destination too. Writing down only a path assumes away that uncertainty.

Rewards specify the objective; they are not the same as values. A reward is received on one transition. A value includes the expected rewards of an entire continuation. Giving a state a large positive reward can make it attractive, but a nearby state can also have high value because it leads there reliably.

For rewards $[1, 2, 3]$ and $\gamma = 1/2$, return is $1 + (1/2)2 + (1/4)3 = 2.75$. The exponent counts how many steps in the future a reward arrives. A reward already included in $R(s, a, s')$ must not be counted again as a terminal continuation value. Otherwise the terminal payoff is accidentally doubled.

8.1.1 Constructing the transition and reward tables

The racing model has states cool (C), warm (W), and overheated (O), with O terminal. Slow driving in C stays cool and pays 1. Fast driving in C pays 2 and reaches C or W with probability $1/2$ each.

Slow driving in W pays 1 and reaches C or W with probability $1/2$ each. Fast driving in W reaches O with certainty and pays -10.

State, action	Next state	Probability	Reward
C, slow	C	1	1
C, fast	C, W	$1/2, 1/2$	2 on either outcome
W, slow	C, W	$1/2, 1/2$	1 on either outcome
W, fast	O	1	-10

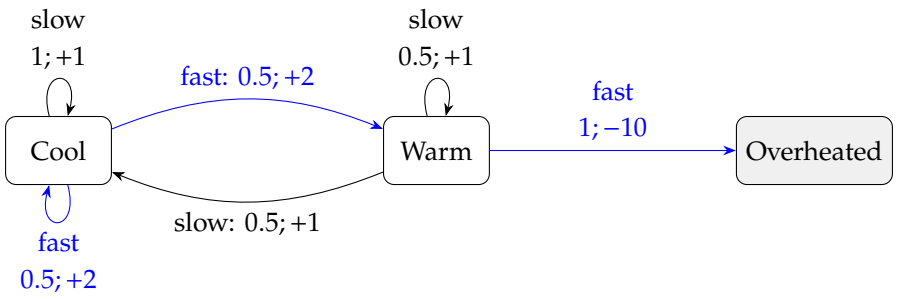


Figure 12: Each edge label gives its probability and reward after the named action.

Each state-action row's probabilities add to one. The choice is not simply "fast gives more reward": repeated fast driving can make the damaging warm state more likely, and fast driving there ends future opportunities entirely. A policy specifies separate actions for C and W.

The Markov state must include temperature because it changes the effect of an action. If the state were just "car is driving," the transition model would depend on hidden temperature history. Adding the relevant variable restores a prediction based on the present state. Markov does not mean that states are independent over time; it means that the present state contains the information from the past needed for the next transition.

Discounting and a living reward affect the policy differently. Discounting reduces the weight of later outcomes. A negative living reward charges each extra transition, encouraging shorter episodes; a sufficiently positive one can make avoiding termination valuable. Changing these quantities changes the problem being optimized, rather than improving an algorithm for the same objective.

8.1.2 Worked example: blocked moves combine into one transition entry

An illustrative grid agent is in a corner. It requests a move north. The intended move has probability 0.8; slips west and east each have probability 0.1. North and west are blocked, while east leads to state e.

1. **Map physical outcomes to successor states.** Both the north attempt and west slip leave the agent at s. They are different random outcomes but the same successor in the chosen state space.
2. **Add their probability mass.** The transition row is $T(s, \text{north}, s) = 0.9$ and $T(s, \text{north}, e) = 0.1$. It sums to one. Storing only one blocked outcome would lose probability mass.
3. **Compute one backup.** Suppose both outcomes give reward -1, $\gamma = 0.5$, and current continuation estimates are $V(s) = 2$, $V(e) = 6$. Then this action has value $0.9(-1 + 0.5 \cdot 2) + 0.1(-1 + 0.5 \cdot 6) = 0.2$.
4. **Interpret the self-loop.** The action can return to s and still consume time and receive a reward. A self-loop is a genuine transition, not a reason to omit its term or to ignore discounting.

If collision outcomes have different rewards, first compute their expected reward conditional on the aggregated successor, or retain enough outcome information to calculate the expectation correctly.

8.2 Bellman optimality and time-limited values

Let $V^*(s)$ be the optimal expected return starting at s, and $Q^*(s, a)$ the expected return after committing to action a and then acting optimally. Decomposing the return into the next reward and the discounted remainder gives

$$Q^*(s, a) = \sum_{s'} T(s, a, s')\big[R(s, a, s') + \gamma V^*(s')\big], \qquad V^*(s) = \max_a Q^*(s, a).$$

Terminal continuation values are zero when their rewards have already been paid on entry. These equations reuse values of repeated states instead of unfolding a separate infinite expectimax tree for every path.

Value iteration begins with $V_0(s) = 0$ and computes

$$V_{k+1}(s) = \max_a \sum_{s'} T(s, a, s')[R(s, a, s') + \gamma V_k(s')].$$

Here V_k is the best value with k steps remaining. Use the previous vector throughout a synchronous iteration. A dense update costs $O(|S|^2|A|)$; sparse transition models require summing only over possible successors.

For a finite-depth problem, enough iterations recover the full value. For bounded rewards and $\gamma < 1$, the tail beyond depth k has magnitude at most $\gamma^k R_{\max}/(1 - \gamma)$. Thus the truncation error vanishes and V_k converges to V^*. The maximizing actions can stabilize much earlier than the numerical values.

8.3 Reading a Bellman backup

The backup has a fixed order of operations. For each possible first action, compute an average over its possible next states. In each term add the immediate reward to the discounted continuation. Only after computing each action's expected return do we choose the largest one.

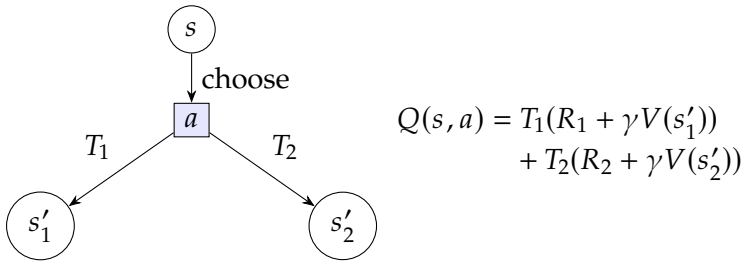


Figure 13: Average outcomes within an action, then compare actions.

Example: In an illustrative backup, action a reaches a state of old value 6 with probability 0.8 and one of value 1 with probability 0.2. If both immediate rewards are -1 and $\gamma = 0.9$, its value is $0.8(-1 + 0.9 \cdot 6) + 0.2(-1 + 0.9 \cdot 1) = 3.5$. Compare 3.5 with the corresponding averages for the other actions.

Algorithm 8.1: Synchronous value iteration

```
Set V(s) = 0 for every state
repeat
    for each nonterminal state s do
        V_new(s) ← max_a Σ_{s'} T(s, a, s')[R(s, a, s') + γV(s')]
    Set terminal entries of V_new to zero
    Replace V by V_new
until the chosen convergence condition is met
```

Each pass adds one more step of lookahead to a finite-horizon calculation. Using a separate new vector makes that interpretation exact. Updating entries in place is a different update schedule, even if it can also converge.

A one-state example makes the limiting process visible. If every step pays 2, returns to the same state, and $\gamma = 1/2$, then $V_0 = 0$, $V_1 = 2$, $V_2 = 3$, $V_3 = 3.5$, and the limit solves $V = 2 + V/2$, giving $V = 4$. The values improve gradually because more of the future reward stream has been included.

8.3.1 Deriving the backup from a return

Write the future return after the next transition as G'. Then $G = r + \gamma G'$. If the first action is fixed to a, conditioning on its possible successor s' gives

$$\mathbb{E}[G \mid s, a] = \sum_{s'} T(s, a, s') \left(R(s, a, s') + \gamma \mathbb{E}[G' \mid s']\right).$$

Under optimal continuation the last expectation is $V^*(s')$. The agent controls a, so it chooses the largest of these expected returns. This separation explains why the maximum lies outside the sum. Writing a maximum inside the successor sum would let the agent choose its first action after seeing which outcome occurred, granting information it does not have.

The finite-horizon recurrence follows by induction. With no steps left, there is no future reward and $V_0 = 0$. If V_k correctly gives the best k-step continuation, one legal action followed by that continuation gives the best $(k+1)$-step value. The same old V_k must be used for every state to preserve this exact interpretation.

8.3.2 Value iteration on the racing model

For illustration choose $\gamma = 1/2$. With $V_k(O) = 0$, the four action backups are

$$\begin{aligned} Q_k(C, \text{slow}) &= 1 + \tfrac{1}{2} V_k(C), \\ Q_k(C, \text{fast}) &= 2 + \tfrac{1}{4}[V_k(C) + V_k(W)], \\ Q_k(W, \text{slow}) &= 1 + \tfrac{1}{4}[V_k(C) + V_k(W)], \\ Q_k(W, \text{fast}) &= -10. \end{aligned}$$

Take the maximum in each state to form V_{k+1}. The resulting passes are

k	$V_k(C)$	$V_k(W)$	$V_k(O)$
0	0	0	0
1	2	1	0
2	2.75	1.75	0
3	3.125	2.125	0
Limit	3.5	2.5	0

For example, the second cool-state pass compares slow value $1+2/2 = 2$ with fast value $2+(2+1)/4 = 2.75$. At the limit, slow is worth 2.75 and fast is worth 3.5 in C; slow is worth 2.5 and fast is worth -10 in W. Thus the optimal policy drives fast while cool and slow while warm. Values continue changing after this action choice becomes apparent.

Finally, the truncation guarantee can be made quantitative. The discarded reward stream after k steps has absolute value at most

$$\sum_{t=k}^{\infty} \gamma^t R_{\max} = \frac{\gamma^k R_{\max}}{1-\gamma}.$$

This bound holds for every policy, hence also bounds the difference between the best truncated and full returns. It provides a sufficient depth for a desired value accuracy. It can be conservative, because it assumes every remaining reward has the largest possible magnitude. Convergence does

not require all MDP trajectories to terminate when $\gamma < 1$ and rewards are bounded; discounting controls even infinite trajectories.

8.3.3 Worked example: finite time remaining can change the best action

At a state s, action cash ends immediately with reward 3. Action invest gives reward 0 and moves deterministically to u, where the only action ends with reward 8. Let $\gamma = 1/2$.

1. **Set the zero-step boundary.** With no actions remaining, $V_0(s) = V_0(u) = 0$. A reward available only after a future action is not yet earned.
2. **Allow one action.** At u, $V_1(u) = 8$. At s, cash gives 3 and invest gives $0 + \gamma V_0(u) = 0$, so $V_1(s) = 3$.
3. **Allow two actions.** At s, invest now gives $0 + \gamma V_1(u) = 4$, exceeding cash's 3. Thus $V_2(s) = 4$ and the action changes.
4. **Keep time in the policy when necessary.** The physical state s is the same, but the correct action differs with the remaining horizon. A finite-horizon policy may be written $\pi(s, k)$, with k the number of actions available. It need not be stationary on physical states alone.

The example explains why V_{k+1} must use V_k in a synchronous finite-horizon backup. Mixing newly computed values into that pass no longer represents one common number of remaining steps for every state.

Source. Dan Klein and Pieter Abbeel, Introduction to Artificial Intelligence, Fall 2018, Lecture 8.

Lecture 9: Markov Decision Processes II

9.1 Policy evaluation and extraction

- **Evaluate a fixed policy.** For a fixed deterministic policy π, the policy value is $V^\pi(s) = \mathbb{E}_\pi[\sum_{t\geq 0} \gamma^t r_t \mid s_0 = s]$. Fixing the action removes the maximum from the Bellman equation:

$$V^\pi(s) = \sum_{s'} T(s, \pi(s), s')[R(s, \pi(s), s') + \gamma V^\pi(s')].$$

  Compute it by iterating this update, at $O(|S|^2)$ per dense pass, or by solving the linear system $(I - \gamma T_\pi)V^\pi = r_\pi$. For a finite MDP with $\gamma < 1$, the system has a unique solution.

- **Extract a policy from values.** Policy extraction from a value estimate requires one-step lookahead:

$$\pi_V(s) \in \arg\max_a \sum_{s'} T(s, a, s')[R(s, a, s') + \gamma V(s')].$$

  Choosing the successor with largest value alone ignores rewards and transition probabilities. From optimal action values, extraction is simply $\pi^*(s) \in \arg\max_a Q^*(s, a)$.

- **Interpret the matrix equation.** The transition matrix T_π has one row for each state and one column for each possible next state. Its entry is $T(s, \pi(s), s')$. The vector r_π contains the expected immediate rewards $r_\pi(s) = \sum_{s'} T(s, \pi(s), s')R(s, \pi(s), s')$. Thus $V^\pi = r_\pi + \gamma T_\pi V^\pi$ is just the collection of the one-state evaluation equations. Solving them together accounts for loops.

One-step policy extraction is not the same as choosing the neighbor with largest V. An action may reach a good state only rarely, charge a large immediate cost, or occasionally fall into a bad state. The full weighted backup includes all three effects.

9.1.1 Solving coupled policy equations

Use the racing model with $\gamma = 1/2$. Under the policy that always drives slowly, the equations are

$$V^\pi(C) = 1 + \tfrac{1}{2}V^\pi(C), \qquad V^\pi(W) = 1 + \tfrac{1}{4}V^\pi(C) + \tfrac{1}{4}V^\pi(W).$$

The first gives $V^\pi(C) = 2$. Substituting it into the second gives $\frac{3}{4}V^\pi(W) = \frac{3}{2}$, so $V^\pi(W) = 2$. The warm state can transition back to itself; solving the equation includes all repeated visits rather than counting only one trip.

In matrix form, restricting to the two nonterminal states,

$$\begin{pmatrix} 1/2 & 0 \\ -1/4 & 3/4 \end{pmatrix} \begin{pmatrix} V^\pi(C) \\ V^\pi(W) \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

The left matrix is $I - \gamma T_\pi$, not just T_π. Iterative evaluation starts with any bounded vector and repeatedly applies the fixed-policy backup. One pass does not generally solve the simultaneous equations; it propagates information one transition farther through the model.

To see uniqueness, suppose two vectors V and W solve the same discounted policy equations. Their difference satisfies $V - W = \gamma T_\pi(V - W)$. A weighted average cannot have a larger absolute value than the largest absolute entry, so $\|V - W\|_\infty \le \gamma \|V - W\|_\infty$. For $\gamma < 1$, this forces $V = W$. The same averaging argument explains why repeated policy backups reduce an initial error.

For policy extraction, suppose an action reaches a state worth 10 with probability 0.1 and a state worth 0 otherwise, while another reaches a state worth 6 with certainty. With equal immediate rewards and the same positive discount, their continuation values are 1 and 6. Choosing the largest possible successor would select the first action incorrectly. Values alone must be combined with how each action reaches those states.

9.1.2 Worked example: solve a policy with a return loop

Under a fixed policy, state A gives reward 1 and moves to B; B gives reward 2 and moves to A. Let $\gamma = 1/2$ and assume the process continues.

1. **Write both equations.** They are $V(A) = 1 + \frac{1}{2}V(B)$ and $V(B) = 2 + \frac{1}{2}V(A)$. Treating either successor value as zero would discard the loop.
2. **Substitute rather than recurse forever.** Replace $V(B)$ in the first equation: $V(A) = 1 + \frac{1}{2}(2 + \frac{1}{2}V(A)) = 2 + \frac{1}{4}V(A)$. Thus $\frac{3}{4}V(A) = 2$, giving $V(A) = 8/3$ and $V(B) = 10/3$.
3. **Check against the reward stream.** Starting at A gives $1, 2, 1, 2, \ldots$, so the discounted return is $(1 + \frac{1}{2} \cdot 2) \sum_{j \ge 0} (1/4)^j = 2/(1 - 1/4) = 8/3$.
4. **Understand the role of discounting.** With $\gamma = 1$ these positive rewards accumulate without bound. The finite discounted solution cannot simply be reused for that changed objective.

Policy evaluation computes the consequences of the policy being followed. It does not compare alternative actions until a separate improvement or extraction step introduces that comparison.

9.2 Policy iteration

Policy iteration alternates two operations: evaluate the current policy to obtain V^π, then replace it by the greedy policy π_{V^π}. The improvement step cannot lower value: it chooses a first action

at least as good as the old policy's action, followed by continuation values under the old policy. Repeated application of the improved policy preserves that inequality. If no action changes, the values satisfy Bellman optimality and the policy is optimal. Retain the old action on ties to avoid cycling between equally good policies.

Value iteration updates values and implicitly reselects every action each pass. Policy iteration spends several cheaper passes evaluating one fixed policy before reconsidering actions. Both solve the same known MDP; the computational advantage depends on the problem and evaluation method.

Example: Consider an illustrative one-state MDP with discount $1/2$. Action a returns to the state and pays 1. Action b returns too and pays 3 with probability $1/2$, otherwise 0. Under the policy always taking a, $V^\pi = 1 + V^\pi/2 = 2$. Improvement compares $Q^\pi(s, a) = 1 + (1/2)2 = 2$ with $Q^\pi(s, b) = 1.5 + (1/2)2 = 2.5$. It switches to b. Evaluating this new policy gives $V = 1.5 + V/2 = 3$.

In that comparison the continuation after the first action is still the *old* policy. Policy improvement asks whether changing the first action is useful when the remaining values are those already evaluated. Once the new policy is adopted everywhere, another evaluation computes the effects of all its future choices.

To justify improvement, let $\mathcal{B}_{\pi'}$ mean taking one backup under the new greedy policy. Then $\mathcal{B}_{\pi'} V^\pi \geq V^\pi$. Applying the same monotone backup repeatedly keeps the inequality, and its limit is $V^{\pi'}$. Therefore $V^{\pi'} \geq V^\pi$. If the greedy improvement changes nothing, the current evaluation is also a solution of the optimality equations.

Policy iteration can save time because many value updates do not require comparing every action again. Exact evaluation can nevertheless be expensive, so there is no universal claim that policy iteration is always faster. The model's size, sparsity, and how accurately policies are evaluated matter.

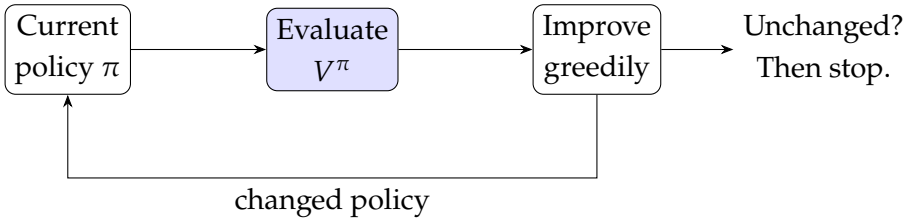


Figure 14: Policy iteration alternates evaluating one policy and choosing a better one.

9.2.1 Completing a policy-iteration cycle

Starting from the slow policy values $(2, 2)$, the cool-state improvement compares

$$Q^\pi(C, \text{slow}) = 2, \qquad Q^\pi(C, \text{fast}) = 2 + \tfrac{1}{4}(2 + 2) = 3.$$

Warm-state slow has value 2 and fast has value -10. The improved policy therefore chooses fast in C and slow in W. Evaluate that policy by solving

$$V(C) = 2 + \tfrac{1}{4}[V(C) + V(W)], \qquad V(W) = 1 + \tfrac{1}{4}[V(C) + V(W)].$$

Subtracting gives $V(C) - V(W) = 1$. Adding gives $V(C) + V(W) = 3 + \frac{1}{2}[V(C) + V(W)]$, hence the sum is 6. The solution is $(3.5, 2.5)$. A new improvement compares 3.5 with 2.75 in C, and 2.5 with -10 in W. No action changes, so the policy is optimal. The intermediate improvement value 3 for fast in C was not its final new-policy value 3.5: it assumed only the first action changed.

Why can an improvement at every state be adopted simultaneously? Define $W_0 = V^\pi$ and $W_{k+1} = \mathcal{B}_{\pi'} W_k$ for the new greedy policy. The greedy choice gives $W_1 \geq W_0$. If $W_k \geq W_{k-1}$, nonnegative transition probabilities give $W_{k+1} \geq W_k$. Thus every additional new-policy step preserves improvement. Discounted evaluation converges to $V^{\pi'}$, proving $V^{\pi'} \geq V^\pi$ rather than just a one-step gain.

For a finite state/action set, there are finitely many deterministic policies. Retaining the current action on ties avoids needless switching among equally good choices. A strict improvement cannot revisit an inferior old policy. At a stable policy, its evaluated values equal their maximal action backups, so they satisfy Bellman optimality. Approximate evaluation changes this exact argument: an action selected from inaccurate values need not be a true improvement, so the evaluation tolerance matters.

9.2.2 Worked example: a high-value successor can be an expensive choice

Suppose two states have accurately known continuation values $V(u) = 10$ and $V(v) = 8$. At s, action a deterministically goes to u with immediate reward -5; action b goes to v with immediate reward 0. Set $\gamma = 1/2$.

1. **Avoid ranking successor values alone.** Choosing a because $10 > 8$ ignores the cost of reaching u.
2. **Compute the action values.** One-step lookahead gives $Q(s, a) = -5 + \frac{1}{2}(10) = 0$ and $Q(s, b) = 0 + \frac{1}{2}(8) = 4$. The greedy improvement chooses b.
3. **Add transition uncertainty.** If b instead reaches v with probability 0.75 and a zero-value terminal state otherwise, still with zero immediate reward, its value becomes $0.75(4)+0.25(0) = 3$. Probabilities affect the comparison even when the most attractive successor has not changed.
4. **State what was held fixed.** This comparison uses the supplied continuation values. In policy iteration they come from evaluation of the old policy; after changing actions, evaluate again before claiming to know the complete value of the new policy.

Policy extraction is an action comparison over expected immediate reward plus discounted continuation. It is not a rule that simply moves toward the largest number displayed on neighboring states.

9.3 From planning to learning

Offline planning computes values and policies from known transitions and rewards without interacting with the environment. When those quantities are unknown, an agent must act and observe. Repeated outcomes estimate their probabilities, but learning incurs mistakes. Exploration gathers information about uncertain actions; exploitation uses current knowledge for reward. Learning a good final policy and collecting good rewards during learning are different objectives.

The double-bandit example highlights the missing information in learning. With known reward probabilities, expected values can be computed before playing. If those probabilities change or are unknown, calculation alone cannot reveal them. The agent needs observations from actions. An unlucky short sequence does not prove an action is poor, so repeated trials are needed to separate chance from a real difference in expected reward.

9.3.1 Why unknown probabilities change the task

In the double-bandit example, blue pays 1 with certainty. Red pays 2 with probability 0.75 and 0 otherwise, so its expected reward is 1.5. With 100 undiscounted rounds and the same choices in both win and lose states, always-red has expected return 150 and always-blue has 100. The outcome label does not affect the next round's opportunities, so those two states have the same remaining-horizon value.

If red's success probability becomes unknown, its expected value cannot be computed from the visible state alone. A short run of zeros might indicate a bad arm, or just bad luck. Trying red gathers information but risks earning less than the known blue reward. Choosing blue exploits a reliable option but leaves uncertainty about red unresolved. This is the central additional burden of reinforcement learning: actions both earn rewards and determine which evidence becomes available.

Source. Dan Klein and Pieter Abbeel, Introduction to Artificial Intelligence, Fall 2018, Lecture 9.

Lecture 10: Reinforcement Learning I

10.1 Learning a model or learning values

- **Estimate the dynamics.** Reinforcement learning retains the MDP objective but learns from observed transitions (s, a, r, s') instead of being given T and R. Model-based learning counts successor outcomes to estimate $\widehat{T}(s, a, s') = N(s, a, s')/N(s, a)$ and estimates rewards from observations. It then solves the estimated MDP. Unvisited state-action pairs need further experience before their model can be trusted.
- **Evaluate a fixed policy directly.** In passive learning, the policy is fixed and the task is evaluation. Direct evaluation averages observed discounted returns following visits to each state. It is model free and simple, but learns states separately rather than exploiting known estimates of their successors; it also waits for a return to become available.
- **Interpret the empirical counts.** In a learned model, counts answer a local question: among the times action a was taken in state s, how often did each successor occur? If four observations of (s, a) lead to u, u, u, v, the transition estimates are $\widehat{T}(s, a, u) = 3/4$ and $\widehat{T}(s, a, v) = 1/4$. A reward estimate is an average if the same transition can have different rewards. Planning on this model still inherits errors from limited data.

Direct return averaging learns a different object. For an episode with rewards $-1, -1, 10$ and $\gamma = 1$, the return from the first state is 8, from the second is 9, and from the third is 10. Each observed return is a noisy sample of a policy value. The method does not automatically force two states with the same successor to share what was learned there.

10.1.1 Comparing methods on the same four episodes

Under a fixed policy, suppose two observed episodes follow $B \to C \to D \to$ terminal, one follows $E \to C \to D \to$ terminal, and one follows $E \to C \to A \to$ terminal. Every move into C, D, A pays -1; exiting D pays 10 and exiting A pays -10. Use $\gamma = 1$ for these terminating episodes. The returns are

Episode type	Starting-state return	Return at C	Exit-state return
B, C, D (twice)	8 at B	9	10 at D
E, C, D	8 at E	9	10 at D
E, C, A	-12 at E	-11	-10 at A

Direct averaging gives $V(B) = 8$, $V(E) = (8 - 12)/2 = -2$, and $V(C) = (9 + 9 + 9 - 11)/4 = 4$. These finite-sample estimates disagree about B and E even though both always reach C at the same cost. Each state was evaluated only from the luck of its own observed episode continuations.

A model-based learner pools the four transitions out of C, estimating $P(D \mid C) = 3/4$ and $P(A \mid C) = 1/4$. Evaluating that learned model gives

$$V(C) = -1 + \tfrac{3}{4}(10) + \tfrac{1}{4}(-10) = 4, \qquad V(B) = V(E) = -1 + 4 = 3.$$

This uses the shared successor structure. It does not prove the true model has probability 3/4; that is still an estimate from four outcomes.

Model-free estimation can also exploit sampling without reconstructing a distribution. For a random variable with observations $x_1, \ldots, x_N$, the empirical mean $N^{-1} \sum_i x_i$ equals the expectation under the empirical frequency distribution. Likely outcomes appear more often, supplying the weights through their frequency. TD uses this idea on Bellman targets while retaining the connection between a state and its successor estimate.

10.1.2 Worked example: estimate a transition model from raw experience

Suppose five illustrative observations of one pair (s, a) are $(u, 2), (u, 4), (v, -1), (u, 3), (v, 1)$, where each tuple lists successor and reward.

1. **Count successors.** Three observations reached u and two reached v, giving $\widehat{T}(s, a, u) = 3/5$ and $\widehat{T}(s, a, v) = 2/5$. These are estimates, not known physical probabilities.
2. **Average rewards within each transition.** The estimates are $\widehat{R}(s, a, u) = (2 + 4 + 3)/3 = 3$ and $\widehat{R}(s, a, v) = (-1 + 1)/2 = 0$. The overall expected immediate reward is $(3/5)3 + (2/5)0 = 9/5$, also the average of all five rewards.
3. **Use the model to plan.** For $\gamma = 1/2$ and continuation estimates $V(u) = 4$, $V(v) = -2$, the action backup is $(3/5)(3 + 2) + (2/5)(0 - 1) = 13/5$.
4. **Separate data from decisions.** This backup can be recomputed when continuation values change without collecting the same transitions again. But limited observations may make the estimated model inaccurate; repeated planning does not itself create additional evidence.

For an untried action the denominator in the empirical transition frequency is zero. A learner needs experience or an explicitly chosen initial model; interpreting missing data as a verified zero transition probability is invalid.

10.2 Temporal differences

Temporal-difference learning updates on each transition under the fixed policy:

$$\delta = r + \gamma V(s') - V(s), \qquad V(s) \leftarrow V(s) + \alpha\delta.$$

The target is one observed reward plus an estimated continuation. Sampling outcomes automatically weights them by their transition probabilities in expectation. This bootstrapping shares information through the connections between states without estimating a transition table.

The update is an interpolation $(1 - \alpha)V(s) + \alpha[r + \gamma V(s')]$. A constant learning rate emphasizes recent samples; suitably decreasing rates allow averages to settle. Terminal continuation values are zero. TD evaluation alone does not supply a model-free improvement rule from state values: deciding among actions would still require their transition and reward models.

The TD error is a mismatch between two estimates of the same return. $V(s)$ is the current prediction before the transition. $r + \gamma V(s')$ is a new target using the reward just seen and the current prediction at the successor. If the target is larger, increase $V(s)$; if it is smaller, decrease it.

For the transition $B \to C$ with reward -2, let $V(B) = V(C) = 0$, $\gamma = 1$, and $\alpha = 1/2$. The target is -2, so $V(B)$ becomes -1. Next observe $C \to D$ with reward -2 and $V(D) = 8$. The target is 6, so $V(C)$ becomes 3. A later visit to $B \to C$ can now use that improved estimate of C. Learning propagates backward through repeated experience.

A constant learning rate is not an ordinary average of all past targets. Expanding the recursion gives weights $\alpha, \alpha(1 - \alpha), \alpha(1 - \alpha)^2, \ldots$ on increasingly old targets. This can be useful while the targets themselves are changing, but it leaves continuing noise. Decreasing rates allow estimates to settle.

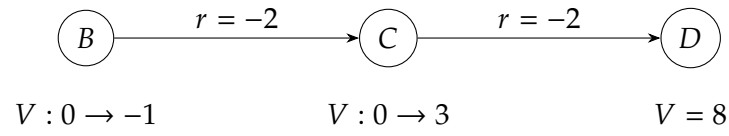


Figure 15: Each observed transition updates the prediction at its starting state.

10.2.1 Following successive TD updates

Continue the $B \to C \to D$ illustration after the two updates already shown. The stored values are now $V(B) = -1$, $V(C) = 3$, $V(D) = 8$. If the next episode again starts with $B \to C$ and reward -2, then

$$\text{target} = -2 + 3 = 1, \quad \delta = 1 - (-1) = 2, \quad V(B) \leftarrow -1 + \tfrac{1}{2}(2) = 0.$$

No new terminal reward was needed for this update. The better estimate of C changed the prediction at B. If the next $C \to D$ again pays -2, its target remains 6 and $V(C)$ moves from 3 to 4.5.

Repeated visits carry these changes backward; one traversal does not instantly propagate a new terminal estimate through all predecessors.

For a fixed value vector, the expected target under the policy is

$$\mathbb{E}[r + \gamma V(s') \mid s, \pi(s)] = \sum_{s'} T(s, \pi(s), s')[R(s, \pi(s), s') + \gamma V(s')].$$

This is the policy-evaluation backup. An individual target can be above or below that expectation, so replacing the old estimate outright can be noisy. The learning rate controls how much to trust one observation.

With constant α and targets $z_1, \ldots, z_t$, repeated interpolation expands to

$$V_t = (1 - \alpha)^t V_0 + \sum_{i=1}^{t} \alpha(1 - \alpha)^{t-i} z_i.$$

The coefficients sum to one including the initial-value term. Old targets receive geometrically smaller weights. Choosing $\alpha_t = 1/t$ for a fixed sequence of targets instead reproduces an ordinary sample mean. In TD the targets themselves change as successor values improve; the update is a bootstrapping process, not simply averaging a fixed batch of episode returns.

10.2.2 Worked example: the same trajectory teaches TD and direct evaluation differently

Take a deterministic episode $A \xrightarrow{0} B \xrightarrow{4}$ terminal, with $\gamma = 1/2$, learning rate $\alpha = 1/2$, and initially zero values.

1. **Process the first transition online.** Its TD target is $0 + \frac{1}{2}V(B) = 0$, so $V(A)$ remains zero. The later reward has not yet been observed or incorporated into B.
2. **Process termination.** At B, the continuation is zero and the target is 4. The update gives $V(B) = 0 + \frac{1}{2}(4 - 0) = 2$.
3. **Repeat the episode.** Its first transition now targets $\frac{1}{2}(2) = 1$, so $V(A)$ becomes 0.5. The second transition raises $V(B)$ from 2 to 3. TD propagates the reward backward through successive predictions rather than revising every earlier state immediately.
4. **Compare direct return targets.** After a complete episode, the observed returns are $G_A = 2$ and $G_B = 4$. Using the same half-step averaging rule would update directly toward these values. Ordinary sample averaging would instead use its visit-dependent learning rate.

Neither target should be confused with an extra reward at the terminal state. The reward 4 belongs to the transition out of B and must be counted once. The distinction between sampled full returns and bootstrapped successor estimates explains the different learning behavior.

10.3 Q-learning

Active learning chooses actions as well as evaluating them. The model-based Bellman update for action values is

$$Q_{k+1}(s,a) = \sum_{s'} T(s,a,s')[R(s,a,s') + \gamma \max_{a'} Q_k(s',a')].$$

Q-learning replaces this expectation by sampled updates:

$$Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s',a') - Q(s,a)].$$

Action selection then uses $\arg\max_a Q(s,a)$ without a transition model. Q-learning is off-policy: the greedy continuation in its target can differ from the exploratory behavior that collected the data. In the finite, discounted tabular setting with bounded rewards, sufficient repeated visits to every state-action pair and per-pair rates satisfying $\sum_t \alpha_t = \infty$ and $\sum_t \alpha_t^2 < \infty$ permit convergence to Q^*. Merely acting greedily from the start need not provide that coverage.

For a terminal successor, use a continuation of zero rather than maximizing over an empty action set. Continued exploration and suitable rates are part of the convergence assumptions. Off-policy learning does not mean that an agent can learn the value of actions it never tries.

10.3.1 A Q-learning trace and the role of the behavior policy

Use an illustrative state s with action a leading to x for reward 0. At x, action good exits for 4, and action bad exits for -2. Let $\gamma = 1/2$, $\alpha = 1/2$, and initialize every Q-entry to zero.

1. Observe $(s, a, 0, x)$. The successor maximum is zero, so $Q(s,a)$ remains zero.
2. Observe $(x, \text{good}, 4, \text{terminal})$. Its target is 4, so $Q(x, \text{good})$ becomes 2.
3. On another visit observe $(s, a, 0, x)$. The target is now $0 + (1/2)\max(2, 0) = 1$, so $Q(s,a)$ becomes 0.5.
4. Suppose exploration then chooses bad at x. Its observed target is -2, so $Q(x, \text{bad})$ becomes -1. This changes that entry, but does not retroactively replace the maximum used in the preceding update.

The last step makes off-policy learning concrete: the target at s assumed the best estimated continuation at x, even though the behavior went on to choose another action. The optimal limiting value of $Q(s,a)$ is 2, since optimal continuation earns 4 one step later. Repeated evidence and updates are needed to approach it.

Source. Dan Klein and Pieter Abbeel, Introduction to Artificial Intelligence, Fall 2018, Lecture 10.

Lecture 11: Reinforcement Learning II

11.1 Exploration and regret

- **Random exploration.** An ϵ-greedy behavior policy selects a random action with probability ϵ and a currently greedy action otherwise. Persistent randomness provides exploration but keeps making avoidable mistakes after learning. Decreasing ϵ can reduce that cost, provided exploration still visits the relevant state-action pairs often enough.
- **Count-based optimism.** An exploration function instead combines value and visit count into an optimistic score $f(u, n)$. For example, give a high score R^+ while a pair has fewer than N_e visits, then use its estimated value. Replacing $\max_{a'} Q(s', a')$ by $\max_{a'} f(Q(s', a'), N(s', a'))$ in a target can propagate the incentive to reach unexplored states backward. The bonus is an exploration device, not a change to the actual reward objective.
- **Learning cost.** Regret measures the expected reward lost relative to optimal behavior over the learning period. Two methods can converge to the same optimal policy and still have very different regret because they make different mistakes on the way.

Exploration must sometimes pay a short-term cost to reach a useful unknown area. Randomly choosing among the immediate actions is easy, but it may be inefficient when learning requires several coordinated exploratory steps. An optimistic score gives the agent a reason to travel toward an uncertain state, not just to randomize after arriving there.

The number of visits matters separately for each state-action pair. An action observed once with a bad result is less well understood than one observed many times with consistently bad results. A bonus expresses that difference; it should gradually stop dominating evidence.

Regret counts the price paid while learning. In a simple bandit illustration, suppose the best action's expected reward is 1.5 and another action's is 1. Taking the second action on 20 trials costs $20(1.5 - 1) = 10$ expected reward units relative to always taking the better action. Individual realized rewards can differ by chance. Some of that loss may be worthwhile to discover which action is better, but unnecessary exploration adds cost after the distinction is already clear.

11.1.1 Quantifying exploration rather than only naming it

With k legal actions and one greedy action, ϵ-greedy selects that action with probability $1 - \epsilon + \epsilon/k$. Each other action has probability ϵ/k. The random branch includes the greedy action. For two actions and $\epsilon = 0.1$, the probabilities are 0.95 and 0.05, not 0.9 and 0.1. If their true expected rewards

are 1.5 and 1, the expected reward after learning is 0.95(1.5) + 0.05(1) = 1.475. The continuing loss of 0.025 per round is the cost of persistent randomness in this simple example.

An optimistic exploration function can be written

$$f(u, n) = \begin{cases} R^+, & n < N_e, \\ u, & n \geq N_e. \end{cases}$$

Here N_e is a visit threshold and R^+ is an optimistic score. With $N_e = 5$, an action worth an estimated 1 after one visit may still receive score $R^+ = 10$, while a well-tested action worth 4 receives score 4. The first is chosen to gather evidence, not because its estimated ordinary reward exceeds 4. Once it is sufficiently visited, the bonus stops masking its actual estimate.

Putting the same optimistic scores inside the successor target can give an incentive to travel toward untested actions. A state leading to an unknown region can acquire a high exploratory value before the region has paid a large actual reward. Counts must still refer to the correct state-action pairs. Counting visits to a state alone would treat an untried action there as familiar merely because a different action was used often.

In a multi-step problem, exploration can require a sequence of choices. If reaching a useful area requires five particular random choices among two actions, their probability under independent uniform choices is 2^{-5}. Making ϵ small reduces it further when those choices are nongreedy. This does not prove a particular exploration function will solve every hard case; it explains why occasional local randomness is not automatically an efficient way to gather distant information.

11.1.2 Worked example: count the actual action probabilities

There are three actions with current estimates 5, 2, 1, and the first is uniquely greedy. An illustrative ϵ-greedy learner uses $\epsilon = 0.3$ and samples uniformly from all three actions when exploring.

1. **Include greedy choices during exploration.** The first action has probability 0.7 + 0.3/3 = 0.8. Each other action has probability 0.1. The exploration probability is not the probability of taking a nongreedy action.
2. **Compute the reward under a known comparison model.** If these numbers also happen to be the true expected rewards, the learner earns 0.8(5) + 0.1(2) + 0.1(1) = 4.3 per step on average.
3. **Measure the loss from continuing exploration.** Always taking the optimal action earns 5, so the expected per-step regret is 0.7. Over 100 identical decision opportunities it is 70 expected reward units.
4. **Separate estimates from truth.** Before learning, the estimated leader might not be optimal. Random actions can reveal that mistake and improve later rewards. The regret calculation

above describes persistent exploration after the ranking is already correct, not its entire learning benefit.

Some implementations explore only among nongreedy actions. Their probabilities would differ. State which convention is used before calculating visitation, expected reward, or regret.

11.2 Approximate Q-learning

A table cannot generalize to unvisited states and can be too large to store. Represent a state-action pair by features $f_i(s, a)$, such as distances to food or ghosts, and use a linear approximation

$$Q_w(s, a) = \sum_i w_i f_i(s, a).$$

Experience now changes a shared set of weights. Define the TD error

$$\delta = r + \gamma \max_{a'} Q_w(s', a') - Q_w(s, a), \qquad w_i \leftarrow w_i + \alpha \delta f_i(s, a).$$

This is a gradient step on squared prediction error when the sampled target is held fixed: the derivative of the prediction with respect to w_i is $f_i(s, a)$. A negative TD error lowers the prediction for the current pair and changes estimates for other pairs sharing its features. With one indicator feature per pair, the rule recovers tabular Q-learning.

Generalization saves data and memory, but states with the same features may have different values. Capacity limits can reduce overfitting, while an inadequate feature representation can prevent a good policy. The tabular convergence guarantee does not automatically carry over to approximation.

Example: Suppose two features of a pair are $f_1 = 1$ and $f_2 = 2$, with weights $w_1 = 0.5, w_2 = 1$. Its prediction is $Q_w = 2.5$. If the sampled target is 1.5, then $\delta = -1$. With $\alpha = 0.1$, the weights become $w_1 = 0.4$ and $w_2 = 0.8$. The new prediction on that pair is 2.0. The larger second feature caused a larger change to its weight.

The same change affects any other pair with those features. A pair with features $(1, 0)$ changes from 0.5 to 0.4 even if it has never been visited. This is the benefit of generalization, and also its danger: shared features can make experiences interfere when the situations should have different values.

The squared-error interpretation treats the target as fixed during the update. For target y, define $L(w) = \frac{1}{2}(y - w^\top f)^2$. Then $\partial L/\partial w_i = -(y - w^\top f) f_i = -\delta f_i$. Gradient descent gives exactly $w_i \leftarrow w_i + \alpha \delta f_i$. This does not make the whole Q-learning process ordinary supervised regression, because later targets also depend on the changing weights.

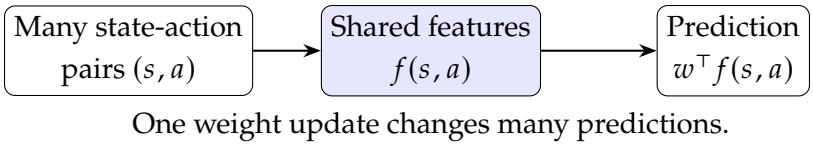


Figure 16: Features tie different experiences to the same learned parameters.

11.2.1 Exactly how one experience generalizes

Let the experienced feature vector be $f = (1, 2)$ and let its update be $\Delta w = (-0.1, -0.2)$ as in the example. For any other feature vector g,

$$\Delta Q(g) = \Delta w^\top g = \alpha \delta f^\top g.$$

The inner product determines the transfer. A pair with $g = (1, 0)$ decreases by 0.1. A pair with $g = (0, 1)$ decreases by 0.2. A pair with $g = (0, -1)$ *increases* by 0.2, because that feature has the opposite sign. A pair orthogonal to f, for example $g = (2, -1)$, is unchanged. Generalization is therefore a precise consequence of the representation and update, not a claim that all vaguely similar states improve together.

If f is an indicator vector with a separate coordinate for every pair (s, a), exactly one component is one and the others are zero. Only its weight changes, reproducing the tabular update. This representation has no transfer to unseen pairs. A compact feature vector saves parameters by making many predictions depend on the same coordinates, at the cost of being unable to distinguish some situations.

For Pac-Man, distance to the nearest ghost can be informative but incomplete. Two states can have the same nearest distance while one is a dead end and the other has several exits. If every supplied feature agrees, a linear function must give the same prediction in both states for the same represented action, even if their true values differ. More training cannot fix an ambiguity that the feature map erases. Adding relevant features changes capacity; adding arbitrary features can also increase the data required to estimate useful weights.

A bias feature equal to one permits a baseline value independent of other measurements. Feature scale affects learning as well: multiplying one feature by 100 multiplies that coordinate's update by 100 at a fixed TD error, and its contribution to the next prediction changes by another factor of 100. Step sizes that behave sensibly for one encoding may be unstable for an otherwise equivalent rescaled encoding.

11.2.2 Worked example: identical features can force contradictory predictions

Suppose two state-action pairs have the same feature vector $f = (1, 0)$, but their desired action values are 4 and -2. A linear approximator predicts $Q_w = w^\top f = w_1$ for both.

1. **Check representability before optimizing.** No weight vector can predict both targets exactly, because the representation makes the two inputs indistinguishable. More training cannot remove this restriction.
2. **Find a compromise on fixed targets.** With equal frequency, the average squared loss is $L(w_1) = \frac{1}{4}[(w_1 - 4)^2 + (w_1 + 2)^2]$. Its derivative is $w_1 - 1$, so the minimizing shared prediction is 1.
3. **Interpret the errors.** Both residual magnitudes are 3. The average target is fitted, while neither distinct situation is represented accurately. Updates from one pair can undo progress on the other.
4. **Add a distinguishing feature.** Use features $(1, 0)$ for the first pair and $(1, 1)$ for the second. Then $w = (4, -6)$ predicts 4 and -2. The change improves what can be expressed, rather than merely changing the learning rate.

This fixed-target calculation isolates the representation issue. In actual Q-learning, targets also depend on current predictions and subsequent action choices, so ordinary regression convergence is not a general guarantee for the whole learning process.

11.3 Policy search

Accurate numerical value estimates are a means to choosing good actions; their ordering can matter more than their absolute errors. Policy search directly adjusts policy parameters to improve expected return. A simple approach perturbs feature weights and compares returns from sampled episodes. Random outcomes require repeated trials, and testing each parameter separately becomes expensive in high dimension. Robotic locomotion and manipulation illustrate why generalization and learning effective policies are needed when enumerating all states is infeasible.

11.3.1 Prediction accuracy versus action quality

Suppose the true action values are $(10, 9)$. Estimates $(100, 90)$ have large numerical errors but choose the better action. Estimates $(9.4, 9.5)$ are much closer numerically yet choose the worse action. This is why squared value error and achieved return need not improve together. An evaluation function used for action selection can be useful without predicting the absolute reward stream accurately.

A simple direct policy search starts from weights w, changes one component to $w_i + h$ or $w_i - h$, and evaluates each resulting policy through repeated episodes. It retains a change if the estimated return improves, then continues with other coordinates. The policy must actually be rerun: a larger linear score does not itself establish better behavior. If changing a weight does not change any selected action, the return can remain unchanged despite a change in predicted values.

Source. Dan Klein and Pieter Abbeel, Introduction to Artificial Intelligence, Fall 2018, Lecture 11.

Lecture 12: Probability

12.1 Distributions, events, and conditioning

- **Represent uncertainty.** A random variable represents an uncertain quantity and has a domain of possible values. A discrete probability distribution assigns nonnegative mass summing to one. A JOINT DISTRIBUTION assigns a probability to every complete assignment of a collection of variables. With n variables of domain size d, its table has d^n entries.
- **Condition and marginalize.** An event is a set of assignments; its probability is the sum of their masses. To obtain a MARGINAL, sum out variables not of interest: $P(x) = \sum_y P(x, y)$. For $P(e) > 0$, conditioning restricts attention to assignments satisfying the evidence and renormalizes:

$$P(x \mid e) = \frac{P(x, e)}{P(e)}, \qquad P(e) = \sum_x P(x, e).$$

  A distribution such as $P(X \mid e)$ is a table over X; an expression with a specified lowercase value, $P(x \mid e)$, is a number. A family $P(X \mid Y)$ contains a normalized distribution for every value of Y.
- **Read a numerical joint table.** For the joint masses $P(\text{hot}, \text{sun}) = 0.4$, $P(\text{hot}, \text{rain}) = 0.1$, $P(\text{cold}, \text{sun}) = 0.2$, and $P(\text{cold}, \text{rain}) = 0.3$, marginalizing gives $P(\text{cold}) = 0.5$. Selecting the cold rows and dividing by 0.5 gives $P(\text{sun} \mid \text{cold}) = 0.4$.

It helps to distinguish three operations on a joint table. Selection removes rows inconsistent with evidence. Marginalization adds rows that differ only in variables being discarded. Normalization divides every retained value by their total. Selection alone does not make probabilities sum to one.

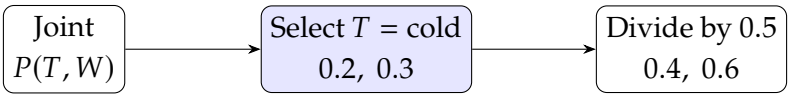


Figure 17: Conditioning first selects compatible outcomes, then rescales them.

The probability of an OR event needs care when its pieces overlap. In the weather table, $P(\text{hot or sun})$ includes the rows (hot, sun), $(\text{hot}, \text{rain})$, and $(\text{cold}, \text{sun})$, giving 0.7. Adding $P(\text{hot}) + P(\text{sun}) = 0.5 + 0.6$ counts the first row twice. Subtracting $P(\text{hot}, \text{sun}) = 0.4$ corrects that double count.

Conditioning is also directional. Here $P(\text{sun} \mid \text{cold}) = 0.4$, but $P(\text{cold} \mid \text{sun}) = 0.2/0.6 = 1/3$. They answer different questions and have different denominators.

12.1.1 Reading tables without confusing their meanings

For the weather model, the complete joint table is

	Sun	Rain	Temperature marginal
Hot	0.4	0.1	0.5
Cold	0.2	0.3	0.5
Weather marginal	0.6	0.4	1

Each interior entry describes an AND event. Row sums remove weather; column sums remove temperature. A conditional row such as $P(W \mid T = \text{cold})$ is obtained by dividing the two entries of the cold row by its row sum 0.5. A conditional column such as $P(T \mid W = \text{sun})$ divides the sun column by its different sum 0.6. The original joint table has one normalization condition over all four entries, while a family of conditional rows has a separate condition for each fixed temperature.

For any events A, B, the disjoint pieces $A \setminus B$, $B \setminus A$, and $A \cap B$ partition their union. Hence

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Marginalization is a simpler addition because assignments to a variable's different values are mutually exclusive. Summing $P(T, W)$ over all W never double-counts one complete outcome. A domain value of probability zero is allowed, but a row of negative probabilities or a distribution whose entries do not sum to one is not a valid model.

12.1.2 A complete enumeration query

Consider the season-temperature-weather joint:

Season	Temperature	Sun	Rain
Summer	Hot	0.30	0.05
Summer	Cold	0.10	0.05
Winter	Hot	0.10	0.05
Winter	Cold	0.15	0.20

To obtain $P(W)$, add all entries in each weather column: sun has mass 0.65, rain has 0.35. To obtain $P(W \mid \text{winter})$, select the two winter rows and sum out temperature: the masses are $0.25, 0.25$. Their sum is $P(\text{winter}) = 0.5$, so the posterior is $(0.5, 0.5)$. To obtain $P(W \mid \text{winter}, \text{hot})$, only one row remains; normalizing $(0.10, 0.05)$ gives $(2/3, 1/3)$.

The same model answers three different questions because the query, evidence, and hidden variables change. Evidence is fixed rather than summed over; hidden variables are summed over rather than

arbitrarily assigned their most likely value. Replacing the sum by a maximum would answer a different question and can produce a different distribution.

12.1.3 Worked example: fill a contingency table before conditioning

An illustrative message collection has 1000 emails. Of 50 spam messages, 40 contain a chosen word. Of 950 ordinary messages, 95 contain it.

	Word present	Word absent	Total
Spam	40	10	50
Ordinary	95	855	950
Total	135	865	1000

1. **Read the likelihood.** Among spam, the word appears with probability $40/50 = 0.8$. This conditions on the message already being spam.
2. **Read the posterior.** Among messages containing the word, only $40/135 = 8/27 \approx 0.296$ are spam. The denominator is the word-present column, not the spam row.
3. **Explain the base-rate effect.** Ordinary messages are much more common. Even their smaller word-occurrence rate $95/950 = 0.1$ produces more word-containing messages than spam does.
4. **Check a joint event.** The fraction that are both spam and word-containing is $40/1000 = 0.04$. This is neither of the preceding conditional probabilities; its denominator is the entire population.

A table makes the denominator visible. Before applying a formula, identify which population remains after the condition is imposed. Bayes' rule then reconstructs precisely this change of denominator from local probabilities.

12.2 Product, chain, and Bayes rules

The definition of conditioning gives the product rule $P(x, y) = P(x \mid y)P(y)$. Repeated application yields the chain rule, valid without independence assumptions:

$$P(x_1, \dots, x_n) = \prod_{i=1}^{n} P(x_i \mid x_1, \dots, x_{i-1}).$$

Writing $P(x, e)$ in both orders gives Bayes' rule:

$$P(x \mid e) = \frac{P(e \mid x)P(x)}{\sum_{x'} P(e \mid x')P(x')}.$$

The prior $P(x)$ is reweighted by the likelihood $P(e \mid x)$ to obtain the posterior. A likely observation under a hypothesis does not by itself make the hypothesis likely: its prior and competing explanations also matter. For a noisy location sensor, multiply each candidate location's prior by the probability of the observed reading at that location, then normalize.

Example: Use a simple illustrative diagnostic model. A condition has prior probability 0.01. A positive test has probability 0.9 when the condition is present and 0.1 when it is absent. Then

$$P(+) = 0.9(0.01) + 0.1(0.99) = 0.108, \qquad P(\text{condition} \mid +) = \frac{0.009}{0.108} = \frac{1}{12}.$$

The test is much more likely to be positive for an affected case, but the many unaffected cases still contribute more positive results in total.

The denominator is a sum over every way the evidence can arise. Omitting the unaffected cases would turn the likelihood $P(+ \mid \text{condition})$ into an incorrect posterior. This is the common error of reversing a conditional probability without accounting for base rates.

The chain rule requires no assumption about which variables cause which. For three variables it reads $P(a, b, c) = P(a)P(b \mid a)P(c \mid a, b)$. Replacing the last term by $P(c \mid b)$ is an additional independence assumption, not another use of the chain rule.

12.2.1 Deriving the chain rule step by step

For three variables, first group (X, Y) as one event:

$$P(x, y, z) = P(z \mid x, y)P(x, y).$$

Then expand $P(x, y) = P(y \mid x)P(x)$. Substitution gives $P(x, y, z) = P(x)P(y \mid x)P(z \mid x, y)$. Any variable ordering works, provided each factor conditions on all preceding variables. This does not reduce storage by itself: the final conditional can still have exponentially many parent contexts. A later independence assumption is what makes local representations smaller.

Bayes' rule is another rearrangement of the same joint: $P(x, e) = P(e \mid x)P(x) = P(x \mid e)P(e)$. Dividing by $P(e)$ gives the posterior. When comparing candidate x values, the denominator is the same for all of them, so one may first compute unnormalized scores $u(x) = P(e \mid x)P(x)$ and then normalize once. A likelihood is a function of the candidate hypothesis with the observation held fixed; it need not sum to one across hypotheses.

The diagnostic illustration can also be read as counts in a hypothetical population of 1000. Ten have the condition, of whom nine test positive. Of the 990 without it, 99 test positive. Among the 108 positive results, only nine have the condition, giving $9/108 = 1/12$. The denominator must include every way of seeing a positive result. This interpretation helps separate the test's sensitivity from the posterior among positive cases.

12.2.2 Updating a location belief from a sensor

For an illustrative three-location ghost model, let the prior over A, B, C be $(0.5, 0.3, 0.2)$. Suppose the observed color has likelihoods $(0.2, 0.6, 0.1)$ at those locations. Multiply coordinate by coordinate:

$$u = (0.10, 0.18, 0.02), \qquad Z = 0.30, \qquad P(G \mid r) = (1/3, 3/5, 1/15).$$

The most likely location changes from A to B, but A remains possible. The sensor reading is more compatible with B; the prior prevents the update from treating all locations as initially equally plausible. A zero posterior at a location requires a zero prior or zero likelihood under this model, not merely a comparatively unlikely reading.

A second independent sensor reading can be incorporated by multiplying its likelihood into the first posterior and renormalizing. Independence here must be conditional on the ghost location. If two readings share additional noise, multiplying their likelihoods as though independent can overstate the evidence. The arithmetic of Bayes' rule is only as appropriate as the model supplying its likelihoods.

12.2.3 Worked example: two successive observations need a dependence model

Let a hidden location be left or right with equal prior probability. Two sensors each report left with probability 0.8 when the location is left and 0.2 when it is right.

1. **Process the first report.** A left report gives unnormalized masses $0.5(0.8) = 0.4$ and $0.5(0.2) = 0.1$, so the posterior for left is 0.8.
2. **State the assumption for combining sensors.** If the two reports are independent given the location, two left reports have likelihoods 0.8^2 and 0.2^2. Their posterior is $0.64/(0.64 + 0.04) = 16/17$.
3. **Consider a duplicate report instead.** Suppose the second "sensor" simply copies the first output. Conditional on the first report, its second left report is certain under either location. The posterior stays 0.8, rather than becoming 16/17.
4. **Apply the chain rule honestly.** Always, $P(e_1, e_2 \mid x) = P(e_1 \mid x)P(e_2 \mid e_1, x)$. Replacing the second term by $P(e_2 \mid x)$ requires the stated conditional independence; it is not a general property of repeated evidence.

The numerical difference is the cost of counting correlated information as fresh evidence. Specifying a dependence model is part of the inference problem.

12.3 Inference by enumeration

Partition variables into query Q, evidence $E = e$, and hidden variables H. Select the joint entries matching e, sum out H, and normalize:

$$P(Q \mid e) = \alpha \sum_{h} P(Q, h, e).$$

Here α is the reciprocal of the sum of the resulting entries, provided the evidence has positive probability. Enumeration is exact, but a full joint table needs exponential storage and generally exponential work. Structured models will express the joint using smaller local tables.

For example, let S be season, T temperature, and W weather. To query weather given $T = t$, sum out the hidden season:

$$P(W \mid T = t) = \alpha \sum_{s} P(s, t, W).$$

For each weather value, add the masses for both seasons. These sums describe weather together with the fixed temperature. Their total is $P(T = t)$. Dividing by it produces a distribution over weather alone.

If the evidence has probability zero under the model, that denominator is zero. The ordinary conditional formula supplies no answer; one must inspect whether the model or observation is inappropriate rather than divide by zero.

Source. Dan Klein and Pieter Abbeel, Introduction to Artificial Intelligence, Fall 2018, Lecture 12.

Lecture 13: Bayes Nets: Representation

13.1 Conditional independence

- **The independence statement.** Two variables are independent when $P(x, y) = P(x)P(y)$ for all values. They are conditionally independent given Z, written $X \perp\!\!\!\perp Y \mid Z$, when, for all z of positive probability,

  $$P(x, y \mid z) = P(x \mid z)P(y \mid z).$$

  Equivalently, knowing Y adds no information about X once Z is known. Conditional independence need not imply unconditional independence. For example, two sensor readings can both depend on a ghost's location while being independent once the location is fixed.

- **Check every conditioning case.** To check conditional independence, the statement must hold for every relevant value of the conditioning variable, not merely for one convenient case. For the cavity example, suppose toothache and a probe catching are generated independently once cavity status is fixed. Knowing the toothache then adds nothing to the catch probability if cavity status is already known. Without that status, a toothache changes belief about the cavity, which in turn changes belief about the probe catching.

- **The modeling commitment.** Independence saves parameters because a table over combinations can be replaced by smaller tables. It also commits the model to treating some information as redundant. The usefulness of the simplification depends on whether those commitments fit the application.

13.1.1 Checking independence with actual probabilities

The weather table has $P(\text{hot}) = 0.5$ and $P(\text{sun}) = 0.6$. If these variables were independent, their joint hot-sun probability would be $0.5(0.6) = 0.3$. Its actual value is 0.4, so the variables are not independent. One failed assignment is enough to disprove independence; confirming one equality is not enough to prove all the required equalities.

Conditional independence concerns an entire family of conditional tables. In the ghost example, let G say the ghost is at the top, and T, B say the top and bottom sensors are red. Take

$$P(g) = \tfrac{1}{2}, \quad P(t \mid g) = 0.8, \quad P(t \mid \neg g) = 0.4, \quad P(b \mid g) = 0.4, \quad P(b \mid \neg g) = 0.8.$$

Assume $T \perp\!\!\!\perp B \mid G$. Then $P(t, b, g) = 0.5(0.8)(0.4) = 0.16$ and $P(t, b, \neg g) = 0.5(0.4)(0.8) = 0.16$. Thus $P(t, b) = 0.32$, whereas $P(t) = P(b) = 0.6$ and $P(t)P(b) = 0.36$. The sensors are conditionally independent but not independent. Their common hidden location induces an association.

This distinction is useful for representation: once G is fixed, each sensor has its own local noise model. Without knowing G, one reading provides evidence about location and thereby about the other reading. The assumption removes a direct dependence between sensor errors; it does not prevent indirect inference through the shared hidden cause.

13.1.2 Worked example: independence can disappear after conditioning

Let X, Y be independent fair bits, and define $Z = 1$ exactly when they differ. The four pairs $(0,0), (0,1), (1,0), (1,1)$ each have probability $1/4$.

1. **Check the unconditional statement.** For example, $P(X = 1, Y = 1) = 1/4 = P(X = 1)P(Y = 1)$. The same product rule holds for all four pairs.
2. **Condition on agreement.** The event $Z = 0$ retains only $(0,0)$ and $(1,1)$, each with conditional mass $1/2$.
3. **Test a conditional product.** Both conditional marginals are $1/2$, but $P(X = 1, Y = 1 \mid Z = 0) = 1/2 \neq (1/2)(1/2)$. Thus X and Y are not independent given $Z = 0$.
4. **Interpret the information.** Once agreement is known, revealing X completely determines Y. Before agreement was known, revealing X said nothing about Y. Conditioning changes which assignments remain possible.

The phrase "given Z" does real work in an independence claim. Neither unconditional independence nor conditional independence can generally be inferred from the other without examining the distribution's structure.

13.2 Graph and local probabilities

A Bayesian network is a directed acyclic graph with one random variable per node and a conditional probability table (CPT) $P(X_i \mid \mathrm{Pa}(X_i))$ per node. It defines

$$P(x_1, \ldots, x_n) = \prod_i P(x_i \mid \mathrm{pa}(x_i)).$$

Choose a topological order. The modeling assumption replaces all earlier variables in the chain rule by the node's parents: the node is independent of earlier nonparents given its parents. The product is normalized because variables can be summed out in reverse topological order, each CPT summing to one for fixed parent values.

In the alarm network, burglary B and earthquake E influence alarm A; the alarm influences calls J and M:

$$P(B, E, A, J, M) = P(B)P(E)P(A \mid B, E)P(J \mid A)P(M \mid A).$$

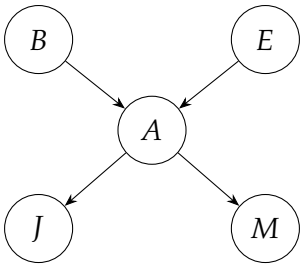


Figure 18: The alarm network's local dependencies. All variables are unobserved.

The absence of an edge does not say that two nodes can never be statistically related. Burglary can be related to a call through the alarm even though there is no direct edge from burglary to the call. Likewise, the graph does not say that an arrow guarantees a large numerical effect; the CPT determines how strongly parent values change a child's probabilities.

A network of independent coin flips has no edges and factors into individual priors. For n Boolean variables with at most k parents each, CPT storage is $O(n2^k)$ rather than $O(2^n)$ for a joint table. This saving depends on the graph's sparsity.

A CPT is a separate distribution for each combination of parent values. For Boolean A with two Boolean parents, there are four parent contexts. Within each context, $P(A = 1 \mid B, E)$ determines $P(A = 0 \mid B, E) = 1 - P(A = 1 \mid B, E)$. A root has no parent context and its CPT is simply a prior.

Example: In the alarm model, take $P(b) = 0.001$, $P(\neg e) = 0.998$, $P(a \mid b, \neg e) = 0.94$, $P(j \mid a) = 0.9$, and $P(m \mid a) = 0.7$. The probability of this *complete* assignment is

$$P(b, \neg e, a, j, m) = 0.001(0.998)(0.94)(0.9)(0.7) = 0.0005910156.$$

No summation is needed because every variable is assigned. Computing $P(b \mid j, m)$ is different: alarm and earthquake are not specified, so their possible values must be summed out before normalization.

For n Boolean variables with at most k parents, the number of independent CPT parameters is at most $n2^k$: one probability for each parent context. A full joint instead needs $2^n - 1$ independent parameters. This saving is large only when the local parent sets remain small.

13.2.1 Specifying the entire alarm model

For Boolean variables, one probability per parent context determines the complement. The alarm model's parameters are

$$P(b) = 0.001, \quad P(e) = 0.002,$$

	e	$\neg e$
$P(a \mid b, E)$	0.95	0.94
$P(a \mid \neg b, E)$	0.29	0.001

and

$$P(j \mid a) = 0.9, \quad P(j \mid \neg a) = 0.05, \qquad P(m \mid a) = 0.7, \quad P(m \mid \neg a) = 0.01.$$

For example, $P(\neg a \mid b, \neg e) = 0.06$. There are ten independent parameters: two root probabilities, four alarm contexts, and two contexts for each call. The unrestricted five-variable joint needs $2^5 - 1 = 31$. This saving comes with assumptions: burglary and earthquake are independent roots, and the calls are independent given alarm status.

A CPT entry is not a posterior over its parents. The large value $P(a \mid b, \neg e) = 0.94$ does not imply that $P(b \mid a) = 0.94$. Burglary is rare, and earthquake or false alarms also explain a. To reverse that conditional one must use priors and sum over alternative causes. The network specifies a joint from which such questions can be computed; it does not place every desired conditional directly in a CPT.

13.2.2 Why local normalization produces a normalized joint

For the alarm product, sum over both call variables first:

$$\sum_j P(j \mid a) = 1, \qquad \sum_m P(m \mid a) = 1.$$

The joint sum reduces to $\sum_{b,e,a} P(b)P(e)P(a \mid b, e)$. Next $\sum_a P(a \mid b, e) = 1$ for every fixed b, e, leaving $\sum_b P(b) \sum_e P(e) = 1$. Every factor is nonnegative, so the product both assigns nonnegative masses and sums to one. In any DAG, reverse topological summation repeats this argument: a remaining leaf appears only in its own conditional factor when it is summed away.

This proof also shows why a directed cycle is a problem for these semantics. There is no topological order supporting this repeated conditional sampling and elimination argument. Arbitrarily multiplying conditionals around a cycle does not inherit the normalization guarantee of a Bayes net.

13.2.3 Worked example: distinguish a joint assignment from a partial event

Consider an illustrative net $A \to B$, $A \to C$, with Boolean variables. Let $P(a) = 0.4$, $P(b \mid a) = 0.8$, $P(b \mid \neg a) = 0.2$, $P(c \mid a) = 0.7$, and $P(c \mid \neg a) = 0.1$.

1. **Complete the CPTs.** Each missing false entry is one minus its true entry. These five free probabilities specify all eight joint assignments.
2. **Evaluate a complete assignment.** The product rule gives $P(a, b, c) = 0.4(0.8)(0.7) = 0.224$. No summation is needed because every variable has a specified value.
3. **Evaluate a partial event.** For $b, \neg c$, the value of A is unknown, so sum its two cases:

$$P(b, \neg c) = 0.4(0.8)(0.3) + 0.6(0.2)(0.9) = 0.204.$$

Multiplying just $P(b \mid a)P(\neg c \mid a)$ would silently assume $A = a$.

4. **Answer a posterior query.** The contribution with $A = a$ is 0.096, so $P(a \mid b, \neg c) = 0.096/0.204 = 8/17$. Both observations matter through their likelihood under each possible cause.

The graph provides a factorization, not a rule that every query is one product. Products evaluate complete assignments; sums handle hidden alternatives; normalization turns retained masses into a posterior distribution.

13.3 Representation and causality

A topology restricts which distributions can be represented by requiring certain conditional independences. A fully connected acyclic graph can express any joint distribution through the chain rule. Causal ordering often makes the model easier to specify, but arrows in a general Bayes net assert a factorization, not necessarily causation. A two-variable joint can be factored in either direction, with appropriately changed priors and CPTs. Observational association alone does not establish which variable causes the other.

The acyclic requirement permits a topological order in which every parent comes before its child. Starting with the chain rule in that order explains the factorization. It also explains a way to generate samples: draw roots first, then draw each child given the parents already generated.

Reversing an edge can preserve a distribution only if the other tables are changed appropriately and the resulting graph still represents the required dependencies. Simply reversing an arrow while retaining the same numbers generally specifies a different model.

13.3.1 Reversing a two-variable factorization correctly

For the traffic example take $P(r) = 1/4$, $P(t \mid r) = 3/4$, and $P(t \mid \neg r) = 1/2$. The joint entries $(r, t), (r, \neg t), (\neg r, t), (\neg r, \neg t)$ are $(3, 1, 6, 6)/16$. Hence

$$P(t) = \frac{9}{16}, \qquad P(r \mid t) = \frac{3/16}{9/16} = \frac{1}{3}, \qquad P(r \mid \neg t) = \frac{1/16}{7/16} = \frac{1}{7}.$$

The graph $T \to R$ with this prior and these conditionals represents the same joint as $R \to T$. It does not assert that traffic physically causes rain. Reusing $P(r) = 1/4$ as the root of the reversed graph or copying the old conditional numbers would fail to reproduce the same distribution.

With more variables, graph choice affects which parent sets are sufficient to express the required dependencies. Causal ordering is often a convenient way to obtain small local tables, but the mathematical object being specified here is a joint distribution and its independence assumptions. Distinct causal stories can be consistent with the same observational factorization.

Source. Dan Klein and Pieter Abbeel, Introduction to Artificial Intelligence, Fall 2018, Lecture 13.

Lecture 14: Bayes Nets: Independence

14.1 Three local configurations

- **Distribution versus graph.** Independence is a property of a distribution. A graph guarantees some independences for every choice of its CPTs; a particular numerical model can have additional independences.
- **Chains and common causes.** In a chain $X \to Y \to Z$, observing the middle variable blocks the connection: $X \perp\!\!\!\perp Z \mid Y$. Without that observation, changes in X can change beliefs about Y and hence Z. A fork $X \leftarrow Y \to Z$ behaves similarly: the effects can be associated through a shared cause, but become independent given that cause, in the isolated three-node model.
- **Colliders and explaining away.** A collider $X \to Y \leftarrow Z$ behaves oppositely. With independent root causes, summing out Y leaves $P(X)P(Z)$, so $X \perp\!\!\!\perp Z$. Observing Y can couple them: evidence for one explanation can reduce the need for the other. Observing a descendant of Y can also activate this connection. This is EXPLAINING AWAY.



blocked chain active collider

Figure 19: Shading marks observation. The arrow pattern determines its effect on a path.

Example: Consider a simple explaining-away model with independent rare causes X, Z, each having probability 0.1, and an effect Y that occurs exactly when at least one cause occurs. Before observing the effect, $P(X) = 0.1$. After $Y = 1$,

$$P(X = 1 \mid Y = 1) = \frac{0.1}{1 - 0.9^2} = \frac{10}{19}.$$

If $Z = 1$ is then learned, it already explains the effect, and $P(X = 1 \mid Y = 1, Z = 1) = 0.1$ again. Information about one cause changes belief about the other only after evidence couples them.

Evidence need not be at the collider itself. If Y causes a report D, observing D can tell us something about Y and thereby connect X and Z. A path test must therefore check observed descendants of colliders, not just the nodes lying directly on the path.

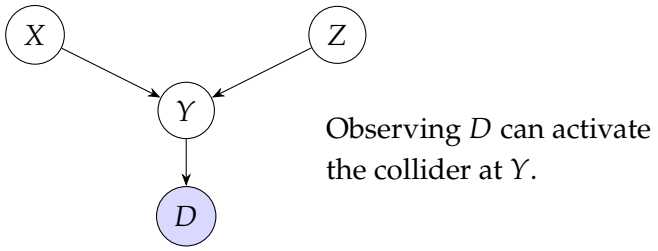


Figure 20: Evidence below a collider can open its path between the causes.

14.1.1 Deriving the three rules algebraically

For the isolated chain, the joint is $P(x)P(y \mid x)P(z \mid y)$. Conditioning on y and dividing by $P(y) > 0$ gives

$$P(x, z \mid y) = \frac{P(x)P(y \mid x)}{P(y)} P(z \mid y) = P(x \mid y)P(z \mid y).$$

For a fork, the joint is $P(y)P(x \mid y)P(z \mid y)$; conditioning on y immediately cancels its prior and gives the same product form. These are conditional-independence proofs, not claims that X and Z are unrelated before the observation.

For a collider, summing its joint over y gives

$$P(x, z) = P(x)P(z) \sum_y P(y \mid x, z) = P(x)P(z).$$

After observing y, however, $P(x, z \mid y) \propto P(x)P(z)P(y \mid x, z)$. The final likelihood generally couples x and z and cannot be split into one factor per cause. Observing a descendant $D = d$ introduces the effective likelihood $\sum_y P(d \mid y)P(y \mid x, z)$, which can couple them in the same way. This is the algebra behind the observed-descendant rule.

In the rare-cause OR example, learning $Y = 1$ raises $P(X = 1)$ from 0.1 to 10/19. If one then learns $Z = 0$, the other cause must be present, so $P(X = 1 \mid Y = 1, Z = 0) = 1$. If instead $Z = 1$, X returns to its prior 0.1. The observation of one cause changes the posterior of the other in opposite directions depending on its value. The graphical rule says dependence is possible; the numbers determine its size and sign.

14.1.2 Worked example: observing a collider's descendant opens dependence

Let independent fair bits X, Z cause $C = X \vee Z$. A noisy report D is true with probability 0.9 if $C = 1$ and 0.1 if $C = 0$.

1. **Draw the relevant structure.** It is $X \to C \leftarrow Z$ with $C \to D$. Without evidence, X and Z are independent.

2. **Observe only $D = 1$.** Three of the four root assignments have $C = 1$. Therefore $P(D = 1) = \frac{3}{4}(0.9) + \frac{1}{4}(0.1) = 0.7$. Also $P(X = 1, D = 1) = \frac{1}{2}(0.9) = 0.45$, so $P(X = 1 \mid D = 1) = 9/14$.
3. **Additionally reveal $Z = 1$.** Then $C = 1$ regardless of X, and D supplies no further information about X. Thus $P(X = 1 \mid Z = 1, D = 1) = 1/2$, different from $9/14$.
4. **Interpret the path rule.** Information below C made the competing causes relevant to one another, even though C itself was not observed. This is why d-separation checks observed descendants of colliders.

A collider need not be observed perfectly to activate a possible dependence. The strength of that dependence depends on the CPTs; the graph determines whether an independence is guaranteed for every compatible choice.

14.2 D-separation

To test whether X and Y are independent given evidence variables E, consider paths in the underlying undirected graph while retaining arrow directions to classify triples. A path is active exactly when:

- every noncollider on it is unobserved;
- every collider is observed or has an observed descendant.

One inactive triple blocks the entire path. If every path is blocked, X and Y are D-SEPARATED by E and $X \perp\!\!\!\perp Y \mid E$ in every distribution represented by the network.

An active path means independence is not guaranteed by the graph, not that dependence must occur for every CPT. Other paths must always be checked: blocking one route does not establish independence if another remains active. The rules depend on which nodes are observed, not on the numerical values of that evidence, for the graph's universal guarantees.

A practical way to apply the rule is to trace one path at a time and inspect each internal node using the two arrows on that path. A node is a collider only if both of those arrows point into it. A node with two incoming edges elsewhere in the graph may be a noncollider on the particular path being tested.

For the alarm graph, $B \rightarrow A \leftarrow E$ is blocked without evidence at A or its descendants, so burglary and earthquake are independent in the prior. Observing John calling can activate the collider because J is a descendant of A. In contrast, the route $B \rightarrow A \rightarrow J$ is a chain: conditioning on A blocks the connection between burglary and the call.

Do not interpret a blocked path as an instruction to erase its probabilities from the model. D-separation answers whether evidence about one variable can change another variable's distribution given the stated observations. Numerical probabilities still come from the complete model.

14.2.1 Checking every path in a larger graph

Consider edges $X \to M \to Z$, $X \to C$, and $Z \to C$. There are two simple undirected paths from X to Z:

$$X \to M \to Z, \qquad X \to C \leftarrow Z.$$

Without evidence, the first is active and the second is blocked at collider C, so independence is not guaranteed. Given M, the first is blocked at its observed noncollider and the second is still blocked: $X \perp\!\!\!\perp Z \mid M$. Given both M and C, the first remains blocked but the second is activated by observing the collider. Independence is again not guaranteed. Additional evidence can open a path rather than merely remove information flow.

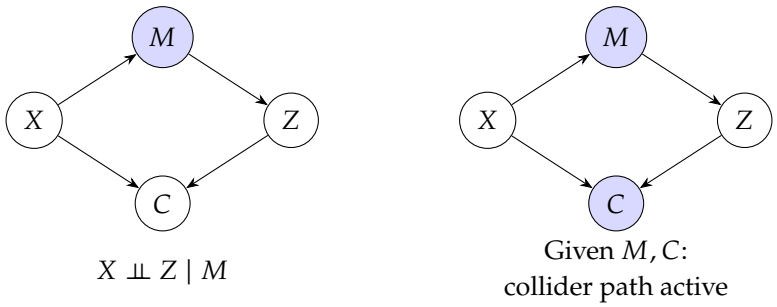


Figure 21: Observing M blocks the upper path; additionally observing C opens the lower path.

A practical path check proceeds in three steps. List the internal nodes of the chosen path. At each one, inspect only the two path edges to decide whether both arrows point inward. Then apply the appropriate evidence rule, including descendants for colliders. Finally repeat for every alternative path. Arrowheads are retained for classifying triples even though reachability is tested using undirected paths. Following arrows forward only would miss common-cause paths and diagnostic influence.

For sets of query variables, d-separation requires that no member of one set has an active path to any member of the other. A single active connection is enough to defeat a graph-wide independence guarantee. It is not necessary to compute a posterior table to establish a guaranteed independence, but checking actual dependence when paths are active may require the CPTs.

14.2.2 A guaranteed independence and an accidental one

Take a chain $X \to Y \to Z$ in which every row of $P(Z \mid Y)$ is the same, say $P(Z = 1 \mid Y = y) = 0.3$ for all y. Then Z is numerically independent of both Y and X even without evidence. The graph allows a dependence that this particular CPT does not use. Changing those rows can remove the extra independence while leaving the graph unchanged.

This is why an active path means "not guaranteed independent" rather than "always dependent." A d-separation proof must hold for every valid choice of CPTs; to disprove a proposed universal independence, one suitable dependent example is enough. This distinction prevents a modeler from reading more numerical information out of the topology than it contains.

D-separation exposes assumptions made by selecting a topology. Adding edges can remove guaranteed independences and enlarge the represented family, but also enlarges CPTs and can make inference more expensive. The graph is a compact description of allowable distributions; numerical inference is a separate task requiring its CPTs.

14.2.3 Worked example: more observations need not block more paths

Use the graph $X \to M \to Z$ and $X \to C \leftarrow Z$. There are two simple paths between X and Z.

Observed set	Path through M	Path through C	D-separated?
$\varnothing$	active	blocked	no
$\{M\}$	blocked	blocked	yes
$\{C\}$	active	active	no
$\{M, C\}$	blocked	active	no

1. **Classify each middle node.** On its path, M is a noncollider and C is a collider. This classification comes from arrow directions along the path, not simply from whether a node has parents somewhere in the graph.
2. **Apply evidence separately to both paths.** Observing M blocks the chain, while observing C opens the collider path.
3. **Combine with an "all paths" test.** Independence is guaranteed only in the second row. One remaining active path defeats d-separation.
4. **Interpret the changed conditioning set.** Adding C to the already observed M removes the guarantee. Conditional independence is not monotone in the amount of evidence.

The table is a procedure for this graph, rather than a claim that every active path produces a large numerical correlation for every parameter choice.

Source. Dan Klein and Pieter Abbeel, Introduction to Artificial Intelligence, Fall 2018, Lecture 14.

Lecture 15: Bayes Nets: Inference

15.1 Factors and elimination

- **The target distribution.** For query variables Q, hidden variables H, and evidence e,

$$P(Q \mid e) = \alpha \sum_{H} \prod_{i} P(X_i \mid \mathrm{Pa}(X_i))\Big|_{E=e}.$$

  A factor is a table over its unassigned variables; it need not be a normalized distribution. Selecting an observed value removes that dimension. Joining factors multiplies their entries over the union of their variables; summing out a variable adds entries that agree on all remaining variables.

- **Eliminate one variable at a time.** Enumeration first constructs the joint and then marginalizes. Variable elimination interleaves these operations. To eliminate hidden variable H, collect every factor containing H, multiply them, sum out H, and return the resulting factor to the collection. Factors not containing H remain unchanged. After eliminating all hidden variables, multiply remaining factors and normalize over Q.

- **Why the rearrangement is valid.** The justification is distributivity: factors independent of the summed variable can be pulled outside its sum. For the chain $R \to T \to L$,

$$P(L) = \sum_{t} P(L \mid t) \underbrace{\sum_{r} P(r)P(t \mid r)}_{f(t)}.$$

  The inner result is stored once for each t, avoiding a full table over R, T, L. With $P(r) = 0.1$, $P(t \mid r) = 0.8$, and $P(t \mid \neg r) = 0.1$, $f(t) = 0.17$. If $P(l \mid t) = 0.3$ and $P(l \mid \neg t) = 0.1$, then $P(l) = 0.17(0.3) + 0.83(0.1) = 0.134$ and $P(\neg l) = 0.866$.

The current factor collection is the algorithm's memory. Eliminating H replaces only the factors containing H because other factors are constant with respect to its sum. If those factors are $f_1(H, A)$ and $f_2(H, B)$, their replacement is

$$g(A, B) = \sum_{h} f_1(h, A) f_2(h, B).$$

Although H disappears, its effect survives as a dependence between A and B. A newly created factor is generally not a CPT for an original node.

Algorithm 15.1: Variable elimination for $P(Q \mid e)$

```
Start with all CPTs, selecting the observed values e
for each hidden variable H in the chosen order do
    Collect factors F_H whose scope contains H
    Form g = Σ_H Π_{f∈F_H} f
    Remove F_H and insert g into the collection
Multiply the remaining factors to obtain u(Q)
return u(Q)/Σ_Q u(Q), if the denominator is positive
```

A factor's *scope* is its set of unassigned variables. Multiplying factors requires matching common-variable values before multiplying entries. It does not mean multiplying arbitrary rows position by position. Summing out a variable groups entries with equal assignments on the remaining scope.

15.1.1 Joining and summing actual factor rows

The rain prior and traffic CPT join to the table

R	T	$P(R)$	$P(R)P(T \mid R)$
r	t	0.1	0.08
r	$\neg t$	0.1	0.02
$\neg r$	t	0.9	0.09
$\neg r$	$\neg t$	0.9	0.81

Summing out R groups the first and third rows for t, and the second and fourth for $\neg t$, producing $f(T) = (0.17, 0.83)$. Multiplying $f(T)$ by $P(L \mid T)$ gives

	l	$\neg l$
t	0.051	0.119
$\neg t$	0.083	0.747

.

Summing the two traffic rows yields $P(L) = (0.134, 0.866)$. At no stage was an eight-entry R, T, L joint needed. If one formed it first, its entries would include $P(r, t, l) = 0.024$ and $P(\neg r, \neg t, \neg l) = 0.729$, and the final answer would be identical.

A selected likelihood factor behaves differently from a conditional row. For example, $P(l \mid T)$ has entries 0.3 and 0.1 over T; their sum is 0.4, not one. They correspond to different parent contexts, not mutually exclusive outcomes under one fixed context. Dividing by 0.4 would change the factor's scale. Some common global rescalings cancel in a final posterior, but row-by-row normalization of an intermediate factor can change relative weights and therefore the answer. Keeping factors unnormalized avoids such unjustified transformations.

The algorithm's invariant is that multiplying the current factors equals the original evidence-selected joint summed over variables already eliminated. For a new hidden variable H, pull outside its

sum all factors that do not contain it, and replace the rest by their summed product. Distributivity preserves the invariant. After all hidden variables are gone, the remaining product is exactly $P(Q, e)$, which proves the final normalization correct.

15.1.2 Worked example: sum a shared cause out of two observation factors

Let H be a fair hidden bit. Given $H = 1$, observations E, F independently have true probabilities $0.8, 0.6$; given $H = 0$, they have true probabilities $0.2, 0.1$. Observe $E = 1$ and query F.

1. **Restrict the evidence factor.** $P(E = 1 \mid H)$ is now a factor over H, with entries $0.8, 0.2$. Retain $P(H)$ and $P(F \mid H)$.
2. **Join all factors containing H.** Their product has one entry for each (h, f). For $F = 1$, the two entries are $0.5(0.8)(0.6) = 0.24$ and $0.5(0.2)(0.1) = 0.01$. For $F = 0$, they are 0.16 and 0.09.
3. **Sum out H.** The new factor is $g(F = 1) = 0.25$, $g(F = 0) = 0.25$. Its total is $0.5 = P(E = 1)$. It is not yet a normalized posterior factor.
4. **Normalize once over the query.** Divide each entry by 0.5, obtaining $P(F = 1 \mid E = 1) = 1/2$ and the same mass for $F = 0$.

The hidden state is eliminated, but its influence is retained in the new factor. Dropping H without multiplying its incident factors would lose the connection between the observations. Summation is the operation that accounts for its alternative values correctly.

15.2 Evidence and normalization

Instantiate evidence in every factor containing an observed variable before elimination. Preserve the query variables. The final unnormalized table is $P(Q, e)$, whose sum is $P(e)$. Dividing by this sum gives the posterior. Normalizing an intermediate factor without tracking its scale can lose an evidence likelihood, so retain factors as unnormalized tables until the desired final normalization.

Continue the traffic example with the evidence $L = l$. Selecting that evidence turns $P(L \mid T)$ into a factor over T with entries $0.3, 0.1$. First eliminating T gives

$$g(r) = 0.8(0.3) + 0.2(0.1) = 0.26, \qquad g(\neg r) = 0.1(0.3) + 0.9(0.1) = 0.12.$$

Multiplying by the rain prior produces masses 0.026 and 0.108. Their sum is 0.134, so

$$P(r \mid l) = \frac{0.026}{0.134} = \frac{13}{67}.$$

The likelihood of being late changes the prior rain probability from 0.1 to about 0.194. The result is not $P(l \mid r) = 0.26$, since the query conditions in the opposite direction.

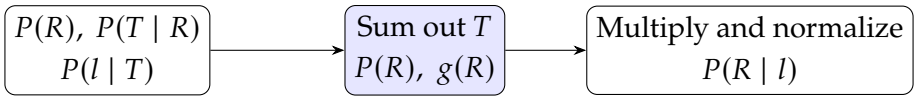


Figure 22: Eliminate the hidden variable while keeping the query variable.

15.2.1 Completing an alarm posterior by elimination

Query burglary given both calls: $P(B \mid j, m)$. The hidden variables are A, E. Select the two calls, then eliminate A by forming

$$g(B, E) = \sum_{a} P(a \mid B, E) P(j \mid a) P(m \mid a).$$

Since $P(j, m \mid a) = 0.9(0.7) = 0.63$ and $P(j, m \mid \neg a) = 0.05(0.01) = 0.0005$, each row is a two-term sum:

B	E	$g(B, E)$
b	e	$0.95(0.63) + 0.05(0.0005) = 0.598525$
b	$\neg e$	$0.94(0.63) + 0.06(0.0005) = 0.59223$
$\neg b$	e	$0.29(0.63) + 0.71(0.0005) = 0.183055$
$\neg b$	$\neg e$	$0.001(0.63) + 0.999(0.0005) = 0.0011295$

Eliminate E next, keeping its prior inside the sum:

$$h(b) = 0.002(0.598525) + 0.998(0.59223) = 0.59224259,$$

$$h(\neg b) = 0.002(0.183055) + 0.998(0.0011295) = 0.001493351.$$

Multiplying by $P(B)$ gives unnormalized masses $u(b) = 0.00059224259$ and $u(\neg b) = 0.001491857649$. Their sum is 0.002084100239, so

$$P(b \mid j, m) = \frac{0.00059224259}{0.002084100239} \approx 0.2842.$$

The posterior is far above the rare prior 0.001, yet below one half. The calls are informative, but the large number of no-burglary situations still contributes appreciable probability to the observed evidence.

15.3 Ordering and complexity

A factor over k variables with domains of size d has d^k entries. Thus time and space depend critically on the largest intermediate factor, which depends on elimination order. Eliminating a

highly connected variable early can join many otherwise separate variables into a large table. The answer is independent of order; the amount of computation is not.

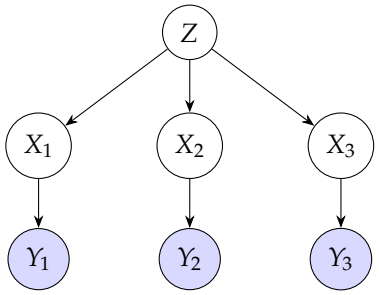
There is no universally efficient ordering: general Bayes-net inference is NP-hard. One can encode a satisfiability problem so that a designated event has positive probability exactly when a satisfying assignment exists. Sparse structures such as polytrees admit efficient inference relative to their local table sizes. Cutset conditioning trades enumeration over selected variables for simpler remaining graphs, paralleling the structural methods for CSPs.

Elimination order changes which variables become coupled in intermediate factors. Eliminating the center of a star can combine all its leaves in one table; eliminating the leaves first can keep each intermediate table small. With k binary variables in a scope, there are 2^k entries, so one added variable can double a table's size.

A compact input network therefore does not guarantee cheap exact inference. Representation size depends on original CPTs; inference cost depends on the intermediate factors needed for the particular query and order. Evidence can shrink scopes by fixing variables and sometimes makes an otherwise difficult query easier.

15.3.1 Seeing the effect of elimination order

Suppose a hidden Boolean Z is a parent of $X_1, \dots, X_n$, and each X_i is a parent of an observed $Y_i = y_i$. Query X_n. Eliminating Z first joins $P(Z)$ with every $P(X_i \mid Z)$, creating a product over $Z, X_1, \dots, X_n$ with 2^{n+1} entries before summation. The replacement factor still has 2^n entries. The small original CPTs do not prevent this large intermediate interaction among all X_i.



Eliminate Z first:
couple X_1, X_2, X_3.

Eliminate X_1, X_2 first:
small messages over Z.

Figure 23: The same sparse network can produce very different intermediate scopes for query X_3.

Instead eliminate each X_i for $i < n$ first:

$$g_i(Z) = \sum_{x_i} P(x_i \mid Z) P(y_i \mid x_i).$$

Each resulting message has only two entries. After these eliminations, join them with $P(Z)$, $P(X_n \mid Z)$, and $P(y_n \mid X_n)$, and sum out Z. Only the two-variable scope (Z, X_n) is needed for

that last join. The posterior is the same, but exponential storage in n was avoided by the order. Preserving the query X_n is essential; it is never summed away.

General networks need not have such a favorable ordering. One can make independent fair Boolean roots represent truth assignments, deterministic nodes evaluate clauses, and a final node report whether all clauses hold. Its true probability is the number of satisfying assignments divided by 2^n. Deciding whether that probability is positive decides satisfiability. This explains why exact inference can encode hard combinatorial work even when the local probabilities are simple. Structural restrictions such as polytrees, or conditioning on a small cutset, can make particular instances manageable without removing the general worst case.

15.3.2 Worked example: estimate factor storage before choosing an order

Suppose all variables are Boolean. An elimination step joins factors whose combined scope is $\{A, B, C, D\}$, then sums out A.

1. **Count the joined entries.** A dense table over four Boolean variables has $2^4 = 16$ entries. The original factors might each have been small; their union of variables determines this intermediate scope.
2. **Count the result.** Summing A leaves a factor over $\{B, C, D\}$ with $2^3 = 8$ entries. Each result adds two product entries.
3. **Compare an alternative step.** If another valid ordering keeps every joined scope to at most two Boolean variables, its largest joined table has only 4 entries. The requested posterior remains the same, while the computation and storage can differ sharply.
4. **Generalize the count correctly.** With domain sizes d_A, d_B, d_C, d_D, replace 16 by $d_A d_B d_C d_D$. Counting variables alone can be misleading when their domains differ greatly.

Avoiding one large intermediate factor can matter more than saving several small arithmetic operations. An ordering heuristic therefore examines how eliminating a variable couples the remaining scopes. It changes efficiency, not which hidden values the exact sum must include.

Source. Dan Klein and Pieter Abbeel, Introduction to Artificial Intelligence, Fall 2018, Lecture 15.

Lecture 16: Bayes Nets: Sampling

16.1 Prior and rejection sampling

- **Generate a prior draw.** To sample a finite distribution, partition $[0, 1)$ into intervals whose lengths equal its probabilities, draw a uniform number, and select its interval. Prior sampling applies this in topological order, drawing each node from its CPT given its sampled parents. The probability of a full sample is the product of these conditionals, exactly the network's joint.
- **Retain evidence-compatible draws.** Estimate an event by its frequency among samples. To estimate $P(Q \mid e)$, rejection sampling discards samples inconsistent with the evidence and tallies query values in the rest. Accepted samples have the desired conditional distribution. If $P(e)$ is small, only about $NP(e)$ of N prior samples survive; impossible evidence yields no posterior estimate.
- **Use the correct denominator.** If $n(q, e)$ is the number of accepted samples with query value q, rejection sampling uses $n(q, e)/n(e)$. Dividing by the total number of original draws would estimate $P(q, e)$ instead of $P(q \mid e)$. This distinction is the same normalization step used in exact inference.

If evidence has probability 0.001, then 100,000 prior draws yield only about 100 accepted samples on average. The apparent size of the simulation is misleading: the conditional estimate is based on those accepted draws. One can stop a draw as soon as it contradicts evidence, saving the remaining sampling work, but this does not improve the acceptance probability.

Prior samples are drawn from the model, not from the unknown real world. They approximate the probabilities already encoded in the CPTs. More samples reduce simulation error; they do not correct a poorly specified model.

16.1.1 Generating a prior sample from start to finish

The sprinkler network uses $P(C = 1) = 0.5$, $P(S = 1 \mid C = 1) = 0.1$, $P(S = 1 \mid C = 0) = 0.5$, $P(R = 1 \mid C = 1) = 0.8$, and $P(R = 1 \mid C = 0) = 0.2$. Wet-grass probabilities for $(S, R) = (1, 1), (1, 0), (0, 1), (0, 0)$ are $0.99, 0.90, 0.90, 0.01$. Complements give the false-value probabilities.

Use independent uniform draws $(0.3, 0.6, 0.4, 0.2)$, with a Boolean event chosen true if its draw is below its conditional true probability. First $0.3 < 0.5$ gives $C = 1$. Then $0.6 > 0.1$ gives $S = 0$. Next $0.4 < 0.8$ gives $R = 1$. Finally $0.2 < 0.9$ gives $W = 1$. The complete assignment's probability is $0.5(0.9)(0.8)(0.9) = 0.324$. This is the probability of that assignment, not of those exact real-valued random draws. The intervals of draws leading to it have precisely that product of lengths.

For evidence $S = 1$, this sample is rejected as soon as $S = 0$ is drawn. The acceptance rate is $P(S = 1) = 0.5(0.1) + 0.5(0.5) = 0.3$. Among accepted samples, $C = 1$ occurs with probability $P(C = 1, S = 1)/P(S = 1) = 0.05/0.3 = 1/6$. This identity explains why filtering samples by evidence produces the desired conditional distribution. Generating enough accepted samples can still be expensive when the acceptance event is rare.

16.1.2 Worked example: rare evidence can exhaust a sampling budget

Suppose an illustrative query has evidence probability $P(e) = 0.001$. A rejection sampler generates $N = 1000$ independent prior samples.

1. **Count expected survivors.** The expected accepted count is $NP(e) = 1$. This is an expectation, not a promise that one sample survives.
2. **Compute the risk of no estimate.** The chance that all draws are rejected is $(1-0.001)^{1000} \approx 0.368$. In that event the denominator of the conditional frequency estimate is zero.
3. **Distinguish budget from effective evidence.** Even if the sampler generated a thousand full assignments, only accepted assignments inform this rejection estimate of $P(Q \mid e)$. One accepted draw would still produce an extremely coarse estimate.
4. **Change the sampling strategy for the actual difficulty.** Likelihood weighting fixes the evidence instead of discarding incompatible draws. It avoids this particular rejection loss, but unequal or nearly zero weights can still leave little useful information.

For comparison, if $P(e) = 0.1$ the same budget has 100 expected survivors. The efficiency difference comes from the evidence event, even before query domain size or network complexity is considered. Never report a conditional estimate from zero accepted samples as a verified probability of zero.

16.2 Likelihood weighting

Likelihood weighting fixes evidence variables to their observed values and samples all other variables in topological order. Initialize $w = 1$. At an evidence node $E_i = e_i$, multiply w by $P(e_i \mid \text{pa}(E_i))$ instead of drawing its value. Return the complete assignment and its weight, and normalize weighted query counts:

$$\widehat{P}(q \mid e) = \frac{\sum_{j=1}^{N} w_j \mathbf{1}\{Q_j = q\}}{\sum_{j=1}^{N} w_j}.$$

Writing z for sampled variables, the proposal contains the non-evidence CPTs, while the weight contains the evidence CPTs:

$$q(z) = \prod_{i \notin E} P(z_i \mid \mathrm{pa}_i), \qquad w(z) = \prod_{i \in E} P(e_i \mid \mathrm{pa}_i), \qquad q(z)w(z) = P(z, e).$$

This identity explains consistency of the weighted estimate. Evidence affects downstream draws immediately, but does not guide upstream draws; a few samples can therefore dominate the total weight.

For the cloudy-sprinkler-rain-wet-grass network, suppose sprinkler and wet grass are observed true. A likelihood-weighted draw might sample cloudy true and rain false. The two fixed observations contribute

$$w = P(S = 1 \mid C = 1)P(W = 1 \mid S = 1, R = 0) = 0.1(0.9) = 0.09.$$

Another draw can receive a different weight because it supplies different parents for those same observations. It is the weighted count, rather than the raw number of draws, that contributes to the query estimate.

Fixing evidence does not mean pretending it was certain under the original model. Its likelihood remains in the weight. If that correction were omitted, upstream variables would be drawn according to their prior regardless of how well they explain the evidence.

Very unequal weights waste effort: a hundred tiny-weight samples can contribute less than one large-weight sample. All-zero weights mean that the current sample collection cannot explain the evidence, so its normalized estimate is undefined. This can happen with finite samples even when the evidence is possible in the full model.

16.2.1 Why the weights correct the changed sampling process

For query $P(C \mid S = 1, W = 1)$, likelihood weighting samples only C, R. Its proposal distribution and weights are

C	R	Proposal $q(C, R)$	Weight	Product $q\,w$
1	1	0.4	0.1(0.99) = 0.099	0.0396
1	0	0.1	0.1(0.90) = 0.09	0.009
0	1	0.1	0.5(0.99) = 0.495	0.0495
0	0	0.4	0.5(0.90) = 0.45	0.18

The proposal frequencies alone give $P(C = 1) = 0.5$, because cloudy was drawn from its prior. Weighted masses for $C = 1$ total 0.0486; for $C = 0$ they total 0.2295. Normalization gives $P(C = 1 \mid S = 1, W = 1) = 18/103$. The evidence makes a cloudy state less plausible because

a running sprinkler is relatively unlikely when cloudy. Weighting restores this upstream effect without changing how C was sampled.

For finitely many draws, replace the proposal probabilities in this table by observed frequencies. The numerator and denominator are both sample averages; their ratio approaches the exact weighted-mass ratio. A finite ratio estimate can be noisy and need not be unbiased. A sample with tiny weight contributes very little, even though producing it took a full draw. This is why keeping every sample is not by itself a guarantee of efficiency.

The evidence probability is the expectation of the weight under the proposal: $\sum_z q(z)w(z) = P(e)$. The *average* weight estimates this probability. The raw sum grows with the number of draws and should not be interpreted as a probability or, by itself, as a count of independent useful observations.

16.2.2 Worked example: weighted counts differ from ordinary counts

Suppose four illustrative likelihood-weighted samples have the following query values and evidence weights.

Sample	1	2	3	4
Query bit	1	0	0	0
Weight	0.9	0.1	0.1	0.1

1. **Add weights by query value.** Total weight for $Q = 1$ is 0.9; for $Q = 0$ it is 0.3. The overall total is 1.2.
2. **Normalize the weighted totals.** The estimate is $\widehat{P}(Q = 1 \mid e) = 0.9/1.2 = 0.75$ and $\widehat{P}(Q = 0 \mid e) = 0.25$.
3. **Contrast ordinary counting.** Counting particles without their weights would give $1/4$ for $Q = 1$, reversing the apparent conclusion. The sampled assignments did not come directly from the posterior; the weights correct that change in the generation process.
4. **Check what a larger list does not guarantee.** Replacing the three small weights by 10^{-12} makes the result almost entirely determined by one draw. The number of stored samples alone does not describe the quality of a weighted estimate. More independent draws are useful only insofar as they represent the important parts of the weighted distribution.

Common scaling of all weights leaves the normalized estimate unchanged. Setting every weight to one does not: that erases the likelihood correction. The weighted estimate is an approximation based on finite draws, not an exact posterior calculation merely because its weights are evaluated exactly.

16.3 Gibbs sampling

Gibbs sampling keeps a complete assignment consistent with the evidence and repeatedly resamples one non-evidence variable given all the others. There is no required topological order. For node X,

$$P(x \mid \text{rest}) \;\propto\; P(x \mid \text{pa}(X)) \prod_{Y \text{ child of } X} P(y \mid \text{pa}(Y)).$$

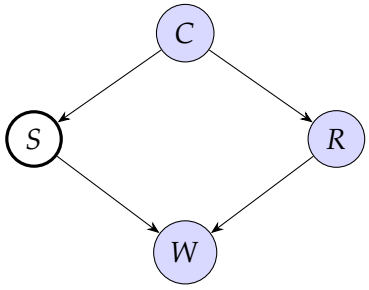
All other CPTs cancel because they do not vary with x. Consequently the parents, children, and other parents of the children supply the needed local information. Both upstream and downstream observations can influence a draw.

Under conditions ensuring that this Markov chain explores the posterior's support and converges, long-run frequencies estimate the posterior. Successive samples are dependent, and slow mixing or deterministic constraints can make convergence difficult. More samples improve accuracy only insofar as they provide useful exploration of the distribution.

To see a Gibbs update, fix $C = 1, R = 0, W = 1$ and resample S. The two unnormalized possibilities are

$$u(S = 1) = P(S = 1 \mid C = 1)P(W = 1 \mid S = 1, R = 0) = 0.09,$$
$$u(S = 0) = P(S = 0 \mid C = 1)P(W = 1 \mid S = 0, R = 0) = 0.009.$$

Normalizing gives $P(S = 1 \mid C = 1, R = 0, W = 1) = 10/11$. The factors $P(C)$ and $P(R \mid C)$ cancel because changing S does not change their entries. Rain and wet grass both matter through the child's CPT, even though rain is not a parent of S.



Resample S.
Keep C, R, W fixed.
Use the CPTs for S and W.

Figure 24: A Gibbs update uses only factors that change with the resampled variable.

16.3.1 Why a Gibbs update preserves its target distribution

Write R for all variables except the currently updated X, with evidence already fixed. Suppose the current joint assignment is distributed as the posterior $\pi(x, r)$. Replacing X by a draw from $\pi(x' \mid r)$ gives

$$\sum_x \pi(x, r)\pi(x' \mid r) = \pi(r)\pi(x' \mid r) = \pi(x', r).$$

Thus one update preserves the posterior if the chain is already at it. Repeated updates can approach that distribution from other starts when they connect its support appropriately. This preservation argument establishes stationarity, while convergence from an arbitrary start requires more.

The conditional is cheap because only CPTs containing X change between candidate values. Its own CPT depends on its parents; each child's CPT depends on that child, X, and the child's other parents. Multiplying these factors for every candidate and normalizing gives the full conditional. A child's other parent can therefore affect the update even though it has no direct arrow into X.

A deterministic constraint can prevent mixing. Suppose two hidden Boolean variables must satisfy $X = Y$, so only $(0, 0)$ and $(1, 1)$ have positive posterior probability. From $(0, 0)$, resampling X given $Y = 0$ cannot change X, and resampling Y given $X = 0$ cannot change Y. Single-variable Gibbs updates remain stuck forever although another supported state exists. A large number of repeated samples of $(0, 0)$ does not establish that it has all the posterior mass. Choosing a sampling method requires considering both computation per step and whether those steps explore the target well.

Source. Dan Klein and Pieter Abbeel, Introduction to Artificial Intelligence, Fall 2018, Lecture 16.

Lecture 17: Decision Networks and Value of Information

17.1 Decisions and expected utility

- **The decision objective.** A decision network augments a Bayes net with decision nodes and a utility node. Chance nodes describe uncertainty, a decision node specifies an action, and utility depends on the action and relevant outcomes. For current evidence e, evaluate each action by

$$EU(a \mid e) = \sum_s P(s \mid e)U(a, s), \qquad MEU(e) = \max_a EU(a \mid e),$$

  in the setting where the action does not change the chance distribution. More general action-dependent outcomes use the corresponding model for each action. Choose the maximizing action after inferring the utility parents.

- **Compare actions under one belief.** For the umbrella example, utilities for leaving it are 100 in sun and 0 in rain; taking it gives 20 in sun and 70 in rain. With sun probability 0.7, leaving has value 70 and taking has value 35. A bad forecast that changes sun probability to 0.34 changes these to 34 and 53, making taking the umbrella preferable.

The shape of each node describes its role. A chance node represents a random quantity, a rectangular decision node represents a choice, and a utility node contains a numerical preference rather than a probability distribution. Utility need not sum to one over outcomes.

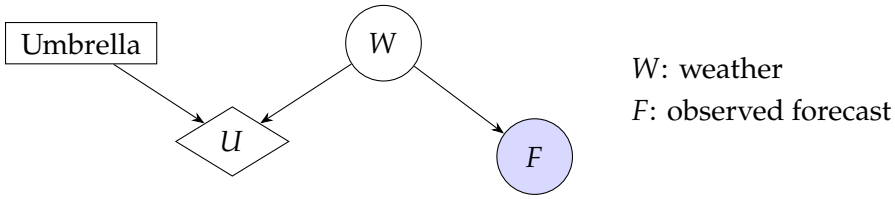


Figure 25: Evidence changes beliefs about weather, while the decision changes utility.

To evaluate taking the umbrella, hold that action fixed and average its two utility entries using the current weather probabilities. To evaluate leaving it, use the same probabilities with the other utility entries. The two actions need not have separate probability models if neither changes the weather. Changing a decision is not the same as observing a random node.

The bad-forecast numbers show the complete calculation: $EU(\text{leave} \mid \text{bad}) = 0.34(100) + 0.66(0) = 34$ and $EU(\text{take} \mid \text{bad}) = 0.34(20) + 0.66(70) = 53$. Inference supplies the distribution; optimization uses that distribution to select the action.

17.1.1 The decision boundary in the weather belief

Let $p = P(\text{sun} \mid e)$. Then

$$EU(\text{leave}) = 100p, \qquad EU(\text{take}) = 20p + 70(1-p) = 70 - 50p.$$

Leaving is optimal exactly when $100p \geq 70 - 50p$, or $p \geq 7/15$. At the threshold the agent is indifferent. Above it, the inconvenience of carrying an umbrella in sunshine outweighs its rain protection; below it, protection is worth more. A posterior probability becomes a decision only after this utility-dependent comparison.

An observation can change the belief without crossing the threshold. If all possible reports keep p above $7/15$, the agent still leaves the umbrella whatever it hears, and that information has zero value for this particular decision. A different rain penalty would move the threshold and could make the same report useful. Information value belongs to a model of outcomes *and* a choice with consequences.

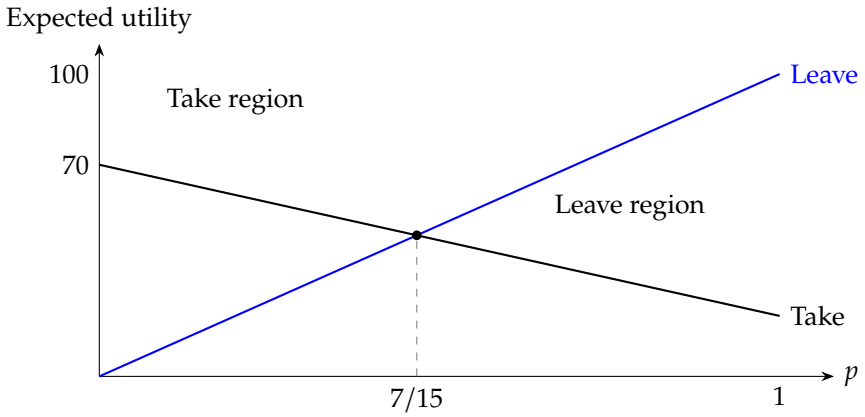


Figure 26: A forecast changes the action only if the resulting sun belief crosses the utility threshold.

17.2 Value of information

Before observing a variable Z, its value is uncertain. The VALUE OF PERFECT INFORMATION about that variable is

$$VPI(Z \mid e) = \sum_z P(z \mid e) MEU(e, z) - MEU(e).$$

The maximum is taken separately after each possible observation: information is useful because the action can adapt. In the oil example, exactly one of two equally likely locations holds oil worth k. Choosing without a survey has value $k/2$; learning its location gives value k, so the information is worth $k/2$ in the same utility units.

VPI is nonnegative because the agent can ignore the observation and keep the old action. It can be zero even for a surprising observation if no action changes. Information values are not generally

additive: two observations can overlap or become useful only together. Total value of observing a set is independent of its order, although the conditional value of the second observation depends on what is already known. Information acquisition is worthwhile when its expected benefit exceeds its cost in utility.

A noisy report is represented by observing a noisy sensor node, not by pretending to observe the underlying state. If the new variable is conditionally independent of the relevant uncertain utility parents given current evidence and every candidate action, it cannot improve the decision.

The timing of the maximum is essential. Without information, choose one action before learning Z. With information, choose a potentially different action for each observed value. In general,

$$\sum_z P(z \mid e) \max_a EU(a \mid e, z) \;\geq\; \max_a \sum_z P(z \mid e) EU(a \mid e, z).$$

The right side equals acting before seeing Z; the left side allows adaptation. This inequality gives the nonnegativity of information value.

For an illustrative noisy oil report that identifies the correct location with probability 0.8, following the report succeeds with probability 0.8 and has expected utility $0.8k$. Without it, the best value is $0.5k$, so this report is worth $0.3k$. Learning the true location itself is worth $0.5k$. Perfect observation of a noisy report is different from perfect observation of the underlying location.

Two observations can be redundant: after the exact oil location is known, a second report about the same location has no value. They can also be complementary if neither alone changes the best action but their combination does. This is why separate information values cannot simply be added.

A variable can have high uncertainty and still zero decision value. If both possible soups would lead to ordering a sandwich, learning which soup is available changes beliefs but not the action. Information is valuable relative to a decision and utility function, not merely because it reduces uncertainty.

17.2.1 A full forecast-value calculation

To complete the weather example, use prior sun probability 0.7 and a forecast with $P(\text{good} \mid \text{sun}) = 0.8$ and $P(\text{good} \mid \text{rain}) = 0.1$. Its good probability is $0.7(0.8) + 0.3(0.1) = 0.59$, and its bad probability is 0.41. Bayes' rule gives

$$P(\text{sun} \mid \text{good}) = \frac{0.56}{0.59} = \frac{56}{59}, \qquad P(\text{sun} \mid \text{bad}) = \frac{0.14}{0.41} = \frac{14}{41}.$$

After a good forecast, leave: its expected utility is 5600/59, exceeding taking. After a bad forecast, take: its utility is $(20 \cdot 14 + 70 \cdot 27)/41 = 2170/41$. Before hearing the forecast, the expected value of this contingent policy is

$$0.59\frac{5600}{59} + 0.41\frac{2170}{41} = 56 + 21.7 = 77.7.$$

Without a forecast the best value was 70, so the information is worth 7.7 utility units. Paying a sensing cost of 5 would leave expected net value 72.7; paying 8 would leave 69.7 and would not be worthwhile. The value is computed before hearing the report, averaging both possible outcomes rather than selecting the most favorable report.

Perfectly observing weather itself is worth more. Leave in sun and take in rain, giving $0.7(100) + 0.3(70) = 91$, an improvement of 21 over acting now. Perfectly reading the forecast node remains imperfect knowledge of weather. This is how noisy information fits the same VPI formula without replacing the uncertain state by a guessed truth.

17.2.2 Nonnegativity, redundancy, and complementarity

Let a_0 be optimal before observing Z. After any result z, optimizing can do at least as well as retaining a_0. Weighting these inequalities by $P(z \mid e)$ and summing gives

$$\sum_z P(z \mid e) MEU(e, z) \geq \sum_z P(z \mid e) EU(a_0 \mid e, z) = EU(a_0 \mid e).$$

The final equality is total expectation. This proves nonnegative value for free information that leaves the same actions available. Acquisition costs are subtracted separately.

Redundant measurements need not add value: observing an exact copy of an already observed report changes nothing. For a complementary example, let two independent fair bits X, Y determine whether a prize is at A or B by their parity. Learning X alone or Y alone leaves either location equally likely, so each alone has zero value. Learning both reveals the location and makes a correct choice certain. Summing the two separate information values would miss this gain. The value of a set of observations is order independent, but the second observation's conditional contribution can depend greatly on the first.

17.2.3 Worked example: accurate information can have zero decision value

Use the umbrella utilities above. Suppose an illustrative report produces posterior sun probability 0.6 or 0.8, each with probability $1/2$. Its average sun probability remains the prior 0.7.

1. **Evaluate actions after each report.** At $p = 0.6$, leaving has utility 60 and taking has $70 - 50(0.6) = 40$. At $p = 0.8$, the corresponding utilities are 80 and 30. Leaving wins in both cases.
2. **Average the optimized utilities.** Expected utility after observing is $\frac{1}{2}(60) + \frac{1}{2}(80) = 70$. Without the report the best value was also 70, so VPI is zero.
3. **Check that the report is informative.** It changes the belief from 0.7 to either 0.6 or 0.8. Zero decision value does not mean the observation is statistically independent of the weather.

4. **Check a measurement cost.** If obtaining the report costs one utility unit, its net value is -1. Nonnegative gross information value assumes the agent may ignore freely available information; it does not make every costly measurement worthwhile.

For consistency, such a report can be generated by a valid joint model: the joint masses for (sun, first report), (rain, first report), (sun, second report), (rain, second report) are $0.3, 0.2, 0.4, 0.1$. The useful question is whether a belief update can alter a consequential choice.

17.3 Partial observability

A PARTIALLY OBSERVABLE MDP adds an observation model to an MDP. The agent maintains a BELIEF STATE, a probability distribution over physical states. An action changes the physical state; an observation updates the belief. Planning can therefore be viewed as an MDP over beliefs, though the belief space is typically continuous and much larger.

In static Ghostbusters, beliefs are determined by accumulated sensor evidence. A truncated expectimax tree can compare acting immediately against sensing once and then acting. This produces a VPI-based sensing rule, while longer planning can account for sequences of information-gathering actions.

A belief state keeps track of what the agent knows, not what the physical state actually is. Two agents at the same physical location can choose different sensing actions because their information differs. After an action and observation, Bayes' rule produces a new belief, and planning continues from that belief.

A one-step VPI policy considers only sensing now and then acting. It can miss a pair of measurements that are useful together or the future benefit of traveling to a better sensing position. Full belief-space planning accounts for those possibilities, at a much greater computational cost.

17.3.1 Updating and planning with beliefs

A belief $b(s)$ is sufficient for predicting future observations in the POMDP model. After action a, first predict

$$\bar{b}(s') = \sum_s T(s, a, s')b(s).$$

For observation o with likelihood $O(o \mid s', a)$, update

$$b'(s') = \frac{O(o \mid s', a)\bar{b}(s')}{\sum_u O(o \mid u, a)\bar{b}(u)}.$$

The denominator is also the probability of seeing o after that action, so it supplies the branch probability in a belief-space expectimax tree. Each observation branch leads to a different posterior

belief, from which a different next action can be chosen. Physical transitions are stochastic, but the belief update is determined by the old belief, action, and observation.

In static Ghostbusters, compare the best immediate bust against each sensing action followed by the best bust for every possible sensor color. Weight those future bust values by the predicted color probabilities and subtract the sensing cost. This is a complete one-sense lookahead. A longer horizon can value a first reading because it makes a second reading useful, or value movement because it reaches a better sensing location. Those benefits cannot be recovered by adding independent one-step VPI scores.

17.3.2 Worked example: infer the state first, then compare the actions

A hidden machine is good or faulty, with prior faulty probability 0.2. An illustrative warning occurs with probability 0.9 when faulty and 0.1 when good. Acting without repair costs 50 if faulty and zero otherwise; repair costs 12 and removes the fault.

1. **Choose before a warning.** Expected cost of acting is $0.2(50) = 10$, below repair's 12, so the current best action is to proceed.
2. **Update after a warning.** The unnormalized faulty and good masses are $0.2(0.9) = 0.18$ and $0.8(0.1) = 0.08$. The posterior faulty probability is $0.18/0.26 = 9/13$.
3. **Recompute the decision under the new belief.** Proceeding now costs $50(9/13) = 450/13 \approx 34.62$ in expectation, so repair is preferred. Using the prior for this comparison would discard the new observation.
4. **Distinguish observation from control.** A warning changes information about the machine. Repair changes the machine's physical state. In a continuing problem, both changes must appear in the appropriate belief transition rather than being treated as interchangeable events.

The belief is a distribution over states. Choosing only its most likely state and ignoring other possibilities can change expected costs, especially when a rare outcome has a large consequence.

Source. Dan Klein and Pieter Abbeel, Introduction to Artificial Intelligence, Fall 2018, Lecture 17.

Lecture 18: Hidden Markov Models

18.1 Markov chains and stationary distributions

- **Prediction and stationarity.** A first-order Markov chain represents a sequence of states with $P(X_{t+1} \mid X_{1:t}) = P(X_{t+1} \mid X_t)$. A stationary transition model uses the same conditional table at every time. Starting from a distribution b_t, prediction pushes probability through the transitions:

  $$b_{t+1}(x') = \sum_x P(x' \mid x) b_t(x).$$

  A stationary distribution satisfies $\pi(x') = \sum_x P(x' \mid x)\pi(x)$. It need not be the limit from every initial distribution for an arbitrary chain; convergence requires additional conditions. In the weather example, $P(\text{sun} \mid \text{sun}) = 0.9$ and $P(\text{sun} \mid \text{rain}) = 0.3$ give $\pi(\text{sun}) = 0.9\pi(\text{sun}) + 0.3\pi(\text{rain})$, hence $\pi(\text{sun}) = 3/4$ and $\pi(\text{rain}) = 1/4$.

- **Connections to other chains.** PageRank uses a random walk on pages with occasional random jumps, assigning larger stationary mass to more reachable pages. Gibbs sampling similarly constructs a chain over complete assignments whose stationary distribution is the desired posterior.

- **Account for every predecessor.** The prediction equation sums over ways to arrive at the same next state. If the current sun probability is 1, the next sun probability is 0.9. After another step it is $0.9(0.9) + 0.3(0.1) = 0.84$: both a sunny current state that stays sunny and a rainy current state that changes to sun contribute.

A stationary *model* and a stationary *distribution* are different. The model is stationary when its transition probabilities do not change with time. The current distribution can still evolve. A stationary distribution is a special input distribution left unchanged by that model. For the example, normalization plus $0.1\pi(\text{sun}) = 0.3\pi(\text{rain})$ gives the $3 : 1$ ratio.

One should not assume every chain forgets its start. Two absorbing states can retain their initial masses forever. A deterministic two-state alternation can oscillate instead of converging. These cases explain why the fixed-point equation alone does not prove convergence from all starts.

18.1.1 Solving the weather evolution explicitly

Write $p_t = P(X_t = \text{sun})$ for the Markov-chain example. The prediction recurrence is

$$p_{t+1} = 0.9p_t + 0.3(1 - p_t) = 0.3 + 0.6p_t.$$

Subtract the stationary value $3/4$ from both sides to obtain $p_{t+1} - 3/4 = 0.6(p_t - 3/4)$. Repeating gives

$$p_t = \frac{3}{4} + 0.6^{t-1}\left(p_1 - \frac{3}{4}\right).$$

From all-sun, the first values are $1, 0.9, 0.84, 0.804$; from all-rain they are $0, 0.3, 0.48, 0.588$. Both approach 0.75 because the multiplier 0.6 shrinks the initial discrepancy. This establishes convergence for this chain, rather than merely identifying a fixed point.

By contrast, a deterministic two-state alternation has recurrence $p_{t+1} = 1 - p_t$. Its stationary distribution is $(1/2, 1/2)$, but starting at one state gives $1, 0, 1, 0, \ldots$. Two absorbing states have every mixture as a stationary distribution and preserve the initial mixture forever. These examples separate existence of a stationary distribution, uniqueness, and convergence toward it.

For PageRank, an occasional uniform jump makes every page reachable from every other and permits staying at the same page with positive probability. These features remove disconnected trapping and deterministic periodicity in the finite random-walk model. The stationary probabilities reflect the whole link structure, not just a count of direct incoming links. A link from a heavily visited page contributes more mass than one from a rarely visited page.

18.2 Hidden states and observations

A HIDDEN MARKOV MODEL (HMM) adds observations E_t emitted from hidden states X_t. It specifies an initial state distribution, transitions $P(X_t \mid X_{t-1})$, and emissions $P(E_t \mid X_t)$:

$$P(x_{1:T}, e_{1:T}) = P(x_1)P(e_1 \mid x_1)\prod_{t=2}^{T} P(x_t \mid x_{t-1})P(e_t \mid x_t).$$

The hidden process is Markov, and the current observation is independent of the other variables given the current state. Observations are not generally independent without conditioning on states.

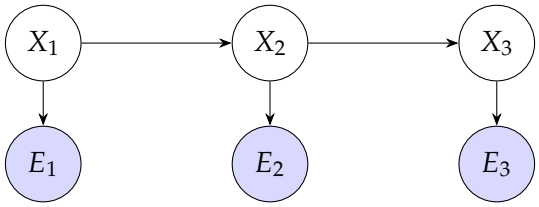


Figure 27: Hidden dynamics and observed emissions in an HMM.

For weather tracking, the state may be whether it rains, while the observed variable is whether someone carries an umbrella. An umbrella is evidence of rain, not rain itself. A person can carry one on a sunny day, so the posterior need not become certain after seeing it.

The two independence assumptions have separate jobs. The transition model lets prediction summarize all past observations by the current belief. The emission model lets an observation update use only the candidate current state. Without these assumptions, the belief needed for an exact online update might have to retain more history.

18.2.1 Worked example: distinguish prediction, filtering, and sequence scoring

Start an illustrative weather HMM with $P(R_1) = 1/2$, $P(R_{t+1} \mid R_t) = 0.7$, $P(R_{t+1} \mid S_t) = 0.3$, and umbrella likelihoods 0.9 in rain and 0.2 in sun.

1. **Predict without observing.** The next rain probability from the initial fair belief is $0.7(0.5) + 0.3(0.5) = 0.5$. This averages transition outcomes and contains no umbrella likelihood.
2. **Filter an observed umbrella.** At the first time, masses are 0.45 for rain and 0.10 for sun. The posterior is $9/11$ for rain. This answers a state question conditioned on a particular observation.
3. **Score the observation itself.** Its probability is the discarded normalizing total 0.55. It is not the posterior rain probability and need not be close to it.
4. **Keep the model's time convention consistent.** An initial prior over X_1 is updated directly by evidence at time 1. An initial prior over X_0 would first require a transition to X_1. The two conventions are both possible but cannot be mixed within a calculation.

Prediction changes the time of the belief; observation updating changes its conditioning information. Tracking these two pieces separately prevents an extra transition or likelihood factor from being inserted accidentally.

18.3 Filtering and the forward algorithm

Filtering maintains $b_t(x) = P(X_t = x \mid e_{1:t})$. Prediction and observation updates are

$$\bar{b}_{t+1}(x') = \sum_x P(x' \mid x) b_t(x), \qquad b_{t+1}(x') = \alpha P(e_{t+1} \mid x') \bar{b}_{t+1}(x').$$

The first equation follows by marginalizing the previous state and using the Markov property. The second is Bayes' rule and emission independence. Initialize by weighting the initial prior with $P(e_1 \mid x_1)$. Predictions often spread belief; informative observations often concentrate it, although an individual observation need not reduce entropy.

The unnormalized forward algorithm instead carries $f_t(x) = P(x_t, e_{1:t})$:

$$f_{t+1}(x') = P(e_{t+1} \mid x') \sum_x P(x' \mid x) f_t(x).$$

Normalize to recover the filtering distribution; scaling each step avoids numerical underflow. With $|S|$ states, a dense update costs $O(|S|^2)$ and stores only the current vector. In robot localization, states are positions and observations are noisy sensor readings; observations disambiguate positions that the motion model alone cannot distinguish.

18.3.1 Deriving the two online updates

At the end of time t, the stored belief conditions on all observations through t. To predict X_{t+1}, introduce and sum out X_t:

$$\begin{aligned} P(x' \mid e_{1:t}) &= \sum_x P(x', x \mid e_{1:t}) \\ &= \sum_x P(x' \mid x, e_{1:t}) P(x \mid e_{1:t}) \\ &= \sum_x P(x' \mid x) b_t(x). \end{aligned}$$

The last step uses the Markov property. To incorporate the new reading, Bayes' rule gives $P(x' \mid e_{1:t+1}) \propto P(e_{t+1} \mid x', e_{1:t}) P(x' \mid e_{1:t})$. Emission independence replaces the first factor by $P(e_{t+1} \mid x')$. The normalizer is the sum over candidate x' values, not a new fitted parameter.

Prediction already preserves normalization: $\sum_{x'} \bar{b}_{t+1}(x') = \sum_x b_t(x) \sum_{x'} P(x' \mid x) = 1$. Observation weighting generally does not, because likelihoods for the same fixed observation across different states need not add to one. Its normalizer is $P(e_{t+1} \mid e_{1:t})$. Confusing these two operations can lead to normalizing the transition model incorrectly or treating a sensor likelihood as though it were itself a state belief.

Only the current belief vector is required because it summarizes the past for future prediction under these assumptions. Keeping an explicit posterior over every possible state history would grow exponentially in time. The forward update merges all histories ending in the same current state by summing their contributions, which is the key dynamic-programming saving.

18.3.2 Worked example: smoothing revises an earlier belief

Continue the same weather HMM and observe umbrellas at both times 1 and 2. Filtering after the first observation gave $P(R_1 \mid U_1) = 9/11$. What does the second umbrella imply about the earlier weather?

1. **Compute the future observation likelihood by earlier state.** If R_1, then $P(U_2 \mid R_1) = 0.7(0.9) + 0.3(0.2) = 0.69$. If S_1, it is $0.3(0.9) + 0.7(0.2) = 0.41$.
2. **Combine past and future evidence.** The unnormalized masses for R_1, S_1 are $0.5(0.9)(0.69) = 0.3105$ and $0.5(0.2)(0.41) = 0.041$.

3. **Normalize at the earlier time.** Their total is 0.3515, so $P(R_1 \mid U_1, U_2) = 621/703 \approx 0.883$. The second umbrella raises confidence that it was raining earlier.

4. **Distinguish the query from filtering at time 2.** $P(R_2 \mid U_1, U_2)$ asks about the current state; the calculation here asks about R_1. Their numerical values happen to agree for these parameters and evidence, but their variables and meanings differ. That equality is not a general property of HMMs.

Smoothing uses observations later than the queried state. It does not mean that an online agent could have used those future observations when choosing an action at the earlier time.

18.4 A complete weather-filtering calculation

Use $P(r_{t+1} \mid r_t) = 0.7$, $P(r_{t+1} \mid \neg r_t) = 0.3$, and umbrella likelihoods $P(u_t \mid r_t) = 0.9$, $P(u_t \mid \neg r_t) = 0.2$. Start with rain probability $1/2$ before the first observation.

After observing an umbrella, the unnormalized rain and sun masses are $0.9(0.5) = 0.45$ and $0.2(0.5) = 0.1$. Dividing by 0.55 gives $P(r_1 \mid u_1) = 9/11$. Predicting the next day gives

$$\bar{b}_2(r) = 0.7\frac{9}{11} + 0.3\frac{2}{11} = \frac{69}{110}, \qquad \bar{b}_2(\neg r) = \frac{41}{110}.$$

Another umbrella weights these by 0.9 and 0.2. The new posterior is

$$b_2(r) = \frac{0.9(69/110)}{0.9(69/110) + 0.2(41/110)} = \frac{621}{703} \approx 0.883.$$

Prediction moved the belief toward the weather dynamics; the new observation then moved it back toward rain.

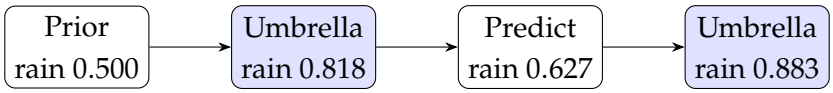


Figure 28: Filtering alternates motion prediction and likelihood reweighting.

Normalization at each step retains a usable belief distribution. If only relative masses are needed for later filtering, any common positive scaling factor can be removed. If the probability of the whole observation sequence is needed, those scaling factors must be retained. Discarding them preserves the posterior shape but loses the sequence likelihood.

18.4.1 Keeping the sequence likelihood while normalizing

For the two-umbrella example, the first unnormalized forward vector is $f_1 = (0.45, 0.10)$ for rain and sun, with total 0.55. Predicting without normalization gives $(0.345, 0.205)$. Weighting by the second umbrella gives

$$f_2 = (0.3105, 0.041), \qquad P(u_1, u_2) = 0.3515.$$

Normalizing this vector reproduces $(621/703, 82/703)$. If beliefs are normalized after every observation, record each normalizer: $Z_1 = 0.55$ and $Z_2 = 703/1100$. Their product is 0.3515, the same observation-sequence probability. This follows from $P(e_{1:T}) = \prod_t P(e_t \mid e_{1:t-1})$.

A contradictory next observation can reverse much of the accumulated belief. After two umbrellas, predict rain probability 4593/7030. A no-umbrella reading has likelihood 0.1 in rain and 0.8 in sun, so the new rain belief is

$$\frac{0.1(4593/7030)}{0.1(4593/7030) + 0.8(2437/7030)} = \frac{4593}{24089} \approx 0.191.$$

The old evidence was not erased; it influenced the predicted prior used in this update. The new reading simply favors sun strongly enough to outweigh much of it. Filtering maintains graded belief rather than committing permanently to the state that was previously most likely.

Source. Dan Klein and Pieter Abbeel, Introduction to Artificial Intelligence, Fall 2018, Lecture 18.

Lecture 19: Particle Filtering and Applications of HMMs

19.1 Particles as a belief representation

Exact filtering is impractical when the state space is enormous or continuous. A particle filter represents a belief with N sampled states $x^{(1)}, \ldots, x^{(N)}$ rather than a table over all possible states. In a discrete space, the fraction equal to x approximates its probability. Store the particles themselves; a table over every possible state defeats the space saving.

One filtering step has three operations:

1. **Predict.** Move each particle independently by drawing $\tilde{x}^{(i)} \sim P(X_{t+1} \mid x^{(i)})$.
2. **Weight.** Weight the predicted particles by the fixed observation: $w_i = P(e_{t+1} \mid \tilde{x}^{(i)})$.
3. **Resample.** Draw N new particles with replacement from the predicted particles, with selection probability $w_i / \sum_j w_j$.

The resulting particles again have equal weights. Prediction approximates the transition sum; weighting and resampling approximate the observation update. Resampling replicates plausible states and removes implausible ones, but introduces sampling noise. If all weights are zero, normalization is undefined and the representation must be recovered rather than silently continuing. More particles help, but difficult beliefs may require many.

With efficient transition draws, likelihood evaluation, and resampling, work is linear in N per step. In localization the map is known and the robot's pose is uncertain. In simultaneous localization and mapping (SLAM), both pose and map are unknown, so the state is substantially larger.

Example: For an illustrative four-particle prediction, suppose the states are $[A, A, B, C]$ and the observation likelihoods are $P(e \mid A) = 0.8$, $P(e \mid B) = 0.2$, and $P(e \mid C) = 0$. The individual weights are $[0.8, 0.8, 0.2, 0]$, totaling 1.8. The resampling probability of selecting a particle at A is $1.6/1.8 = 8/9$, at B it is $1/9$, and at C it is zero. A possible resampled collection is $[A, A, A, B]$; another draw could produce four copies of A.

Repeated particles are not an error. Their multiplicities represent probability mass. Replacing the list by its set of distinct states would discard that information. Conversely, particle filtering does not say that every unrepresented state is impossible in the true model: it may simply have been missed by the finite sample.

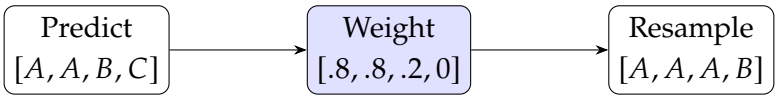


Figure 29: Resampling turns unequal weights back into an equally weighted particle list.

Algorithm 19.1: One bootstrap particle-filter step

Input: Particles $x^{(1)}, \ldots, x^{(N)}$ and observation e
for $i = 1, \ldots, N$ **do**
  Sample $\tilde{x}^{(i)}$ from the transition model given $x^{(i)}$
  Set $w_i = P(e \mid \tilde{x}^{(i)})$
if $\sum_i w_i = 0$ **then**
  return a failure indication for this particle representation
Normalize the weights
Draw N indices with replacement using the normalized weights
return the predicted particles at those indices, now with equal weights

Resampling focuses effort where the latest data says the state is plausible. It can also lose diversity. Once every particle has left a region, a future observation favoring that region may be hard to handle unless the transition model can move particles back there.

19.1.1 Why the three particle operations approximate filtering

The empirical belief is $\widehat{b}(x) = N^{-1} \sum_i \mathbf{1}\{x^{(i)} = x\}$. After transition sampling, the expected fraction of particles at x' is

$$\frac{1}{N} \sum_i P(x' \mid x^{(i)}) = \sum_x P(x' \mid x) \widehat{b}(x).$$

Thus prediction implements the exact transition update for the empirical input belief in expectation. Weighting then applies the observation likelihood to those predicted samples. Conditional on the weighted collection, the expected number of copies of particle i after resampling is $N w_i / \sum_j w_j$. The resampled empirical distribution approximates the normalized weighted belief.

For $[A, A, B, C]$ with weights $[0.8, 0.8, 0.2, 0]$, the probability intervals for the four particle indices are $[0, 4/9)$, $[4/9, 8/9)$, $[8/9, 1)$, and an empty interval. Uniform draws $0.1, 0.5, 0.7, 0.95$ select $[A, A, A, B]$. The expected counts are $32/9$ at A, $4/9$ at B, and zero at C; one particular resample need not equal these fractional expectations. Sampling without replacement would not implement this update: it would prevent high-weight particles from receiving the multiple copies that encode increased belief.

A fixed particle count fixes memory, but not approximation quality. If a posterior has several separated plausible regions, a finite sample can miss one. Repeated resampling can eliminate a region by chance even before later evidence makes it important. All-zero weights can mean either impossible evidence under the model or an inadequate current particle collection. Those cases require investigation rather than an arbitrary division by zero.

For continuous robot poses, a particle is a concrete position and orientation, not one entry in an exhaustive position grid. Motion noise spreads the poses; range readings weight their agreement with the known map. In SLAM, each hypothesis must also account for an uncertain map. This increases the state being represented and couples the interpretation of readings with location.

19.1.2 Worked example: resampling can eliminate a possible state

Suppose four predicted particles are $[A, A, B, C]$ with observation weights $[0.8, 0.8, 0.2, 0]$, as above. Resample four times independently with replacement.

1. **Group mass by state.** State A has probability $8/9$ on each draw and B has $1/9$. State C cannot be selected with this observation.
2. **Compute expected multiplicities.** The expected counts are $4(8/9) = 32/9$ for A and $4/9$ for B. They need not be integers because they describe averages across repeated resampling runs.
3. **Quantify losing B.** The chance of drawing no B is $(8/9)^4 = 4096/6561 \approx 0.624$. Thus the new particle list often has no B even though its updated probability was positive.
4. **Interpret a missing state.** Absence from a finite sample is not proof of impossibility. If later evidence strongly favors B, recovery may be difficult unless subsequent transitions can generate it again or the algorithm explicitly restores lost support.

Resampling converts weights into multiplicities but adds randomness. Keeping a set of distinct states would erase those multiplicities, while treating an absent state as logically ruled out would overstate what the approximation knows. The independent-draw probability above does not automatically describe every alternative resampling scheme.

19.2 Dynamic Bayes nets

A DYNAMIC BAYES NET repeats a structured collection of variables at each time, with dependencies within a slice and from the preceding slice. It generalizes an HMM whose single state variable encodes the whole slice.

Exact inference unrolls the network and eliminates old variables, retaining factors needed for the present. These factors may become large. For a particle filter, each particle must be a complete joint assignment to the slice, such as both ghosts' positions. Sample its successor from the joint transition model and weight it by the observation likelihood. When emissions are conditionally independent, multiply their likelihood factors. Sampling each tracked entity independently would lose dependencies represented by the joint model.

19.2.1 Why a DBN particle must be a joint assignment

Suppose two ghosts are equally likely to occupy (L, R) or (R, L), and they cannot occupy the same square. Each ghost's marginal is uniform on L, R. Sampling them independently from those marginals assigns probability $1/4$ each to (L, L) and (R, R), states that have no probability in the joint model. A particle containing both positions preserves the correlation.

The transition sampling for a slice must obey its factorization. If the second ghost's next position depends on the first ghost's newly sampled position, sample the first and then condition the second on that value. For conditionally independent sensor readings, the weight of the complete joint particle is the product of the likelihoods given the represented positions. Independent measurements do not imply independent hidden states.

Exact online inference has a related issue: eliminating the previous slice can create a factor coupling the current variables. Keeping only separate marginals can discard that coupling and is an approximation, even if the initial slice looked sparse. A DBN supplies a compact transition model; it does not guarantee that its filtering posterior remains equally compact.

19.3 Most likely state sequences

Filtering asks for a marginal distribution at the current time. A MOST LIKELY EXPLANATION instead asks for the single sequence $\arg\max_{x_{1:T}} P(x_{1:T} \mid e_{1:T})$. The evidence normalizer does not depend on the sequence, so maximize the joint.

The VITERBI ALGORITHM replaces forward sums over paths by maxima:

$$v_1(x) = P(x)P(e_1 \mid x), \qquad v_{t+1}(x') = P(e_{t+1} \mid x') \max_x [P(x' \mid x)v_t(x)].$$

Store the maximizing predecessor for each state and time. Choose the best final state, then trace back these pointers. A state trellis makes the distinction clear: forward inference sums the weights of all paths reaching a node, whereas Viterbi retains only the best path. Choosing the most likely state separately at each time need not give the most likely valid sequence.

For Viterbi, $v_t(x)$ is the probability of the *best complete prefix* ending in x, including observations through time t. To extend to x', compare every predecessor's best prefix times the transition probability. The emission probability for x' then multiplies the winning result.

A small illustration shows why marginal modes are insufficient. Suppose the only two-state paths with nonzero posterior masses are $(A, B) : 0.4$, $(B, A) : 0.35$, and $(B, B) : 0.25$. The most probable first state is B and the most probable second state is B, but the path (B, B) has probability only 0.25. The most likely path is (A, B) with probability 0.4.

Backpointers preserve the choices that made the best prefix. Keeping only the final scores would identify the best last state but lose the earlier states needed for the explanation. In a dense model, both forward and Viterbi updates cost $O(|S|^2)$ per time step; storing a complete Viterbi trace requires predecessor information over time.

19.3.1 A complete Viterbi table with backpointers

Use the umbrella HMM with initial rain/sun probabilities $(1/2, 1/2)$, transition stay probability 0.7, switch probability 0.3, and umbrella likelihoods $(0.9, 0.2)$. After the first umbrella, $v_1(r) = 0.45$ and $v_1(s) = 0.10$. For the second umbrella, compare the predecessor contributions before multiplying the current emission:

Next state	Via rain	Via sun	Final best-prefix score	Predecessor
Rain	$0.45(0.7) = 0.315$	$0.10(0.3) = 0.03$	$0.9(0.315) = 0.2835$	Rain
Sun	$0.45(0.3) = 0.135$	$0.10(0.7) = 0.07$	$0.2(0.135) = 0.027$	Rain

The best final state is rain. Its backpointer goes to rain at the first time, so the best path is (r, r) with joint probability 0.2835. Forward inference instead adds the two predecessor contributions and gives $f_2(r) = 0.3105$, $f_2(s) = 0.041$. These are larger because they include all paths into each endpoint, not only one best prefix.

If a third reading is no umbrella, the best rain endpoint has score $0.1 \max(0.2835 \cdot 0.7, 0.027 \cdot 0.3) = 0.019845$. The best sun endpoint has score $0.8 \max(0.2835 \cdot 0.3, 0.027 \cdot 0.7) = 0.06804$. Trace back from sun to the rain endpoint at time two, and then to rain at time one: the decoded path is (r, r, s). Saving an endpoint's predecessor at every time is what makes this reconstruction possible.

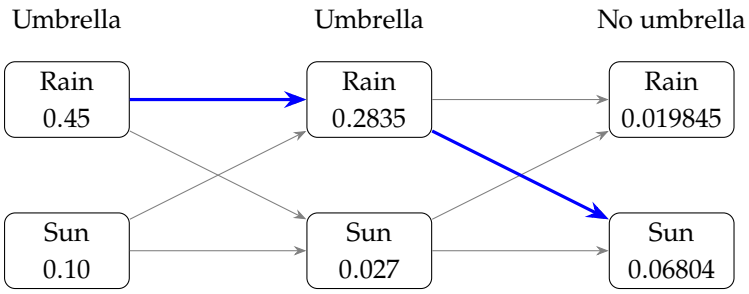


Figure 30: Node scores retain the best prefix; tracing back the best endpoint gives the blue rain-rain-sun path.

The recurrence is correct because all paths ending at a predecessor share the same transition and new emission when extended to a particular next state. A worse prefix to the same predecessor can never become better by multiplying by those same nonnegative factors. Only its best prefix need be retained. In implementation, logarithms turn products into sums and avoid underflow; impossible transitions receive log score $-\infty$. The final posterior normalizer can be omitted for selecting the path because it is the same for every complete candidate sequence.

19.3.2 Worked example: marginally best states can form a worse path

An illustrative two-time posterior gives the following probabilities to the four complete hidden paths:

Path	(A, A)	(A, B)	(B, A)	(B, B)
Posterior probability	0.35	0.05	0.30	0.30

1. **Find each time's marginal mode.** At time 1, $P(B) = 0.60$, so choose B. At time 2, $P(A) = 0.65$, so choose A. These two separate maximizations produce path (B, A).
2. **Find the joint mode.** The largest single table entry is $P(A, A) = 0.35$, exceeding $P(B, A) = 0.30$. Thus the most likely complete path is (A, A), even though its first state is not the first marginal mode.
3. **Locate the mathematical difference.** Marginals sum over the other time's states before maximizing. Sequence decoding maximizes a joint assignment. In general, maximizing after a sum is not the same as maximizing the complete joint table.
4. **Connect to Viterbi.** Viterbi retains the best prefix for each possible endpoint, then follows backpointers from the best final endpoint. It does not concatenate separately most probable states from filtered or smoothed beliefs.

The posterior table can arise from a two-state Markov model, so the distinction is already present in a minimal sequential setting.

Source. Dan Klein and Pieter Abbeel, Introduction to Artificial Intelligence, Fall 2018, Lecture 19.

Lecture 20: Machine Learning: Naive Bayes

20.1 Classification and the model

- **The classifier assumption.** Supervised classification learns from labeled inputs (x, y) to predict labels for new inputs. Features encode properties such as word occurrences, sender identity, or image pixels. A naive Bayes classifier assumes that features are conditionally independent given the label:

$$P(y, f_1, \ldots, f_n) = P(y) \prod_i P(f_i \mid y), \qquad \hat{y} = \arg\max_y P(y) \prod_i P(f_i \mid y).$$

  Normalize the class scores for a posterior distribution. Conditional independence is a modeling assumption, not a claim that features are unconditionally independent. The model has only linearly many local feature tables rather than an exponential table over all features.

- **The text representation.** For text, a bag-of-words model makes word positions conditionally independent and identically distributed given the class, tying $P(W_i \mid Y)$ across positions. The score is insensitive to order. Compute log scores, $\log P(y) + \sum_i \log P(w_i \mid y)$, to avoid underflow.

- **Derive the prediction rule.** The prediction rule comes from Bayes' rule: $P(y \mid f) \propto P(y, f)$. Substitute the naive Bayes factorization, then compare class scores. The evidence probability is common to every candidate class and can be omitted when only the maximizing label is needed. It must be restored by normalization if posterior probabilities are requested.

For a binary pixel, both an on value and an off value are evidence. Ignoring off pixels changes the model. In a word-position model, by contrast, the observations are the words present at the positions. A binary word-occurrence model and a repeated-token bag-of-words model are different representations and need different counts.

Conditional independence permits strong simplification but can exaggerate confidence when several features repeat the same evidence. Two nearly identical word-pattern features might each multiply in a similar likelihood ratio, effectively counting the clue twice.

20.1.1 What the feature representation commits the model to

For a binary-pixel digit model, every pixel contributes either $P(F_i = 1 \mid y)$ or $1 - P(F_i = 1 \mid y)$. An off pixel can distinguish a 1 from an 8 just as an on pixel can. For a repeated-token text model,

write $c_v(x)$ for the number of occurrences of vocabulary word v in input x. The class score can be collected as

$$P(y) \prod_v P(v \mid y)^{c_v(x)}.$$

Changing the word order leaves every count unchanged. Repeating a word changes its exponent and therefore its contribution. A binary word-present feature instead contributes once per vocabulary item, including absence; it cannot be substituted into the token-count formula without changing the model.

If some feature is genuinely unobserved, sum it out rather than treating it as observed false. In naive Bayes this produces $\sum_{f_i} P(f_i \mid y) = 1$, so a missing feature can be omitted from the product. Observed absence and missing information are different cases. For example, knowing a pixel is off supplies a likelihood; not seeing that part of the image supplies no pixel evidence in the model.

With K classes and n categorical features of d values, the model uses $(K-1) + nK(d-1)$ independent probabilities. Each feature table has a separate normalization for each class. Conditional independence is what makes this linear in n; a full feature distribution per class would require $K(d^n - 1)$ probabilities.

20.1.2 Worked example: an absent feature is evidence, a missing feature is not

Consider equal-prior classes spam and ordinary with one Boolean feature "contains offer." Let its true probabilities be 0.8 for spam and 0.1 for ordinary.

1. **Observe presence.** Class masses are $0.5(0.8) = 0.4$ and $0.5(0.1) = 0.05$. The spam posterior is $8/9$.
2. **Observe absence.** Use complementary probabilities, giving masses $0.5(0.2) = 0.1$ and $0.5(0.9) = 0.45$. The spam posterior is $2/11$. Omitting this factor would discard a known negative observation.
3. **Leave the feature unobserved.** Marginalizing its two values contributes $P(F = 0 \mid y) + P(F = 1 \mid y) = 1$ for each class. With no other evidence, the posterior remains the equal prior.
4. **Check the representation.** A Bernoulli word-presence model includes evidence about both present and absent vocabulary words. A model of observed word tokens is a different representation and does not acquire the same absent-word factors by default.

Encoding an unavailable feature as zero silently changes "unknown" into "observed absent." Missingness may itself carry information in a richer model, but that assumption must be specified rather than introduced by an accidental data encoding.

20.2 Parameter estimation and smoothing

Maximum likelihood estimates categorical probabilities by relative frequencies. For a feature value v in class y,

$$\widehat{P}(F_i = v \mid y) = \frac{N(F_i = v, y)}{N(y)}.$$

An unseen event then receives probability zero, eliminating a class no matter how strong the other evidence is. Laplace smoothing adds $k > 0$ pseudo-counts to each of the d_i possible feature values:

$$\widehat{P}(F_i = v \mid y) = \frac{N(F_i = v, y) + k}{N(y) + kd_i}.$$

Smooth each conditioning context separately. In a word-position model, count tokens within each class and use the vocabulary size in the denominator. Larger k pulls toward a uniform distribution. Linear interpolation is an alternative: mix the empirical conditional with an overall empirical distribution, reducing reliance on sparse conditional counts.

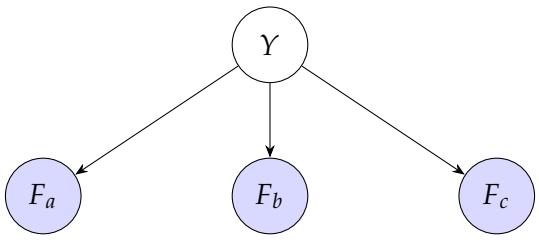
Example: Consider an illustrative one-feature spam model. Among 4 spam emails, 3 contain a chosen word; among 6 ham emails, 1 contains it. With add-one smoothing, $P(\text{word present} \mid \text{spam}) = (3+1)/(4+2) = 2/3$ and $P(\text{word present} \mid \text{ham}) = (1+1)/(6+2) = 1/4$. Using unsmoothed class priors $0.4, 0.6$, the scores for a word-containing email are $0.4(2/3) = 4/15$ and $0.6(1/4) = 3/20$. The spam posterior is

$$\frac{4/15}{4/15 + 3/20} = \frac{16}{25} = 0.64.$$

The choice to smooth the prior separately is independent of smoothing the feature tables.

The denominator adds k once for *every possible value*. A binary feature adds $2k$ to its class count, not merely k. For a vocabulary of size V, a token distribution adds Vk to the token total. The numerator and denominator must describe the same kind of observation.

When using logarithms, sum the log prior and feature log likelihoods. To recover normalized probabilities, subtract the largest log score before exponentiating. This common shift leaves relative scores unchanged while avoiding extremely small raw products.



Feature values are observed.
Compare a class score for each possible Y.

Figure 31: Naive Bayes connects each observed feature to the unknown label.

20.2.1 Deriving frequency estimation and interpreting smoothing

For one binary feature in a fixed class, let a observed examples have value one and b have value zero. Its likelihood as a function of $\theta = P(F = 1 \mid y)$ is $\theta^a(1-\theta)^b$. For $a, b > 0$, maximizing its logarithm gives

$$\frac{d}{d\theta}[a \log \theta + b \log(1-\theta)] = \frac{a}{\theta} - \frac{b}{1-\theta} = 0, \qquad \theta = \frac{a}{a+b}.$$

If $a = 0$ or $b = 0$, the optimum is on a boundary, explaining the zero probabilities that cause trouble on new examples. Categorical frequency estimation follows the same principle with probabilities constrained to sum to one. Each conditioning class supplies its own counts.

Add-k smoothing has the useful mixture interpretation

$$\frac{N_v + k}{N + kd} = \frac{N}{N + kd}\frac{N_v}{N} + \frac{kd}{N + kd}\frac{1}{d} \qquad (N > 0).$$

It blends empirical frequency with the uniform distribution. With little data, the uniform part has greater influence; with much data, observed frequencies dominate. At $N = 0$, smoothing gives $1/d$ directly. Interpolation instead can blend with a nonuniform background distribution: $\alpha\widehat{P}(v \mid y) + (1-\alpha)\widehat{P}(v)$. The mixture weights are hyperparameters, not new observations.

20.2.2 Training and classifying a complete toy document

Use an illustrative vocabulary {free, meeting, other} and token counts

Class	Free	Meeting	Other	Total tokens
Spam	3	1	0	4
Ham	1	3	4	8

With add-one smoothing, spam's probabilities are $(4/7, 2/7, 1/7)$; ham's are $(2/11, 4/11, 5/11)$. Each denominator adds the vocabulary size 3, and each row now sums to one. Use class priors $0.4, 0.6$. For the three-token message "free free other," the class scores are

$$u_S = 0.4(4/7)^2(1/7) = \frac{32}{1715}, \qquad u_H = 0.6(2/11)^2(5/11) = \frac{12}{1331}.$$

The spam posterior is $u_S/(u_S + u_H) = 10648/15793 \approx 0.674$. Without smoothing, the unseen spam token "other" would make its entire score zero, regardless of the two occurrences of "free."

For a long document, compute $\ell_y = \log P(y) + \sum_v c_v \log P(v \mid y)$. Subtract $m = \max_y \ell_y$ and normalize $e^{\ell_y - m}$. The common factor e^{-m} cancels between numerator and denominator; the largest exponent is now zero, preventing every class score from underflowing together.

20.3 Generalization and evaluation

Training learns parameters. A separate held-out set selects hyperparameters such as smoothing strength; the test set estimates final performance and must not guide repeated tuning. Overfitting occurs when training peculiarities or noise are fitted at the expense of future prediction. More data, simpler representations, and regularization can help.

Compare against a baseline such as always predicting the most frequent training label. High accuracy on an imbalanced problem can conceal a weak classifier. Inspecting errors can reveal missing features, such as whether the sender is known or whether a displayed link matches its destination.

A probabilistic classifier's top posterior is a confidence estimate, but the model can be wrong about that confidence. Calibration asks whether predictions made with a given confidence are correct at the corresponding rate; a high numerical posterior alone is no guarantee of reliability.

A parameter is estimated within a chosen model, such as a feature probability. A hyperparameter chooses how that estimation is done, such as smoothing strength. Trying many smoothing values on the test set would gradually fit the evaluation procedure to that same test set. Held-out tuning protects the final test from this feedback.

Accuracy also needs context. If most messages are ham, always saying ham can obtain a high score while missing every spam message. Compare with that baseline and inspect which kinds of errors remain. A confident mistake can reveal a stronger modeling problem than a near-tie mistake.

20.3.1 Evaluating what was actually learned

For each candidate smoothing strength, estimate its probabilities from the training set, evaluate classification on held-out data, and select the best candidate there. Then evaluate the chosen procedure on the untouched test set. A test set reused to choose features or smoothing becomes another tuning set, even if the individual probability estimates were fit elsewhere.

A classifier can memorize a document-ID feature and achieve perfect training accuracy while knowing nothing about a new ID. Bag-of-words shares parameters across many documents and therefore allows transfer, but its independence assumption can still count correlated clues repeatedly. If two identical features each supply a likelihood ratio of 4 in favor of spam, the model multiplies them to 16 despite their conveying the same evidence. The top label can be right while its posterior confidence is too extreme.

Calibration checks that distinction empirically. Among many predictions near confidence 0.8, roughly 80% should be correct for a well-calibrated model on the evaluation distribution. Accuracy alone asks only whether the top label was right. A model can keep the same labels while changing every confidence and hence its calibration. Both quantities must be interpreted alongside class

imbalance and the baseline, since a high overall accuracy can hide failure on a rare but important label.

20.3.2 Worked example: accuracy can hide a poor spam filter

On an illustrative held-out set of 1000 messages, 50 are spam. A classifier flags 40 messages, of which 30 are spam and 10 are ordinary.

1. **Complete the confusion counts.** There are 30 true positives, 10 false positives, 20 false negatives, and 940 true negatives. Every message belongs to exactly one of these four cases.
2. **Compute accuracy.** It is $(30 + 940)/1000 = 0.97$. An always-ordinary classifier already obtains 0.95 because spam is rare.
3. **Ask what happens to flagged and missed messages.** Precision is $30/40 = 0.75$; recall is $30/50 = 0.6$. The filter catches only 60% of spam despite its high overall accuracy.
4. **Separate evaluation from threshold selection.** A posterior threshold can trade false alarms against missed spam. Choose that threshold using validation data and the application objective, then evaluate the chosen system on untouched test data. Repeatedly adapting to test results turns the test set into another source of training decisions.

The right metric depends on consequences. Classification accuracy describes how often the chosen label is correct, but it does not by itself establish calibrated probabilities or low expected cost for an asymmetric decision task.

Source. Dan Klein and Pieter Abbeel, Introduction to Artificial Intelligence, Fall 2018, Lecture 20.

Lecture 21: Machine Learning: Perceptrons and Logistic Regression

21.1 Linear scores and perceptron updates

- **Represent an input.** A feature vector $f(x)$ maps an input to numbers. A linear classifier scores it by $z = w^\top f(x)$ and predicts the sign. A constant feature represents a bias, allowing the separating hyperplane to move away from the origin.
- **Correct a binary mistake.** The binary PERCEPTRON starts with zero weights. For each example with gold label $y^* \in \{-1, +1\}$, predict using the current score. On a mistake, update $w \leftarrow w + y^* f(x)$; otherwise leave it unchanged. The update raises the gold label's signed score on that example.
- **Correct a multiclass mistake.** For multiple classes, keep a vector w_y for each class and predict $\hat{y} = \arg\max_y w_y^\top f(x)$. On a mistake,

  $$w_{y^*} \leftarrow w_{y^*} + f(x), \qquad w_{\hat{y}} \leftarrow w_{\hat{y}} - f(x).$$

  Only the correct and incorrectly predicted class vectors change. The relative score of the correct class increases on the mistaken input.

If the binary training set is linearly separable, perceptron training converges after finitely many mistakes; the margin affects the mistake bound. Nonseparable data can make weights keep changing. Averaging successive weight vectors can stabilize predictions. A separating boundary need not generalize well, and repeated passes can overfit; held-out performance helps choose when to stop.

The weights tell how a feature changes the score while other features are held fixed. A positive weight raises the positive-class score; a negative weight lowers it. The bias is the weight of an always-one feature. For the rule $4f_{\text{free}} + 2f_{\text{money}} - 3 > 0$, the boundary is $4f_{\text{free}} + 2f_{\text{money}} = 3$.

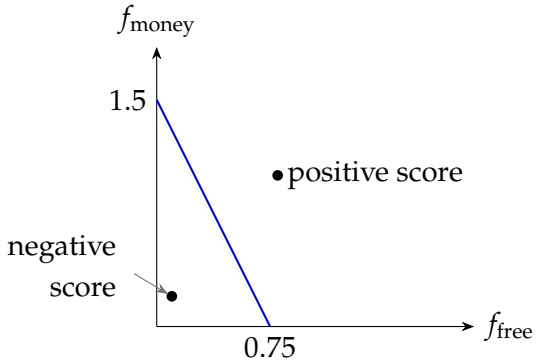


Figure 32: A linear score separates feature vectors by a straight boundary.

For a binary update, take an illustrative mistaken positive example $f = (2, 1)$ and current weights $w = (-1, 0)$. The score is -2, so the prediction is negative. Adding the positive example gives $w' = (1, 1)$ and changes this example's score to 3. In general, the gold signed score changes by $y^*(w + y^* f)^\top f - y^* w^\top f = \|f\|^2$. This explains the direction of the update, but an update can also change scores on other examples.

Algorithm 21.1: Multiclass perceptron training

```
Initialize each class vector w_y = 0
for each training pass do
    for each example (x, y*) do
        Predict ŷ ∈ arg max_y w_y^⊤ f(x) using a fixed tie rule
        if ŷ ≠ y* then
            w_{y*} ← w_{y*} + f(x)
            w_ŷ ← w_ŷ − f(x)
```

The separability assumption means that one linear boundary can classify every training example correctly, with ties handled consistently. A positive margin means the examples lie a nonzero distance from that boundary after the separating direction is normalized. A narrow margin allows more mistakes before convergence. If labels conflict or the needed boundary is nonlinear in the chosen features, repeated updates need not settle. More passes are not a substitute for an adequate representation.

21.1.1 A multiclass training trace

Use two-dimensional features and classes A, B, C, with ties resolved in that order. Initially all vectors are zero. The first example $((1, 0), A)$ is predicted A and causes no change. The next example $((0, 1), B)$ is incorrectly predicted A, so

$$w_A = (0, -1), \qquad w_B = (0, 1), \qquad w_C = (0, 0).$$

For $((1, 1), C)$, the scores are $(-1, 1, 0)$, so the mistaken class is B. The update gives $w_B = (-1, 0)$ and $w_C = (1, 1)$, leaving w_A unchanged. The gold score rises from 0 to 2, and the mistaken score falls from 1 to -1. Their relative gap improves by $2\|f\|^2 = 4$.

Now revisit $((1, 0), A)$. Scores are $(0, -1, 1)$, so the new prediction is C. Update w_A to $(1, -1)$ and w_C to $(0, 1)$. A correction on one example can create a mistake on another because weights are shared. Training therefore uses repeated passes or a stopping rule rather than assuming one pass fixes every example permanently.

The binary bias can be handled in the same vector arithmetic by appending an always-one coordinate. Its weight changes on a mistake just like every other active feature. Without it, the zero-score boundary must pass through the origin of the supplied feature space. A rule requiring "at least

three occurrences" rather than merely a positive count generally needs such an intercept or an equivalent feature construction.

21.1.2 Why separability yields finitely many mistakes

Suppose all feature vectors have norm at most R, and there is a unit vector u with $y_i u^\top f_i \geq \rho > 0$ for every training example. The positive number ρ is a margin in the augmented feature space. After M mistaken updates from zero, the projection onto u satisfies

$$u^\top w_M = \sum_{j=1}^{M} y_j u^\top f_j \geq M\rho.$$

On a mistake, $y_j w^\top f_j \leq 0$, so

$$\|w + y_j f_j\|^2 = \|w\|^2 + 2y_j w^\top f_j + \|f_j\|^2 \leq \|w\|^2 + R^2.$$

Consequently $\|w_M\| \leq R\sqrt{M}$. By Cauchy–Schwarz, $M\rho \leq u^\top w_M \leq \|w_M\|$, giving $M \leq (R/\rho)^2$. This proves a finite mistake bound under the stated assumptions. It does not identify the best generalizing separator, nor apply when no positive-margin separating direction exists. Averaging or held-out stopping addresses noisy practical behavior rather than making nonseparable labels satisfy the theorem.

21.1.3 Worked example: contradictory labels make the perceptron cycle

Let two training examples have the identical one-dimensional feature $f = 1$, but labels +1 and −1. Break a zero-score tie in favor of +1 and start at $w = 0$.

1. **Process the positive example.** The zero score predicts +1, so no update occurs.
2. **Process the negative example.** It is misclassified, and $w \leftarrow 0 + (-1)(1) = -1$.
3. **Repeat in the same order.** The positive example is now misclassified, giving $w \leftarrow -1 + 1 = 0$. The negative example then returns the weight to −1. Each subsequent pass repeats these corrections.
4. **Check the theorem's assumption.** No classifier using this feature can assign opposite labels to identical inputs. The data are not linearly separable, so the finite-mistake guarantee does not apply. The cycling is not evidence that its proof failed under its own assumptions.

With one positive and one negative observation, logistic regression instead can represent an empirical conditional probability of 1/2 at this feature. That does not make both labels deterministically correct. It represents the uncertainty or unresolved distinctions remaining in the chosen input description. Improving features and changing the optimization method solve different problems.

21.2 Logistic probabilities

Perceptron scores are not probabilities. Binary LOGISTIC REGRESSION uses the sigmoid

$$\sigma(z) = \frac{1}{1+e^{-z}}, \quad P(y=+1 \mid x; w) = \sigma(w^\top f(x)), \quad P(y=-1 \mid x; w) = 1 - \sigma(w^\top f(x)).$$

The sign still determines the most likely class, but the score's magnitude controls confidence. Learn weights by maximizing conditional log likelihood:

$$\ell(w) = \sum_i \log P(y^{(i)} \mid x^{(i)}; w).$$

This objective uses degrees of confidence rather than changing weights only on mistakes. It can learn probabilistic predictions even when no linear boundary perfectly separates the labels.

The sigmoid maps score 0 to probability $1/2$, score $\log 3$ to $3/4$, and score $-\log 3$ to $1/4$. Its output is always between zero and one, but that alone does not prove the probabilities are calibrated on real data.

For a positive training label, the log-likelihood contribution is $\log p$; for a negative label it is $\log(1-p)$. A confident wrong prediction receives a much larger penalty than an uncertain one. For example, a positive label with $p = 0.01$ contributes about -4.605, while $p = 0.5$ contributes about -0.693. This gives a graded training signal even when two models make the same hard prediction.

Separating data perfectly does not by itself give finite optimal logistic weights without further constraints or regularization: enlarging separating scores can keep increasing the unregularized likelihood. Validation and regularization concern how the probabilities generalize, not only whether training labels fall on the correct side.

21.2.1 Scores as log odds and confidence

The sigmoid satisfies

$$\frac{p}{1-p} = e^z, \qquad \log \frac{p}{1-p} = z = w^\top f(x).$$

A unit increase in score multiplies the modeled odds by e; it does not add a fixed amount to probability. For example, increasing z from 0 to $\log 3$ moves p from $1/2$ to $3/4$. Doubling that score gives odds 9 and probability $9/10$, not probability 1.

Multiplying all weights by a positive constant preserves the sign boundary but changes probabilities. A point scoring 1 has probability about 0.731; scoring 2 raises it to about 0.881. A hard perceptron prediction cannot distinguish these levels of confidence. Logistic training can: it rewards higher probability on observed labels and penalizes confident errors.

For labels $t_i \in \{0, 1\}$, conditional log likelihood is

$$\ell(w) = \sum_i \left[t_i \log \sigma(w^\top f_i) + (1 - t_i) \log(1 - \sigma(w^\top f_i)) \right].$$

The features are treated as given; the model is trained to predict labels conditioned on them. Naive Bayes instead models the feature distribution within each class and uses Bayes' rule to obtain the conditional. Both can supply a classifier, but they impose different probability models and use different training objectives.

On strictly separable data, scaling a separating w upward makes every gold-label probability approach one and each log term approach zero. No finite scale attains exactly one. Thus unregularized likelihood can have a supremum approached by diverging weights rather than a finite maximizer. This is separate from perceptron convergence to some finite separator.

21.2.2 Worked example: the best decision threshold depends on costs

A logistic classifier outputs $p = P(\text{spam} \mid x) = 0.3$. Suppose deleting an ordinary message costs 20 units, while keeping spam costs 1 unit. Correct actions have zero cost.

1. **Average the cost of deletion.** It is $20(1 - p)$ because cost is incurred only when the message is ordinary.
2. **Average the cost of keeping.** It is p, the probability of spam times its unit cost. At $p = 0.3$, the costs are 14 and 0.3, so keep it.
3. **Derive the threshold for all inputs.** Delete only when $20(1 - p) < p$, or $p > 20/21$. A $1/2$ threshold instead corresponds to equal error costs under this two-action setup.
4. **Relate the threshold to the score.** Since $p/(1 - p) = e^z$, the threshold is $z > \log 20 \approx 2.996$. The probability model can be retained while the decision rule changes to reflect the application objective.

This calculation assumes that the output probabilities are suitable estimates for the population in which the decisions will be made. Good classification accuracy alone does not establish calibration. Probability estimation, threshold selection, and evaluation under costs are separate questions.

21.3 Multiclass logistic regression

For class scores $z_y = w_y^\top f(x)$, SOFTMAX gives

$$P(y \mid x; w) = \frac{e^{z_y}}{\sum_{y'} e^{z_{y'}}}.$$

The largest score remains the most probable label, and the probabilities sum to one. Optimize the sum of the gold labels' log probabilities on training data. All class scores enter the normalizer, so increasing one class's score changes the probabilities of the others. Subtracting a common constant from the scores preserves these probabilities and permits stable computation.

For an illustrative three-class score vector $(\log 2, 0, 0)$, exponentiation gives weights $(2, 1, 1)$, so softmax produces $(1/2, 1/4, 1/4)$. These are relative probabilities: increasing one score changes the shared denominator. Subtracting $\max_y z_y$ before exponentiating leaves the ratios unchanged and prevents overflow from unnecessarily large exponents.

The most likely class still has the highest linear score. Turning scores into probabilities changes the training objective and the confidence interpretation, not the shape of the hard decision boundaries in the same feature space.

21.3.1 Understanding softmax through score differences

For any two classes a, b,

$$\frac{P(a \mid x)}{P(b \mid x)} = e^{z_a - z_b}.$$

Only score differences matter. Adding a common constant c multiplies every unnormalized score by e^c, which cancels. A three-class example with scores $(\log 2, 0, 0)$ gives probabilities $(1/2, 1/4, 1/4)$, whereas boosting the second score to $\log 3$ gives $(1/3, 1/2, 1/6)$. Even the unchanged third score loses probability because the shared denominator has grown.

A boundary between classes a and b satisfies $(w_a - w_b)^\top f(x) = 0$. It remains linear in the given features. Probabilistic output does not make an inadequate linear representation capable of separating every possible pattern. Nonlinear features or learned representations change that expressive limitation; softmax changes how scores are interpreted and fitted.

Source. Dan Klein and Pieter Abbeel, Introduction to Artificial Intelligence, Fall 2018, Lecture 21.

Lecture 22: Machine Learning: Optimization and Neural Networks

22.1 Gradient optimization

- **Choose the ascent direction.** To maximize a differentiable objective $g(w)$, gradient ascent uses $w \leftarrow w + \alpha \nabla g(w)$. The first-order approximation $g(w + \Delta) \approx g(w) + \nabla g(w)^\top \Delta$ explains the direction: among steps with fixed Euclidean length, the gradient has greatest predicted increase. The learning rate α controls distance; an excessively large step can overshoot, while a small step can make progress slow.
- **Choose the data in an update.** For conditional log likelihood $\ell(w) = \sum_i \log P(y^{(i)} \mid x^{(i)}; w)$, batch ascent sums gradients over the entire training set before updating. Stochastic ascent updates from a randomly selected example, and mini-batch ascent uses a small random subset, allowing parallel computation with more frequent updates than a full batch. Their gradient scaling and learning rate must be used consistently.
- **Interpret the logistic signal.** For a binary label $t \in \{0, 1\}$ and $p = \sigma(w^\top f)$, differentiating $t \log p + (1 - t) \log(1 - p)$ gives $(t - p)f$. This displays the training signal: raise scores where the observed label exceeds the predicted probability.

The gradient is a vector of partial derivatives, one for each parameter. For a unit-length proposed direction d, the predicted change is $\alpha \nabla g(w)^\top d$. By the Cauchy–Schwarz inequality this is at most $\alpha \|\nabla g(w)\|$, attained in the gradient direction. This statement is local: it does not guarantee that a long step keeps increasing the actual function.

For the illustrative objective $g(w) = -(w - 2)^2$, the derivative is $4 - 2w$. Starting at $w = 0$ with step size $1/4$ gives $w = 1$, then 1.5, then 1.75. The steps shrink as the slope shrinks. A very large step could jump back and forth across the optimum instead.

The logistic gradient can be derived rather than memorized. Since $dp/dz = p(1 - p)$ and $z = w^\top f$,

$$\frac{\partial \ell}{\partial z} = \left(\frac{t}{p} - \frac{1 - t}{1 - p} \right) p(1 - p) = t - p, \qquad \frac{\partial \ell}{\partial w_i} = (t - p) f_i.$$

For softmax with gold label y^*, the corresponding class-vector gradient is $(\mathbf{1}\{y = y^*\} - P(y \mid x)) f(x)$. The correct class is pushed up, while competing classes are pushed down in proportion to their current probabilities.

Stochastic gradients are noisy estimates of a full-data direction. They can make useful progress before an expensive full pass is finished, but the objective need not increase on every individual step. Mini-batches reduce some noise and suit parallel computation.

22.1.1 Learning-rate behavior on a solvable objective

For $g(w) = -(w-2)^2$, ascent gives $w_{t+1} = w_t + \alpha(4 - 2w_t)$. Write the error as $e_t = w_t - 2$. Then $e_{t+1} = (1-2\alpha)e_t$. If $0 < \alpha < 1$, the absolute multiplier is below one and the iterates converge to 2. With $\alpha = 1/4$, the error halves each step. With $\alpha = 3/4$, its sign alternates but its size also halves. At $\alpha = 1$ it alternates without shrinking; above 1 it grows. A correct derivative is therefore not sufficient for a stable update.

In many dimensions, the same rate can move too slowly along one direction and too aggressively along another. The first-order approximation explains the local direction, while curvature controls how long that approximation stays useful. Objective values, parameter changes, and held-out behavior can inform stopping; a fixed number of iterations is a computational budget, not a proof that the optimum was reached.

22.1.2 Following a logistic gradient numerically

For a positive example with $f = (2, 1)$ and $w = (0, 0)$, the score is zero and $p = 1/2$. The log-likelihood gradient is $(1-p)f = (1, 1/2)$. An ascent step with $\alpha = 0.1$ gives $w' = (0.1, 0.05)$. The example's new score is 0.25, so its probability rises to about 0.562. A negative example at the same feature vector supplies the opposite direction at the initial weights. Conflicting labels therefore produce a compromise rather than a separating solution that cannot exist.

For softmax, write a gold example's log probability as $\ell = z_{y^*} - \log \sum_y e^{z_y}$. Differentiating a class score gives

$$\frac{\partial \ell}{\partial z_y} = \mathbf{1}\{y = y^*\} - \frac{e^{z_y}}{\sum_u e^{z_u}}.$$

Multiplication by $\partial z_y / \partial w_y = f$ yields the stated class weight gradient. Unlike the perceptron, it updates competing classes even when the gold class already has the largest score, because probabilities can still improve.

If the objective is an average over N examples, the average gradient of a uniformly sampled mini-batch is an unbiased estimate of that full gradient. For a sum objective, a single-example gradient estimates the sum divided by N; the scaling can be absorbed into the learning rate. Switching silently between batch sums and means changes effective step size. Random selection also prevents a fixed ordering from repeatedly supplying a systematically unrepresentative part of the data early in each pass.

22.1.3 Worked example: sums and averages produce different update sizes

Consider two illustrative logistic-training examples with one feature and current weight $w = 0$. Their (feature, target) pairs are $(1, 1)$ and $(2, 0)$. Both predicted probabilities are $1/2$.

1. **Differentiate each example at the same weight.** The log-likelihood gradients $(t - p)f$ are $+1/2$ and -1.
2. **Choose an objective convention.** The summed objective has gradient $-1/2$; the mean objective has gradient $-1/4$. With learning rate 0.2, their batch updates give weights -0.1 and -0.05.
3. **Make a fair comparison.** Doubling the learning rate for the mean objective reproduces the sum-objective step for this batch. Reporting only the learning rate without the reduction convention leaves the update size ambiguous.
4. **Do not confuse a batch with sequential updates.** If the first example updates w before the second is processed, the second probability is no longer $1/2$. The second gradient must be recomputed at the new weight; the result need not equal a full-batch update at the old weight.

For a batch of m examples, the sum is m times the mean. Batch size and learning rate therefore interact with the objective convention. This is an arithmetic issue independent of the additional sampling noise in stochastic or mini-batch optimization.

22.2 Learning representations

Logistic regression applies a learned linear score to fixed features. A NEURAL NETWORK learns intermediate features as well:

$$h^{(0)} = x, \qquad h^{(k)} = g(W^{(k)}h^{(k-1)} + b^{(k)}), \qquad P(y \mid x) = \text{softmax}(W^{\text{out}}h^{(L)} + b^{\text{out}})_y.$$

Nonlinear activations such as sigmoid, hyperbolic tangent, or $\text{ReLU}(z) = \max(0, z)$ permit nonlinear input-output functions. Without nonlinearities, composing affine layers would remain affine.

Training again maximizes conditional log likelihood, now over all layers' weights. Large networks can fit complex structure and can also memorize training data. Early stopping uses held-out performance to limit overfitting; training accuracy alone does not identify the best stopping point.

A hidden unit first computes a weighted combination of inputs, then applies a nonlinearity. A later layer combines those learned intermediate values. Thus a feature does not need to be supplied as a handcrafted rule before training; its parameters can change to make the final prediction better.

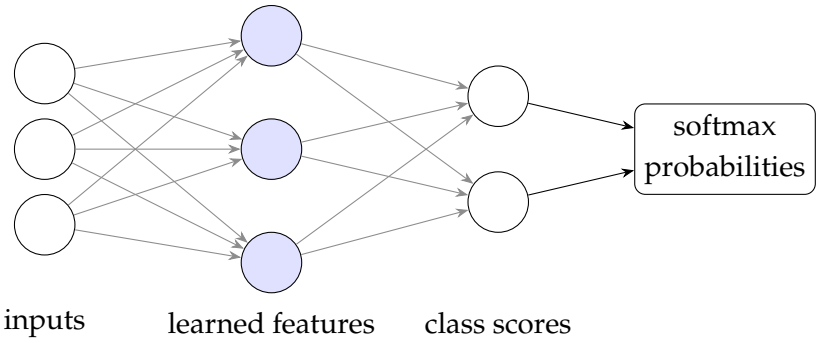


Figure 33: Hidden layers learn features that feed the final classifier.

For ReLU, negative inputs become zero and positive inputs pass through. Sigmoid compresses large positive and negative inputs toward its extremes. The nonlinearity changes which input patterns a layer can represent. Merely adding more linear layers would not create nonlinear boundaries, because a product of linear maps is still linear.

Depth and width control expressive capacity differently, but neither removes the need for data and optimization. An approximation theorem says some weights exist for a sufficiently large model; it does not give those weights or show that a finite dataset identifies them.

22.2.1 Why nonlinear hidden units add expressive power

Two affine layers without an intervening nonlinearity give

$$W_2(W_1x + b_1) + b_2 = (W_2W_1)x + (W_2b_1 + b_2).$$

This is still one affine map, however large the hidden layer. A nonlinearity makes the effective response depend on the input region. ReLU units, for example, activate only where their weighted input is positive; different active subsets can implement different affine pieces.

For a small illustrative Boolean construction, let $h_1 = \max(0, x_1 + x_2)$ and $h_2 = \max(0, x_1 + x_2 - 1)$. Then $h_1 - 2h_2$ equals $0, 1, 1, 0$ on inputs $(0,0), (0,1), (1,0), (1,1)$. Thresholding this score at $1/2$ represents XOR. The hidden features make a pattern separable that a raw-input linear threshold cannot represent. The construction shows existence of useful weights; a learning algorithm must still discover suitable features from examples.

For a layer with d inputs and h hidden units, W has shape $h \times d$ and b has h entries. An output layer for K classes has a $K \times h$ weight matrix and K biases. Keeping these dimensions explicit helps distinguish a feature vector from a layer's collection of parameters. The same hidden features are reused by all output classes, so training one class's probability can change representations used by the others.

22.3 Approximation and automatic differentiation

With suitable nonlinear activations, a network with one sufficiently wide hidden layer can approximate continuous functions on a compact domain to arbitrary accuracy. This is an expressiveness result; it does not guarantee that training finds the weights, that the network is small, or that it generalizes from finite data.

A network is a composition of simple operations, so the chain rule computes its derivatives. Automatic differentiation records intermediate values on a forward pass and propagates derivative information backward. This backpropagation computes gradients with respect to many parameters at a cost comparable in order to evaluating the network, rather than separately perturbing every weight. Learned features support tasks such as object recognition, speech recognition, translation, and captioning.

For a scalar chain-rule illustration, let $u = wx$, $h = \sigma(u)$, and $z = vh$. A forward pass computes and stores u, h, z. A derivative with respect to w follows the chain:

$$\frac{\partial z}{\partial w} = \frac{\partial z}{\partial h}\frac{\partial h}{\partial u}\frac{\partial u}{\partial w} = v\, h(1-h)x.$$

In a large network, many parameters share intermediate computations. The backward pass reuses those stored results rather than recomputing the entire model separately for each parameter. Training still needs a learning rate, data selection, and a stopping rule; automatic differentiation supplies derivatives, not those other decisions.

22.3.1 A forward and backward calculation on one hidden unit

Consider $u = wx + b$, $h = \sigma(u)$, $z = vh + c$, and $p = \sigma(z)$ for a binary output. Use the illustrative values $x = 3$, $w = b = c = 0$, $v = 2$, and gold label $t = 1$. The forward pass gives $u = 0$, $h = 1/2$, $z = 1$, and $p = \sigma(1) \approx 0.7311$. Store these intermediate values.

For log likelihood, the output derivative is $\partial \ell / \partial z = t - p \approx 0.2689$. The chain rule then gives

$$\frac{\partial \ell}{\partial c} = t - p, \qquad \frac{\partial \ell}{\partial v} = (t-p)h,$$
$$\frac{\partial \ell}{\partial b} = (t-p)v\, h(1-h), \quad \frac{\partial \ell}{\partial w} = (t-p)v\, h(1-h)x.$$

Numerically these are about 0.2689, 0.1345, 0.1345, and 0.4034. The residual at the output is propagated through the downstream weight, the hidden activation derivative, and the input. A gradient step updates all four parameters from the derivatives at the same old forward values. Updating v first and then using its new value to differentiate w would mix two parameter points and would not be this gradient.

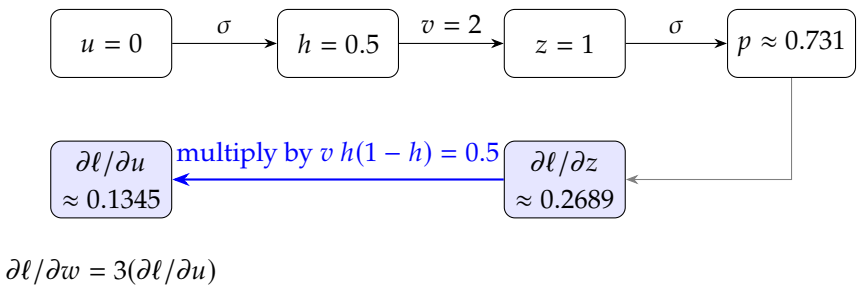


Figure 34: The forward pass stores values; the backward calculation reuses them to propagate derivatives.

This scalar example explains what is reused by automatic differentiation without requiring a separate finite-difference evaluation for every weight. Large-network software applies the chain rule across its computation graph, accumulating contributions where a value feeds multiple later operations. It computes the gradient of the implemented function. It cannot determine whether the labels, architecture, loss, or training/test split express the intended learning problem.

Learned representations replace some manual feature design in vision, speech, and translation. Handcrafted visual features encode chosen patterns such as edge orientations; a network can adjust early representations to support its prediction objective. Greater expressive capacity can improve fit and also permit memorization, so the distinction between representability, optimization, and generalization remains necessary.

22.3.2 Worked example: shared computations require adding derivative paths

Consider a small computation $u = x^2$ and $v = u + 3x$, evaluated at $x = 2$. The question is the derivative of v with respect to the shared input x.

1. **Run the forward calculation.** Store $u = 4$ and $v = 10$. The input contributes both through u and directly through $3x$.
2. **Differentiate the final operation.** Its local derivatives are $\partial v/\partial u = 1$ and a direct contribution of 3 to $\partial v/\partial x$.
3. **Propagate through the square.** Since $du/dx = 2x = 4$, the route through u contributes $1 \cdot 4 = 4$. Add the two routes to obtain $dv/dx = 4 + 3 = 7$.
4. **Check the result independently.** Expanding the whole expression as $v = x^2 + 3x$ gives derivative $2x + 3 = 7$ at $x = 2$. Overwriting the direct contribution with the square's contribution would incorrectly return 4.

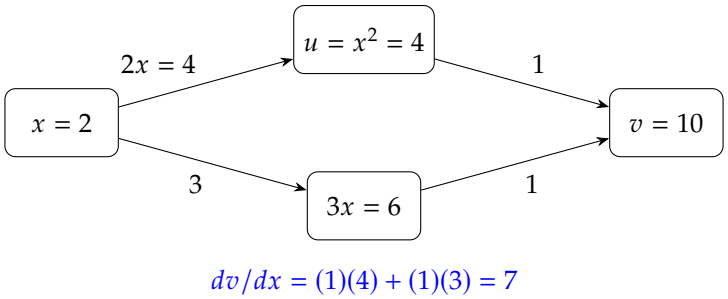


Figure 35: Add the derivative contributions from both routes through which an input affects the output.

Reverse-mode differentiation accumulates derivatives when a value influences the objective along several routes. The same rule applies when a parameter is reused across training cases or network computations: each use contributes to its total gradient. A computational graph records dependencies, not just a sequence of unrelated derivative assignments.

Source. Dan Klein and Pieter Abbeel, Introduction to Artificial Intelligence, Fall 2018, Lecture 22.

Lecture 23: Machine Learning: Neural Networks II and Decision Trees

23.1 Induction, consistency, and simplicity

- **What learning selects.** Learning chooses a hypothesis h from a hypothesis space H using examples of an unknown target function. Classification predicts discrete labels; regression predicts real values. A hypothesis consistent with all observed examples can still be wrong elsewhere. Expressive models, including neural networks, can represent more targets but also fit more accidental patterns.
- **Control the hypothesis space.** The tradeoff between consistency and simplicity can be controlled by reducing the hypothesis space, selecting features, adding modeling assumptions, or regularizing selection within a large space. These choices balance underfitting against sensitivity to the sample, often described as bias and variance. Held-out performance evaluates that balance.

23.1.1 What examples do and do not determine

Suppose two hypotheses agree on every observed input but disagree on an unseen one. Training consistency cannot choose between them: both explain all available labels. A hypothesis-space restriction or a preference for simplicity supplies an *inductive bias*, a rule favoring some possible extensions beyond the data. Without such a preference, an expressive model can complete the unseen labels in arbitrary ways.

An overly restricted space can exclude the target and underfit even with unlimited training. A very large space admits many hypotheses that fit a small sample, making the chosen solution sensitive to which examples happened to be collected. This is the bias–variance motivation in this setting. It is not a rule that a larger model is always worse: sufficient data and appropriate regularization can support its extra flexibility.

Naive Bayes supplies a structural independence bias. A linear perceptron supplies a geometric restriction in the chosen feature space. A decision tree with a depth limit restricts the number of tests on one path. Smoothing, early stopping, and pruning instead influence which fit is chosen within a potentially rich family. These methods address related generalization problems through different mechanisms.

23.2 Decision trees and expressiveness

A DECISION TREE tests an attribute at each internal node, follows the matching branch, and predicts at a leaf. Each root-to-leaf path represents a conjunction of tests. A feature can have different effects in different branches, automatically capturing interactions that a linear classifier would need explicit conjunction features to represent.

For finitely many discrete attributes, a sufficiently large tree can represent any function by distinguishing complete assignments. With n Boolean attributes, there are 2^{2^n} Boolean functions, so unrestricted expressiveness is enormous. The practical aim is a small useful tree rather than a table memorizing every assignment.

A tree can reuse a feature in different contexts because each branch represents a different subset of examples. A linear classifier on raw features adds fixed contributions. To express that one feature matters only when another is present, it needs an appropriate interaction feature. A tree obtains such conjunctions through its paths.

Expressing a function and finding it from examples are separate problems. The full truth table of n Boolean attributes has 2^n rows. A tree that distinguishes all rows can represent any labeling, but the number of possible labelings is 2^{2^n}. A learner needs a preference for some hypotheses over others; the training examples alone may leave many possibilities.

23.2.1 Counting functions and representing interactions

A truth table for n Boolean attributes has 2^n input rows. Each row can independently receive one of two labels, yielding 2^{2^n} different Boolean functions. Distinct tree drawings can represent the same function, so this is a count of functions expressible by unrestricted trees, not a count of all possible syntactic tree structures.

For a one-attribute stump, there are n attribute choices and four ways to label its two leaves, giving $4n$ syntactic choices. Constants are repeated across attributes: as distinct functions there are only $2n + 2$ for $n \geq 1$ (the two literals for each attribute and the two constants). This distinction matters when comparing expressive families rather than encodings.

A root-to-leaf path such as "patrons full AND wait short" is a conjunction. Another branch can test different attributes and assign the same attribute a different practical role. A raw linear score gives each input coordinate a fixed additive contribution; adding conjunction features can recover such interactions, but the feature designer must specify them. Trees create them through recursive partitioning, although greedy search may fail to find a compact useful partition.

23.3 Information gain and recursive learning

Grow a tree recursively by choosing an informative attribute, splitting examples by its values, and repeating within each subset. A pure subset can become a leaf. When no useful split remains, use a label based on the examples reaching the leaf.

For label proportions $p_1, \dots, p_k$, ENTROPY is

$$H(Y) = -\sum_j p_j \log_2 p_j, \qquad 0 \log_2 0 := 0.$$

It measures average information in the label: a fair binary label has one bit, while a certain label has zero. The INFORMATION GAIN of attribute A is

$$IG(A) = H(Y) - \sum_a \frac{N_a}{N} H(Y \mid A = a).$$

Use subset frequencies N_a/N to weight the post-split entropies. Choose the largest gain and recurse. This greedy rule need not find the globally best tree: useful interactions can be invisible at an individual split.

For a worked split, suppose eight examples have four positive and four negative labels. Their entropy is 1 bit. Attribute A splits them into two equal groups, with label counts $(3, 1)$ and $(1, 3)$. Each group's entropy is

$$-\frac{3}{4}\log_2\frac{3}{4} - \frac{1}{4}\log_2\frac{1}{4} \approx 0.8113.$$

The weighted post-split entropy is also 0.8113, so the gain is about 0.1887 bits. A different attribute separating the four positives from the four negatives has zero entropy in both children and gain 1 bit. This explains why pure children are preferred.

The weights N_a/N are important. A split that isolates one easy example while leaving a large mixed group should not be judged by an unweighted average of its children. Information gain uses the label uncertainty of a randomly selected training example after its branch is known.

Algorithm 23.1: Recursive decision-tree learning

Input: Examples E, remaining attributes A, and a fallback label
if E is empty **then**
  return a leaf with the fallback label
if all labels in E agree **then**
  return a leaf with that label
if A is empty or a stopping rule applies **then**
  return a leaf with the majority label in E
Choose an attribute X with largest information gain
Create a decision node testing X
for each value v of X **do**
  Recursively learn a child from $\{e \in E : X(e) = v\}$ and $A \setminus \{X\}$, using the current majority label as fallback
return the decision node

23.3.1 Entropy as expected information

An outcome of probability p contributes information $-\log_2 p$ bits. Averaging this quantity over possible labels gives entropy. Four equally likely labels have entropy 2 bits; proportions $(1/2, 1/4, 1/4)$ give $(1/2)1 + (1/4)2 + (1/4)2 = 1.5$ bits. A certain label contributes zero. The convention $0 \log 0 = 0$ is the limiting value as p approaches zero, so absent categories do not create undefined terms in the sum.

After an attribute test, the learner knows which branch was taken. Its remaining label uncertainty is the entropy within that branch. Since branch a occurs with frequency N_a/N, the expected remaining uncertainty is the weighted sum in information gain. A tiny pure branch contributes little to the average if most examples remain in a large mixed branch.

23.3.2 Growing a complete two-level tree

Use the illustrative dataset below, with repeated examples summarized as counts:

A	B	Positive	Negative
0	0	3	0
0	1	0	1
1	0	0	3
1	1	1	0

There are four labels of each class, so initial entropy is one. Splitting on A creates counts $(3, 1)$ and $(1, 3)$, giving gain 0.1887 as above. Splitting on B instead creates $(3, 3)$ and $(1, 1)$, both still balanced, so its gain is zero. The greedy root is A.

Within branch $A = 0$, testing B separates three positives from one negative, so both leaves are pure and the local gain is 0.8113. Within $A = 1$, B likewise separates three negatives from one positive. The final tree predicts positive exactly when $A = B$. Its root split was useful because the remaining test had different label associations in the two branches. Each recursive call uses only examples reaching it; reusing the global label counts inside a child would compute the wrong gain.

If a possible branch receives no training examples, the training data supply no label distribution there. The recursive algorithm uses a fallback, such as the parent's majority label. If all attributes are exhausted but labels still conflict, it also uses a majority leaf. Recursion terminates because an attribute is removed after each test along a path. These cases are part of a usable algorithm, not exceptions to be left unspecified.

23.3.3 Worked example: weight child entropies by the number of examples

A node has 8 positive and 2 negative examples. An illustrative binary attribute splits them into a pure child with 2 positives and a child with 6 positives and 2 negatives.

1. **Compute the parent uncertainty.** It is $H(0.8) = -0.8 \log_2(0.8) - 0.2 \log_2(0.2) \approx 0.7219$ bits.
2. **Compute both child uncertainties.** The pure child's entropy is 0. The larger child's positive fraction is $6/8 = 0.75$, so its entropy is $H(0.75) \approx 0.8113$ bits.
3. **Weight by branch frequency.** Remaining uncertainty is $(2/10)0 + (8/10)(0.8113) \approx 0.6490$. Information gain is therefore $0.7219 - 0.6490 \approx 0.0729$ bits.
4. **Reject an unweighted average.** Averaging the two children as $(0 + 0.8113)/2$ would pretend half of future cases enter the small branch. The resulting apparent gain of about 0.3163 is based on the wrong branch probabilities.

A pure child does not make an attribute highly informative if most examples remain in a mixed child. Information gain measures average reduction in uncertainty under the observed branch frequencies, not the best-looking individual branch.

23.4 Overfitting and pruning

Small subsets can appear pure by chance. Continuing to split them can improve training accuracy while harming generalization. One approach builds a tree and prunes bottom-up, removing splits whose apparent predictive relationship is not sufficiently supported, for example under a chi-squared significance criterion. Later structure can redeem a split that looks uninformative by itself, as with XOR, so premature stopping can miss interactions.

A pruning threshold is a regularization hyperparameter chosen on held-out data. A maximum-depth restriction instead directly limits the hypothesis space. Smaller trees can underfit, while larger trees can fit noise; the test set is reserved for evaluating the final choice.

XOR is a useful warning about greedy stopping. Let the label be one when two Boolean attributes differ and zero when they agree. At the root, either attribute alone leaves a balanced pair in each branch, so its information gain is zero. Splitting on both attributes nevertheless represents the function perfectly.

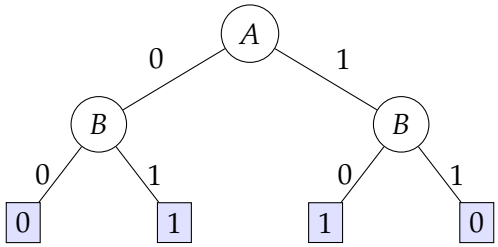


Figure 36: A two-level tree captures an interaction invisible to either root split alone.

Pruning asks whether a more complicated subtree is justified by evidence, not whether every split improved training fit. Tiny sample counts often produce apparent patterns by accident. A significance-based rule compares observed label counts with what would be expected if the attribute and label were unrelated. Its threshold is tuned on held-out data.

A simpler tree is easier to estimate from limited data, but simplicity is not always correctness. An overly aggressive pruning rule can remove a real interaction. The goal is to select a complexity supported by data, rather than to maximize or minimize tree size for its own sake.

23.4.1 What a split-significance test compares

The null hypothesis is that the proposed attribute and label are independent in the examples reaching that node. If child a has N_a examples and the parent's label-j proportion is p_j, its expected count is $E_{a,j} = N_a p_j$. A chi-squared discrepancy uses

$$\chi^2 = \sum_{a,j:E_{a,j}>0} \frac{(O_{a,j} - E_{a,j})^2}{E_{a,j}}.$$

For the $(3,1)$ and $(1,3)$ split, every expected count is 2, so the statistic is $4(1^2/2) = 2$. For a perfect $(4,0),(0,4)$ split with the same margins, it is $4(2^2/2) = 8$. The latter provides stronger evidence against independence at this sample size. A significance threshold translates such evidence into a pruning decision; it is not the probability that a learned subtree is correct. Very small counts also limit the quality of large-sample significance approximations.

Bottom-up pruning asks whether the subtree's added distinctions are supported well enough to keep. Stopping as soon as a root has zero information gain would rule out XOR, where the second test makes the first useful. Growing and then pruning permits later structure to provide that justification. Choosing the pruning threshold on held-out data balances lost real structure against accidental patterns, while reserving the test set for the final assessment.

23.4.2 Worked example: a perfect training split can lose on validation data

At a tree node, the unsplit prediction is positive. A candidate split creates two leaves, predicting positive on one and negative on the other. It classifies all 10 training cases correctly, while the unsplit node gets 8 correct.

1. **Evaluate a separate validation set.** Suppose validation gives 4 correct and 1 incorrect prediction in the positive leaf, and 1 correct and 4 incorrect predictions in the negative leaf. The split gets 5/10 correct.
2. **Evaluate the replacement on the same cases.** The first leaf has four actual positives and the second has four actual positives. Predicting positive for all ten therefore gets 8/10 correct.
3. **Choose according to the validation criterion.** Replacing the split by its parent prediction improves validation accuracy from 0.5 to 0.8, despite reducing training accuracy from 1 to 0.8.
4. **Interpret the evidence carefully.** The split fitted a pattern that did not transfer to these held-out cases. A small validation sample is itself noisy; this example illustrates the selection rule rather than proving that every pruned tree will perform better on all future data.

Model selection requires evaluating the competing models on comparable data. Comparing the split's training accuracy with the parent's validation accuracy would confound model quality with the choice of sample. Keep the final test set separate from these pruning decisions.

Source. Dan Klein and Pieter Abbeel, Introduction to Artificial Intelligence, Fall 2018, Lecture 23.

Lecture 24: Advanced Applications: Robotics, Vision, and Control

24.1 Combining search and learning

- **The components of game playing.** Large applications combine the course's methods. In Go, exhaustive minimax is infeasible because of branching and depth. A policy network narrows plausible moves; a value network estimates positions without reaching the end. Supervised learning from expert play, reinforcement learning through self-play, and Monte Carlo rollouts contribute complementary information.
- **The questions they answer.** A policy network and a value network answer different questions. The policy suggests which moves to consider; the value estimates how promising the position is if play continues. A rollout instead samples a continuation and uses its outcome as evidence about the position. Search combines these signals to spend computation where it is most useful.
- **The sources of experience.** Learning from expert moves provides an initial policy, while self-play can generate experience beyond those examples. The learned models guide search; they do not make the game's legal moves or terminal rewards unnecessary. The application therefore connects learned estimates with an explicit decision problem.

24.1.1 How the learned components change search

Exhaustive Go search is difficult because both the branching factor and useful planning depth are large. A policy network supplies a distribution over legal moves. Search can examine its more plausible moves first or focus limited expansion there. A value network predicts the quality of a nonterminal position, replacing the need to finish every continuation. A rollout follows a sampled policy to an outcome and supplies one noisy estimate rather than a full enumeration of the future.

These are different approximations. Restricting candidate moves reduces breadth but can omit a strong move. Cutting a branch at a learned value reduces depth but inherits prediction error. Rollouts reduce enumeration but introduce sampling variance and depend on the rollout policy. Combining them directs computation toward promising decisions; it does not inherit an exact full-tree minimax guarantee simply because search remains involved.

The training pipeline also separates roles. Expert positions and moves train an initial policy by supervised classification. Self-play provides outcomes under the system's developing behavior and can improve the policy through reinforcement learning. Positions paired with eventual outcomes provide data for a value model. A value target must reflect the continuation behavior used to obtain

it; a probability of winning under one policy is not automatically an exact game-theoretic minimax value.

24.1.2 Worked example: a policy suggestion and a value estimate can disagree

Suppose a learned policy assigns candidate moves A, B, C probabilities $0.7, 0.2, 0.1$. A limited search later estimates their continuation win probabilities as $0.45, 0.65, 0.40$ under its model.

1. **Use the policy for search guidance.** Examining A first is reasonable because the policy strongly suggests it. This determines where computation begins, not the result of every later comparison.
2. **Use continuation estimates for the final comparison.** Among the evaluated alternatives, B has the largest estimated win probability. A search that is permitted to reconsider its initial suggestion can select it.
3. **Identify the cost of hard restriction.** Keeping only the policy's top move would remove B before its continuation could be evaluated. This saves branching at the price of potentially excluding a strong move.
4. **Interpret the numbers as estimates.** A value of 0.65 depends on the learned model, search, and continuation behavior. It is not automatically a rigorous probability of beating every possible opponent.

The policy and value models supply different quantities. Multiplying their outputs or replacing one by the other needs a specified algorithmic reason; a large numerical score from one model does not settle the other model's question.

24.2 Helicopter estimation and control

A helicopter's state includes position, orientation, and their rates of change. Cameras and inertial sensors provide noisy measurements; a dynamics model predicts motion under control inputs:

$$s_{t+1} = f(s_t, a_t) + w_t.$$

Tracking combines prediction with observation updates, as in an HMM. Control chooses inputs affecting the main rotor and tail rotor. A hovering objective penalizes deviation from a target pose, but complex maneuvers also require specifying a useful target trajectory.

Demonstrations can be observations of an underlying desired trajectory. Their speeds differ, so directly averaging equal time indices blurs the maneuver. Probabilistic alignment combines a dynamics model with temporal alignment and state estimation to recover a cleaner latent trajectory.

A controller can then track the learned maneuver. This connects imitation, sequence inference, and control rather than treating demonstrations as independent input-output examples.

The estimation and control problems form a feedback loop. First predict the next state using the previous estimate and the control just applied. Then compare new sensor readings with what candidate states would predict, and update the estimate. The controller chooses its next input using that updated information.

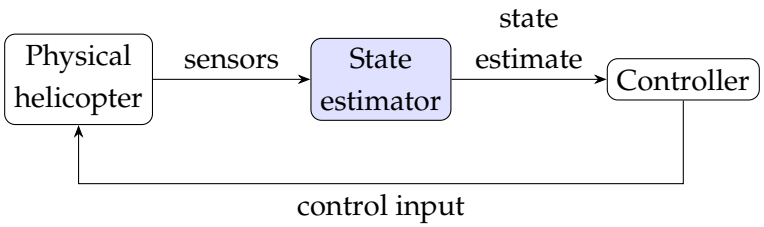


Figure 37: Control uses an estimated state and receives new feedback after acting.

An inertial sensor measures motion-related quantities, while vision provides external position information. Each is imperfect. Combining them through a dynamics and observation model is more useful than treating every reading as an exact state. The same position can require different controls when velocities or orientations differ, explaining why the state includes these quantities.

Demonstration alignment solves a different uncertainty: which moments in different demonstrations correspond to the same part of a maneuver? A slow demonstration and a fast one should not be averaged at identical clock times. A latent trajectory plus a timing alignment can pool evidence about the maneuver's actual shape.

24.2.1 The state, actions, and reward of helicopter control

One state representation is

$$s = (x, y, z, \phi, \theta, \psi, \dot{x}, \dot{y}, \dot{z}, \dot{\phi}, \dot{\theta}, \dot{\psi}),$$

combining three-dimensional position, orientation angles, and their rates. Two helicopters with the same position but different velocity need different inputs to hover there. A position-only state would leave this essential predictive difference hidden.

The four control inputs have distinct physical roles: longitudinal cyclic pitch affects pitch motion, lateral cyclic pitch affects roll motion, collective main-rotor pitch affects thrust, and tail-rotor pitch affects yaw-related control. The dynamics $f(s, a)$ couples these effects, so choosing one input cannot generally be treated as solving an independent scalar task. Vision supplies external position measurements; gyroscopes, accelerometers, and magnetometers supply complementary motion and orientation information.

For hovering, a simple illustrative reward is a negative weighted sum of squared deviations from target position, orientation, and rates. The weights express how the objective trades those errors and must respect their different units. For a flip, penalizing deviation from one fixed upright pose would actively discourage the desired maneuver. A target trajectory provides the changing desired state, showing why reward design requires specifying what successful motion should look like.

24.2.2 Why aligning demonstrations changes the inferred target

Suppose one demonstration has a distinctive maneuver peak at time index 2 and another at index 4. Averaging their states at equal clock times can produce two weak peaks or blur the maneuver, even if both are good executions of the same underlying movement. Alignment associates moments of similar progress rather than assuming equal sampling indices mean the same phase.

Dynamic time warping makes this correspondence explicit. For sequences $a_1, \ldots, a_m$ and $b_1, \ldots, b_n$, let $d(a_i, b_j)$ measure disagreement. A cumulative alignment cost can use

$$D(i, j) = d(a_i, b_j) + \min\{D(i-1, j), D(i, j-1), D(i-1, j-1)\},$$

with suitable start boundaries. A path through this table pairs samples in order while permitting one sequence to progress more slowly. It cannot arbitrarily reverse the maneuver's order. A dynamics-aware probabilistic model adds the requirement that the inferred latent trajectory resemble feasible helicopter motion, rather than merely matching observations pointwise.

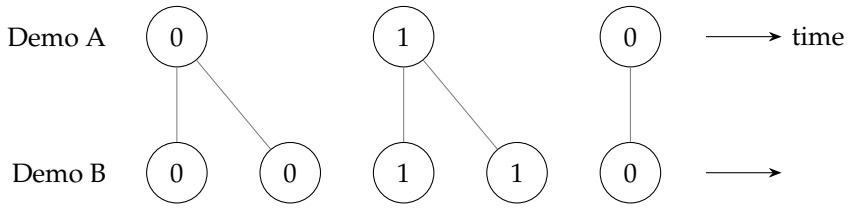


Figure 38: An illustrative alignment matches maneuver phases even when a demonstration spends more samples in a phase.

The estimation and alignment tasks interact: a cleaner trajectory improves timing correspondence, and better correspondence allows demonstrations to be pooled into a cleaner trajectory. Filtering or smoothing in the nonlinear dynamics model handles sensor noise; temporal alignment handles different execution speeds. Neither addresses the other's uncertainty by itself.

24.2.3 Worked example: why velocity belongs in a control state

For a deliberately simplified one-dimensional model, use state (x, v) and updates $x' = x + v$, $v' = v + a$ for a unit time interval. The target is to remain at $x = 0$ with zero velocity. This arithmetic model is not a flight simulator.

1. **Compare equal positions with different motion.** States $(0, 0)$ and $(0, 2)$ have the same location. With $a = 0$, their next positions are 0 and 2, so a position-only description cannot predict both correctly.
2. **Account for the timing of control.** At $(0, 2)$, choosing $a = -2$ gives next state $(2, 0)$ in this discretization. It cancels the next velocity, but does not retroactively cancel motion during the current step.
3. **Interpret the modeling implication.** A controller must act using motion and the specified dynamics, rather than assume that setting a velocity command changes position instantly. Different dynamics conventions would require different predictions.
4. **Connect back to hovering.** A real helicopter has coupled positions, orientations, and rates. Including rates makes the state more predictive; estimating them from noisy sensors is part of the feedback loop.

The example explains a state-representation requirement without claiming that this toy control input is safe or sufficient for an actual vehicle. A realistic controller must use its own dynamics and actuator constraints.

24.2.4 Worked example: compute a complete temporal alignment table

Take illustrative scalar demonstrations $a = [0, 1]$ and $b = [0, 0, 1]$, with local cost $d(a_i, b_j) = (a_i - b_j)^2$. Use the warping recurrence above, with $D(0, 0) = 0$ and other row-zero or column-zero entries infinite.

	$b_1 = 0$	$b_2 = 0$	$b_3 = 1$
$a_1 = 0$	0	0	1
$a_2 = 1$	1	1	0

1. **Fill the first row.** Matching zero to the first two zeros adds no cost; extending to the final one adds 1.
2. **Fill the second row.** The first two entries each have local cost 1 and a cheapest predecessor of cost 0, so both are 1. The final entry has local cost 0 and predecessor minimum 0, so it is 0.
3. **Trace the alignment.** A zero-cost path pairs a_1 with both b_1, b_2, then pairs a_2 with b_3. The repeated zero is a longer phase, not evidence that the demonstrations perform different maneuvers.
4. **Respect ordering.** The permitted predecessor moves preserve time order while allowing different speeds. Arbitrarily pairing closest values without an ordered path would solve a different matching problem.

24.3 Locomotion and rewards

Continuous-control policies can map joint angles and velocities through a neural network to joint torques. Learning in simulation permits repeated trials, while the representation must support coordinated motion. Hierarchical control separates a high-level choice, such as where to place a foot, from low-level motor commands that execute it.

For terrain traversal, a reward of the form $R(s) = w^\top f(s)$ can express preferences over terrain properties. Demonstrated paths help infer reward weights; planning with those weights then chooses paths across a new height map. Learning the objective can matter as much as optimizing a fixed one.

Learning a reward can generalize a preference rather than merely copy an action sequence. A demonstrated route across one terrain reveals something about which local properties are desirable. On a new terrain, the original moves may be impossible, but the learned feature weights can still rank new candidate routes.

Hierarchical control reduces the scale of each decision. A foot-placement planner compares a manageable set of future contact locations, while a lower-level controller handles motor commands at finer time scales. Failure at either level affects the other: a beautiful high-level plan is unhelpful if its low-level actions cannot reliably be executed.

24.3.1 Learning a transferable terrain preference

A high-level foot-placement or route planner can score candidate terrain using features such as roughness, slope, and route length. For illustration, let a route's accumulated feature vector be (r, s, ℓ) and its reward $-2r - 3s - \ell$. Route 1 with features $(1, 2, 5)$ scores -13; route 2 with $(2, 0, 7)$ scores -11. The longer second route is preferred because it avoids the costly slope. Merely following the shortest path would optimize a different objective.

Demonstrations can reveal such tradeoffs on training terrain. On a new height map, planning uses the learned feature weights to compare new routes, rather than trying to replay old motor commands. The low-level controller must then execute the chosen contacts reliably. A planner whose successor model assumes impossible foot placements can produce an attractive numerical plan that the robot cannot follow.

A demonstrated path need not uniquely identify reward weights: several weight vectors can rank it above the observed alternatives. Learning an objective therefore involves representation and modeling choices in addition to optimization. Its value is the ability to express preferences that transfer to new configurations where an exact action sequence does not.

24.4 Autonomous driving and manipulation

An autonomous vehicle combines positioning, perception, planning, and feedback steering. Laser height differences can flag obstacles, but raw measurements contain errors. A temporal probabilistic model combines uncertain position and repeated sensor observations, reducing spurious detections relative to isolated threshold tests. Camera-based perception can learn from other sensors' reliable labels, an example of self-supervision. Neural semantic segmentation assigns a class to each image pixel.

Robotic laundry, object manipulation, and learning several physical skills add high-dimensional perception, contact dynamics, and generalization to the control problem. Learned visual and control representations connect observations to action, while interaction supplies data unavailable from static labeled images alone. These examples show why sensing uncertainty, reward design, computation, and learning must be addressed together.

In obstacle detection, a height difference can result from an object or from measurement and pose error. Temporal inference considers whether a sequence of observations is more consistent with a persistent obstacle than with noise. Its benefit comes from modeling how errors and states relate across observations; blindly averaging contradictory readings is not the same operation.

Self-supervision uses a reliable signal from one part of the system to train another. A range sensor can identify nearby drivable ground and provide labels for camera appearance, allowing vision to extend beyond the range sensor's immediate coverage. The resulting predictor still depends on the quality and coverage of its training labels.

The central engineering difficulty is keeping the components' assumptions compatible. A planner needs a transition model that reflects what the controller can do, and a controller needs an estimate containing the variables that affect its dynamics. Better optimization cannot fix a state representation that leaves out essential information.

24.4.1 Why temporal perception helps an autonomous vehicle

A simple laser obstacle test compares nearby estimated heights Z_i, Z_j and flags a difference exceeding a threshold, such as 15 cm. A small pose error can shift many height estimates together, producing apparent obstacles on flat ground. These are correlated errors: treating every laser return as independent evidence would overstate the case for an obstacle.

A temporal model keeps uncertain vehicle pose x_t, motion information from GPS and inertial sensors, and terrain observations z_t. Predict the pose distribution through the dynamics, compare observations with the terrain that candidate poses predict, and update. An obstacle that persists at a consistent world location is different evidence from a transient discrepancy explained by pose error. The model's state and dependencies make this distinction possible; averaging raw heights without accounting for motion does not.

Self-supervised vision can use confident laser-based ground labels in the nearby region to train a camera predictor of drivable appearance farther ahead. This reduces manual labeling but transfers label errors if the teaching sensor is wrong. Semantic segmentation supplies a class per pixel rather than one label for the entire scene, providing spatial information for planning. The planner must still turn perceived geometry into actions that the controller can execute and continually revise them as the vehicle moves.

Manipulation adds changing contacts and deformable objects. For laundry, an image may not reveal all cloth folds or which grasp will untangle them. The robot's action changes what can be observed next, so learning from interaction supplies information unavailable from static image labels alone. These applications require compatible representations between perception, state estimation, planning, and control; each component's output is part of another component's assumed input.

Source. Dan Klein, Pieter Abbeel, and Anca Dragan, Introduction to Artificial Intelligence, Fall 2018, Lecture 24.

Lecture 25: Advanced Topics and Final Contest

25.1 Cooperative game strategies

- **The cooperative objective.** Cooperative Pac-Man requires an agent to collect food while coordinating with another agent and responding to ghosts. The contest strategies illustrate several ways to combine the course's tools.
- **A local feature policy.** A feature-based reflex policy can weigh distance to food, teammate separation, and ghost danger. Seeking food farther from a nearby teammate reduces duplicated effort. Approximate Q-learning can instead learn weights for distance and score features, including a term that discourages deadlock.
- **A combined planning policy.** A reward-density strategy first scores areas by food concentration, lowers scores where a teammate is likely to go, discounts distant targets, and penalizes nearby ghosts. Particle filtering maintains beliefs about teammate positions. A* selects routes to attractive areas when safe, while minimax handles immediate ghost danger. The central lesson is that estimation, search, local evaluation, and learning can serve different parts of one policy.

The components solve different failures of a purely local food-seeking rule. Teammate separation avoids both agents chasing the same reward. Ghost features discourage a locally attractive move that is unsafe. An exploration term can discourage repeatedly revisiting the same state without progress. A belief about an unobserved teammate matters because coordination depends on what that teammate is likely to do, not only on its last visible position.

A reward-density map expresses a spatial objective, while A* supplies a route toward a selected target. Switching to adversarial search near a ghost changes how immediate uncertainty is modeled. These choices must share a consistent notion of what is valuable, or one component can undo another's progress. A complicated collection of methods is useful only when each addresses a concrete part of the task.

25.1.1 Making the coordination tradeoff explicit

A nearby teammate creates a reason to seek different food, but maximizing distance from that teammate alone can send an agent away from every useful target. The feature policy must trade separation against food and ghost risk. A teammate can also draw a ghost's attention, changing the value of an otherwise dangerous action. The cooperative objective is team reward, so individually cautious or individually greedy choices need not combine into the best joint behavior.

For an illustrative reward-density score, let

$$D(v) = \frac{\text{food}(v)}{1 + \text{distance}(v)}[1 - P(\text{teammate targets } v)] - \text{danger}(v).$$

This is one possible realization of the qualitative strategy. If region A has food score 8, distance 3, and teammate probability 0.75, its pre-danger score is 0.5. Region B with food 6, distance 2, and teammate probability 0.1 scores 1.8. Coordination therefore favors B even though A contains more food. A danger penalty of 3 at B changes that comparison again. Each term addresses a different reason why raw food density can be misleading.

A belief about the teammate can be predicted through a motion model and reweighted when observations arrive, as with a particle filter. The target selection then uses expected teammate presence rather than assuming the last observed position is current. A* plans a route after the target is chosen; minimax near a ghost addresses immediate adversarial replies. These components operate at different stages, so the route planner need not optimize the raw density expression on every edge. Their objectives must still agree about what progress toward the selected target is worth.

Caching a corridor's actions can save repeated search when there is only one meaningful way forward. Such a cache is conditional on the relevant situation remaining unchanged. A newly observed ghost or a changed target can invalidate the cached continuation. Search reuse saves computation only when the state information on which the plan depended still applies.

25.1.2 Worked example: individually attractive targets can waste team reward

Two agents choose region A or B. Region A contains food worth 10 and costs each visiting agent 1 to reach; B contains food worth 8 and costs 2 to reach. Food in one region is collected only once even if both agents visit.

Joint choice	Food collected	Total travel cost	Team score
(A, A)	10	2	8
(A, B) or (B, A)	18	3	15
(B, B)	8	4	4

1. **Evaluate an isolated greedy choice.** Each agent compares $10 - 1 = 9$ with $8 - 2 = 6$ and prefers A if it assumes the reward is available independently of the teammate's choice.
2. **Evaluate the joint consequence.** Both choosing A earns only 8 for the team. Splitting targets earns 15 because both regions contribute and duplicated travel is avoided.
3. **Use a teammate model.** If one agent is committed to A, the other's marginal food gain from also choosing A is zero. Beliefs about teammate destinations change the value of a target, not just its distance.

4. **Keep risk in the comparison.** Adding a ghost-risk cost at B can make splitting unattractive again. Coordination is a joint reward calculation, not an unconditional instruction to maximize separation.

25.2 Speech as sequential inference

Speech recognition maps an acoustic waveform to words. Short-time spectral analysis describes energy across frequencies. Vocal-cord vibration supplies harmonics; the vocal tract amplifies selected frequency bands according to its shape. Different speakers, sounds, speaking rates, and transitions between sounds make matching waveforms directly unreliable.

An HMM uses acoustic feature vectors as observations and a PRONUNCIATION CURSOR as its hidden state: a position within a word's sequence of sounds. Emissions model how likely an acoustic vector is for that sound. Transitions advance within words and between words, allowing variation in duration and pronunciation.

A language model scores transitions between words. A bigram uses $P(w_t \mid w_{t-1})$; a trigram uses $P(w_t \mid w_{t-2}, w_{t-1})$. More context captures additional correlations but requires more data or parameter sharing. For example, the probability of "door" after "close the" can be far larger than after "the" alone.

Decoding finds the most likely state sequence given the acoustics, then reads the words from that sequence. This is a Viterbi query rather than a sequence of independent classifications:

$$\hat{x}_{1:T} = \arg\max_{x_{1:T}} P(x_1)P(e_1 \mid x_1) \prod_{t=2}^{T} P(x_t \mid x_{t-1})P(e_t \mid x_t).$$

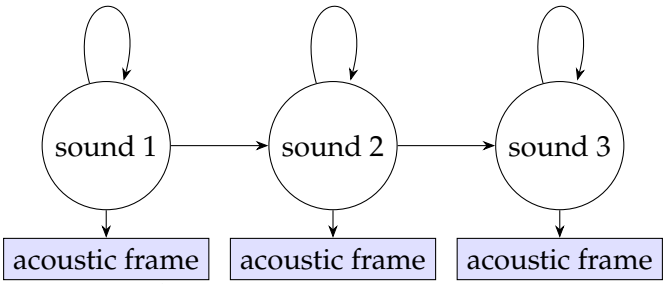
Acoustic and language models contribute different evidence. Neural networks can parameterize these models and learn richer representations than fixed local tables. Robust recognition at scale still differs from full contextual understanding of language.

The hidden state must carry a position within a pronunciation, rather than just a word label for the entire utterance. A word lasts many acoustic frames, and different frames correspond to different sounds or transitions. Self-transitions allow a state to persist for multiple frames; forward transitions advance through the pronunciation.

The acoustic model scores how well a sound state explains the current frame. The language model scores how plausible a word continuation is. A locally plausible sound match can belong to an implausible word sequence, while a common phrase can have poor acoustic support. Decoding balances both kinds of evidence.

Increasing language-model order preserves more context but fragments counts. The contexts for a trigram are word pairs, so many specific pairs may be rare even in a large dataset. Smoothing or learned parameter sharing is therefore needed to avoid treating every unseen sequence as

impossible. This is the same sparse-count problem seen in naive Bayes, now applied to a sequence model.



Schematic pronunciation states;
a self-loop allows variable duration.

Figure 39: Speech decoding follows a path through sound states that explain the frames.

Viterbi stores the best prefix ending at each pronunciation state. Many different word sequences can share possible sound prefixes, so searching a trellis reuses computation rather than expanding every full sequence independently. Taking logarithms turns path-probability products into sums, which makes very long sequences easier to score numerically.

A neural acoustic or language model changes how probabilities are estimated. It does not remove the distinction between observed acoustics, hidden linguistic structure, and a decision about the final transcription. Those roles are useful for understanding what each part of the system contributes.

25.2.1 From a waveform to observations

Digitization samples air-pressure variation at regular times. At an illustrative 16 kHz rate, a 25 ms segment contains 400 samples. Short-time spectral analysis decomposes such a segment into frequency contributions; a spectrogram places time on one axis, frequency on the other, and uses intensity to show energy. The acoustic observation is a vector describing a short interval, not a letter or a complete word.

Vocal-fold vibration produces a fundamental frequency and harmonics. The vocal tract's shape amplifies some frequency regions more than others, producing resonances that help distinguish vowels. Raising pitch changes harmonic spacing without necessarily changing the vowel. Volume, pitch, and the spectral envelope therefore carry different information. Transitions between neighboring sounds also change the spectrum, so cutting a waveform into fixed independent pieces does not isolate clean word labels.

An ideal tube closed at one end and open at the other illustrates the resonance mechanism. A pressure maximum at the closed end and minimum at the open end fit a quarter wavelength, giving $f_1 = c/(4L)$ and odd multiples for higher modes. With illustrative $c \approx 350$ m/s and $L = 0.175$ m, these are about $500, 1500, 2500$ Hz. A real vocal tract is not a uniform tube; its changing shape

shifts the resonances. The schematic explains why vocal tract geometry affects the acoustic features available to a recognizer.

25.2.2 Building a pronunciation-state graph

A word with a three-sound pronunciation gets states for sound 1, sound 2, and sound 3. A self-loop allows a sound to last several frames; a forward transition advances to the next sound. At the end, transitions enter the first sound state of a possible next word. Two words sharing a sound need not share the same state, because their pronunciation position and allowed continuations can differ.

For self-loop probability q and advance probability $1 - q$, remaining in a sound for exactly $d \geq 1$ frames before advancing has probability $q^{d-1}(1 - q)$. Thus duration variation is represented by paths through the same state graph. A single word label without a pronunciation cursor would not say which sound should explain the next frame. The state must summarize the information needed for its emission and legal continuation.

At word boundaries, a bigram language model contributes the probability of the next word given the preceding word. A trigram retains two previous words, which requires additional state context or an equivalent expanded decoding graph. For the example probabilities $P(\text{door} \mid \text{the}) = 0.0006$ and $P(\text{door} \mid \text{close the}) = 0.05$, the longer context raises the word's probability by a factor of about 83. It captures a phrase-level regularity that the shorter context averages together with many unrelated uses of "the." Sparse contexts require smoothing or shared learned parameters.

25.2.3 Why decoding must combine acoustic and language evidence

For each candidate pronunciation path, multiply the initial-state probability, its transition probabilities, and the emission likelihood of every acoustic frame. The language model is incorporated at the corresponding word transitions. Two word hypotheses can explain local frames similarly but differ substantially in sequence plausibility; a very common phrase can still lose if its acoustic likelihood is too poor. Neither acoustic matching nor language frequency alone answers the full inference query.

Viterbi compares competing prefixes ending at each pronunciation state and retains the best, with a backpointer. Future extensions depend on the current state, so the best prefix can be reused rather than enumerating every complete utterance separately. At the end, trace back the winning path and read off the words at its boundary transitions. Taking the highest-scoring sound at each frame independently can combine incompatible pronunciations, just as marginal state modes need not form a most likely HMM sequence.

Neural models improve the estimation of acoustic or language probabilities by sharing parameters across many observations and contexts. The relevant question remains what conditional quantity each model supplies and how it enters decoding. A fluent likely word sequence is evidence

of language-model fit; robust recognition must also account for the actual sound. Producing a transcript does not by itself establish deep understanding of its meaning.

25.2.4 Worked example: acoustic evidence can outweigh a common phrase

Compare two illustrative word-sequence hypotheses W_1, W_2, with language priors $0.9, 0.1$. Initially their acoustic likelihoods for an observed utterance are $0.01, 0.20$.

1. **Combine the two sources.** Unnormalized posterior scores are $0.9(0.01) = 0.009$ and $0.1(0.20) = 0.020$. The rarer word sequence wins.
2. **Normalize if probabilities are needed.** Among these two hypotheses, the posterior of W_2 is $20/29$. These are candidate-normalized probabilities; omitted hypotheses would also contribute to a complete denominator.
3. **Change the acoustic support.** If $P(e \mid W_1)$ rises to 0.05, its score becomes 0.045, now exceeding 0.020. Neither prior frequency nor acoustic likelihood alone determines the ranking.
4. **Express the comparison as odds.** Initially the prior odds favor W_1 by $9 : 1$, but the likelihood ratio is $0.01/0.20 = 1/20$. Posterior odds are $9/20$, favoring W_2. Log-domain decoding adds these log contributions instead of multiplying tiny path probabilities.

A pronunciation graph supplies the legal hidden paths used to compute these sequence scores. A language model can help resolve ambiguity, but fluent output without adequate acoustic support would optimize a different objective.

25.3 Connecting inference to action

A physical crawler's action can specify motor positions or a timed motion; its resulting displacement supplies reward. If the state omits the joint configuration, the same command may appear to have inconsistent effects because it starts from different mechanical situations. If observations are noisy, the agent may need an estimate of that configuration rather than treating a single reading as exact. The representation determines whether learning can distinguish those cases.

Likewise, a physical Pac-Man platform must connect a planned direction to actual wheel motion and then measure where it arrived. Slippage changes the transition distribution; delayed sensing changes what information is available when an action is selected. Repeated estimation and feedback let the agent respond to the realized outcome. A mathematically optimal plan in an inaccurate model remains conditional on assumptions that the physical system may violate.

25.3.1 Worked example: one physical transition updates one action estimate

For an illustrative discretized crawler, state s records two joint-angle bins and a contact-status flag. Action a commands a short motor movement. An observed transition gives reward 0.03 meters of forward displacement and successor state s'. Let $Q(s, a) = 0.10$, $\max_b Q(s', b) = 0.08$, $\gamma = 0.9$, and $\alpha = 0.2$.

1. **Calculate the target.** It is $0.03 + 0.9(0.08) = 0.102$. The displacement is immediate reward; the discounted successor estimate represents later displacement under the modeled objective.
2. **Calculate the correction.** The temporal-difference error is $0.102 - 0.10 = 0.002$, so the updated estimate is 0.1004.
3. **Check what the experience established.** One successful motion does not determine the exact value of the action. Slipping, changed contact, and sensor noise can alter later transitions and rewards.
4. **Check the state before reusing the value.** If identical angle bins hide different contact modes with different motion outcomes, the state can alias situations requiring different actions. Adding a relevant contact variable addresses a representation failure that changing the learning rate would not repair.

The numerical update is simple; obtaining consistent state, action, and reward measurements is what connects it to a physical learning problem. A transition model or learned value is useful only for situations matching the information on which its predictions depend.

Source. Dan Klein and Pieter Abbeel, Introduction to Artificial Intelligence, Fall 2018, Lecture 25.